

Engineered Substrates for Coplanar Integration of Lattice-Mismatched Semiconductors with Silicon

by

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B.E. Materials Science and Engineering
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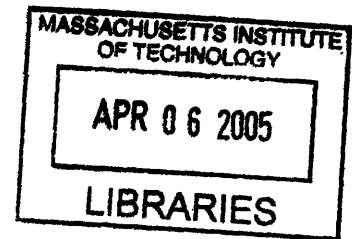


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ABSTRACT

As we approach the end of traditional CMOS scaling, further improvements in integrated circuit performance and functionality will become limited by the inherently low carrier mobility and indirect bandgap of silicon. These performance shortcomings can be supplemented with high performance semiconductors such as Ge and GaAs, which have respectively improved carrier mobilities and a direct bandgap for efficient light emission. However, due to the economic superiority of Si-based microelectronics, it is unlikely that the CMOS industry will abandon Si entirely. Instead, it will be necessary to integrate materials such as Ge and GaAs with the Si platform by means of engineered substrates. In this thesis, thin Ge layers were transferred to Si by wafer bonding of compositionally graded structures. This approach combines the beneficial aspects of graded buffers with those of wafer bonding to provide a coplanar integration platform for lattice-mismatched semiconductors.

The various innovations that were necessary to realize epitaxial layer transfer from virtual substrates stem from the fact that thin films of Ge are difficult to planarize. The large surface roughness of graded buffers requires smoothing of the surface prior to bonding. The poor surface passivation of GeO_2 in aqueous chemo-mechanical planarization (CMP) slurries necessitates that Ge virtual substrates be planarized indirectly, using a deposited CMP layer. Furthermore, H-induced exfoliation is the only practical method of separating a thin Ge layer from the surface of a virtual substrate, leading to extensive surface damage of the transferred layer. This damage is traditionally removed using a CMP step for exfoliated Si layers. However for Ge transfer, a $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop layer was incorporated for damage removal using a selective chemical etch. These techniques have enabled transfer of epitaxial Ge-on-insulator (GOI) structures to large diameter Si wafers.

Tensilely strained layers have the ability to attract interstitially-dissolved hydrogen and accelerate the nucleation of platelets- both of which contribute to the layer exfoliation process. As a result, a strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer was used to enhance the

exfoliation kinetics of Ge by providing a gettering site for ion-implanted hydrogen. During 250 °C annealing of hydrogen-implanted $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ gettering structures, preferentially-nucleated platelets are made to grow within the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer with minimal loss of hydrogen to surface effusion. Subsequent annealing at a temperature exceeding >300 °C yields significantly improved surface blistering kinetics over samples which do not contain a gettering layer.

A platelet growth model was formulated accounting for both chemical and strain energy contributions to the free energy of platelet formation. Microstructure and strain relaxation data corroborate the free energy computations, revealing two kinetically-limited regimes of platelet growth within tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layers. Low temperature annealing allows the platelets to grow in the strain-limited regime, resulting in a local platelet density of $>10^{10} \text{ cm}^{-2}$ and significantly improved exfoliation kinetics. Incorporation of strained layers has the potential of reducing the implantation dose necessary for layer transfer. Combined with virtual substrate bonding, gettering structures provide a promising solution for economical integration of high performance materials with silicon.

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Chapter 1: Introduction

1.1: Background and motivation

The overwhelming success of silicon in modern microelectronics is evident from the countless number of telecommunication, medical, manufacturing, military and consumer electronics functions that are served by the metal-oxide field effect transistor (MOSFET). The ubiquity of silicon is attributed to the economics of fabricating integrated circuits on a silicon wafer. Modern manufacturing facilities are capable of producing billions of transistors on substrates of ever-increasing diameter at a cost of only ~1 microcent per device. These economics of scale are enabled as a result of two key material characteristics of silicon. First, the mechanical strength of Si allows wafers to be manufactured in diameters significantly larger than any other semiconductor. Since integrated circuits are manufactured using batch processes, the number of devices that can be fabricated on a single wafer can be increased by enlarging the size of the substrate. Secondly, silicon's high quality native oxide plays a vital role in the dimensional scalability of a MOSFET, leading to dense integrated circuits of complimentary MOS devices (CMOS) on large diameter substrates at an economical price.

Advances in IC performance have been historically achieved through dimensional scaling of device geometries. The scaling parameter that is generally used for assessing integrated circuit performance is the gate length of the MOSFET. A decrease in gate length increases the speed at which the transistor can operate while also reducing power consumption. This trend has resulted in a steady increase of the number of transistors per integrated circuit over the last 30+ years. The frequently referenced Moore's Law¹ states that the number of transistors contained on a chip doubles every 18 months. This prediction has been remarkably accurate in modeling technology trends since the 1960s. However, according to the 2003 International Technology Roadmap for Semiconductors (ITRS)², traditional scaling will finally end shortly after the 45 nm technology node. At this point, further improvements in circuit speed and functionality will become limited by the inherently low carrier mobility of Si and its inefficiency at generating and detecting light.

Many of the III-V compounds (comprised of an element from each of the group III and V columns of the periodic table) have direct bandgaps and high carrier mobility. While these high performance semiconductors can fill in many of the performance shortfalls of silicon, it is doubtful that the microelectronics industry will abandon silicon entirely. Instead, it will be necessary to integrate lattice-mismatched semiconductors with silicon in such a way that their technological benefits are synergistically combined. For example, integration of direct bandgap semiconductors enables optical interconnects for high speed data transmission for both on and off chip level applications.³ This approach utilizes the existing CMOS processing infrastructure, thus preserving its economics of scale while increasing the functionality of the Si platform.

1.2: The engineered substrate

Semiconductor materials find applications ranging from InP heterojunction bipolar transistors⁴ for high-speed RF amplification to InSb photodiodes⁵ for infrared detection and Si MOSFETs for high-volume/low-cost digital circuitry. Each of these applications is tailored to a particular set of material properties including carrier mobility, bandgap and mechanical rigidity which are inextricably linked to the internal structure of the semiconductor. Nature rarely allows one property to be altered without simultaneously influencing another. The most frequently occurring example of this is illustrated in Figure 1.1 which depicts the general decrease of bandgap with lattice constant for several technologically relevant semiconductor materials. As a result, combining the functionality of multiple semiconductors is no easy task as many of these materials have differing lattice constants, leading to unacceptably high defect density in heteroepitaxially-grown layers. While this thesis will primarily address the challenge of lattice-mismatch, other matters pertaining to materials integration include thermal expansion mismatch and anti-phase disorder which arises during growth of compound semiconductors on elemental semiconductor substrates.^{6,7}

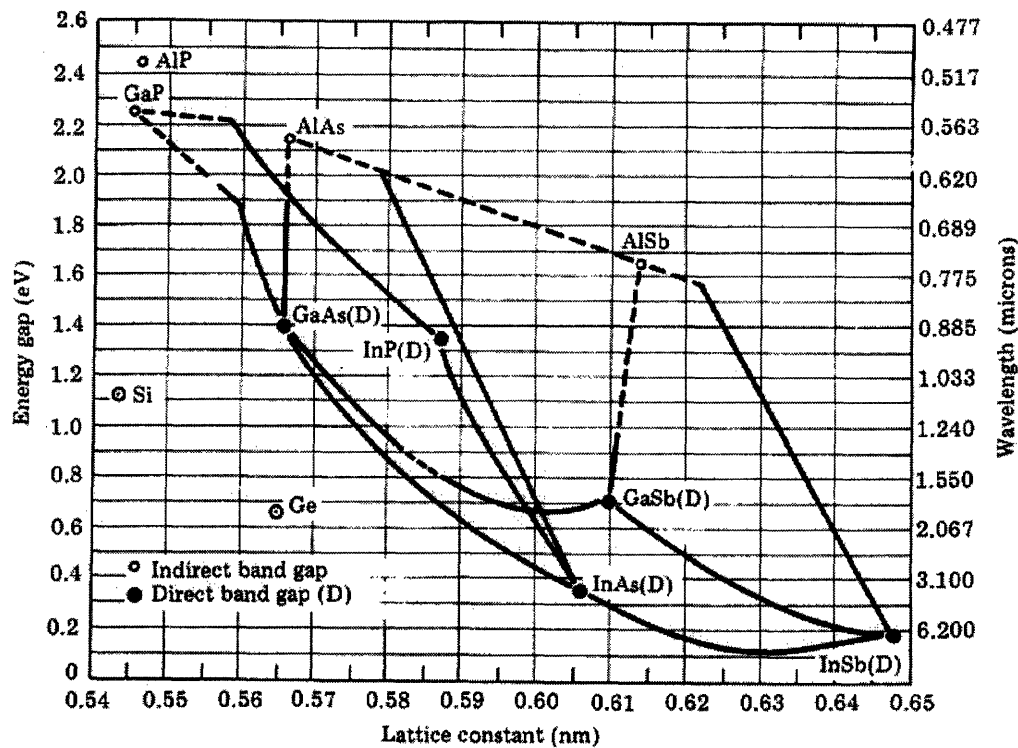


Figure 1.1: Diagram showing bandgap vs. lattice constant for a variety of semiconductor materials. Figure adapted from Mayer and Lau.⁸ Note that many technologically relevant semiconductors have a lattice constant that is larger than that of Si.

The challenges to material integration can be addressed by engineering a substrate to provide a platform where multiple semiconductors can be physically combined on a single wafer. Two highly successful technologies that have been introduced thus far are the compositionally graded buffer and the wafer bonding and layer transfer process.

The graded buffer enables fabrication of an arbitrary lattice constant by compositional grading of epitaxial layers. High quality films have been integrated with lattice-mismatched substrates using this approach for a number of semiconductor systems, including SiGe/Si⁹⁻¹¹, InGaAs/GaAs¹², InGaP/GaP¹³ and GaAsP/GaAs¹⁴. The SiGe/Si system is the natural choice for bridging the lattice-constant gap between Si and the III-V semiconductors since Ge is closely lattice-matched to GaAs, as seen in Figure 1.1. Furthermore, compositionally graded In_{1-x}Ga_xAs layers can be subsequently grown to access even larger lattice constants for integration of infrared emitter and

detector materials with silicon. Unfortunately, graded buffers require growth of thick epitaxial layers to maintain low defect density in the resulting films. This inevitably leads to large thermal strains and a device layer that is not coplanar with the Si substrate, thus complicating device integration with the underlying CMOS circuitry.

Wafer bonding and layer transfer is a second technique that can be used to integrate lattice-mismatched semiconductors. This method begins with the formation of a metallurgical bond between two dissimilar wafers followed by separation and transfer of a thin film of material to the silicon wafer.¹⁵ Wafer bonding allows fabrication of the mismatched material directly on the silicon substrate, without the need for thick epitaxial layers. However, due to the invariably small diameter of III-V substrates, this approach will always be limited to bonding wafers that have a smaller diameter than those being used in leading-edge CMOS fabrication facilities. Both of these methods will be discussed in detail in Chapter 2, leading to an engineered substrate which combines the advantageous features of compositionally graded buffers and layer transfer.

1.3: Thesis organization

The format of this thesis is a didactic narrative of the technical issues pertaining to coplanar integration of lattice-mismatched semiconductors. The primary focus is on wafer bonding and layer transfer from compositionally graded buffers. Each chapter is devoted to solving a particular technical problem and concludes with the introduction of an unresolved issue which is addressed in subsequent fashion.

Chapter 2 lays down the theoretical background necessary for tackling the general issues of lattice-mismatched integration. Two approaches to integration are discussed in detail: compositional grading and wafer bonding. The chapter concludes with a novel method of layer transfer from compositionally graded buffers, emphasizing the new challenges specific to this combined approach. The remainder of the thesis addresses these issues in effort to fabricate lattice mismatched layers directly on large diameter silicon wafers.

Chapter 1: Introduction

Chapter 3 presents the experimental background for the thesis. This chapter provides an overview of ultra high vacuum chemical vapor deposition- the epitaxial technique that is employed in this work. The experimental details pertaining to preparation of $\text{Si}_{1-x}\text{Ge}_x$ graded buffers are discussed including growth on double-side polished wafers to minimize thermally-induced wafer bow. The chapter concludes with a description of the material characterization techniques that are used throughout the thesis.

Chapter 4 addresses the challenge of chemo-mechanical planarization (CMP) of virtual substrates- a process that is necessary to provide a smooth surface for wafer bonding. This chapter explains the fundamental limitations to CMP of SiGe compositionally graded layers leading to the implementation of deposited Si and SiO_2 layers for indirect planarization of Ge virtual substrates. The chapter concludes with a demonstration of epitaxial Ge transferal from a virtual substrate to Si using an epitaxial Si CMP layer and the hydrogen-induced layer exfoliation process. This result highlights the limitations of CMP techniques for removal of exfoliation-induced damage in transferred Ge layers, thereby motivating the work presented in Chapter 5.

Chapter 5 presents a novel CMP-less approach for the removal of surface damage in exfoliated Ge layers by using an etch-stop layer and selective chemical etching. The focus of this chapter is growth and characterization of tensilely strained SiGe films for etch-stop applications. This chapter also investigates initiation of homoepitaxial growth of Ge for preparation of etch-stop structures on bulk Ge wafers.

Chapter 6 describes the hydrogen-gettering behavior of tensilely strained SiGe layers and their applicability for lowering the thermal budget of the layer exfoliation process. This chapter presents experimental evidence of two independent gettering mechanisms: preferential accumulation of implanted hydrogen and nucleation of (100) and (010) platelets in biaxially strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ on Ge ($\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$). The gettering behavior of stained $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ is investigated in terms of hydrogen distribution, platelet morphology and strain relaxation as a function of annealing temperature. The gettering behavior of tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ is shown to accelerate the exfoliation kinetics of hydrogen-implanted Ge.

Chapter 7 provides the theoretical basis for the experimental observations made in Chapter 6. A free energy model is formulated for nucleation and growth of platelets in tensilely strained layers. Platelet growth simulations are then compared to experimental data revealing strain and hydrogen diffusion-limited regimes that occur during platelet nucleation and growth.

Finally, Chapter 8 summarizes the major results of the thesis and suggests direction for future work.

Chapter 2: Integration of lattice-mismatched semiconductors

The goal of this work is to develop a technology which enables integration of lattice-mismatched semiconductors on large-diameter Si substrates. A review of existing integration technologies motivates the approach taken here, namely wafer bonding of compositionally graded buffers. This chapter presents the material issues pertaining to lattice-mismatched epitaxy and wafer bonding, thus providing the groundwork for the thesis.

2.1: Introduction

Every technique for lattice-mismatched integration can be categorized based on the point at which the differing materials are combined within the processing sequence. Figure 2.1 illustrates the two general approaches that can be used. Figure 2.1a depicts a hybrid integration technique where prefabricated Si and III-V components are individually assembled at the back end of device production. One example of this method is flip-chip bonding of III-V optical emitters and detectors to pre-fabricated CMOS circuitry for telecommunication applications. This integration scheme requires two fabrication facilities since III-V manufacturing techniques are generally incompatible with CMOS. While hybrid techniques have their place in low-volume assembly of specialty products, they cannot be integrated with high-volume batch processing of wafers. An alternate approach, depicted in Figure 2.1b, requires integration of the mismatched III-V layer on the Si substrate prior to device fabrication. These so-called monolithic platforms allow both materials to be co-processed using a single mask set for lithographic patterning of device features. By blanketing the III-V layers with a passivation layer such as Si, SiO₂ or Si₃N₄, both III-V and CMOS device layers can be co-processed in a single fabrication facility. Most importantly, the monolithic method capitalizes on the state-of-the-art processing infrastructure already in place for CMOS circuit fabrication, allowing for high volume integration of high performance materials with CMOS circuitry on large Si wafers.

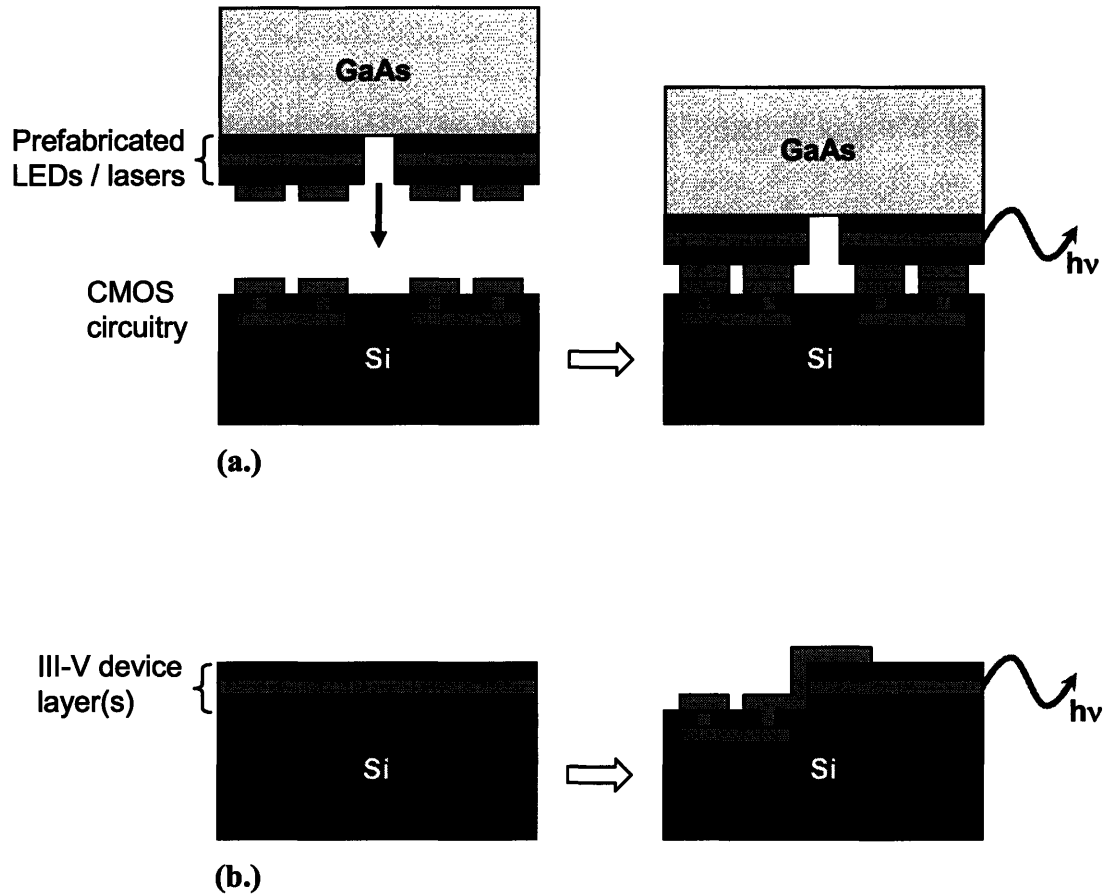


Figure 2.1: A comparison of hybrid and monolithic integration schemes. (a.) The hybrid method combines pre-fabricated components at the back end of device production. (b.) Monolithic integration utilizes a single mask set to co-process the lattice mismatched and Si device layers on the same substrate.

The full economic and performance benefits of material integration with silicon can be realized only with the monolithic approach, requiring that the mismatched layer be fabricated directly on the Si wafer. Unfortunately, monolithic integration techniques have been limited by an assortment of technical challenges related primarily to defect nucleation. Many of these areas have been an active area of research since the 1960s.¹⁴ The result of these efforts has led to two independently developed technologies: lattice-mismatched epitaxy and layer transfer which are now reviewed.

2.2: Lattice-mismatched epitaxy

The most direct approach to monolithic integration is epitaxial growth of the mismatched layer directly on the Si substrate. The main difficulty with this method is caused by the differing lattice constant between Si and many other technologically-relevant semiconductors. Lattice mismatch can induce considerable strain in the epitaxial layer leading to rampant nucleation of deleterious dislocations. For example, the lattice constant between Si and GaAs is mismatched by 4.1%, resulting in a threading dislocation density as high as 10^8 cm^{-2} in GaAs films deposited directly on Si.¹⁶ These defects behave as scattering and non-radiative recombination centers and must be therefore minimized to attain high-performance majority and minority carrier devices.

The subject of strain relaxation can be classified by the thermodynamic and kinetic aspects of dislocation formation. Thermodynamics yields equilibrium models for strain relaxation based on energy balance considerations. From thermodynamics the well-known Matthews-Blakeslee equation is derived, defining the condition for which dislocation formation at the interface between coherently-strained epitaxial layers is favored.¹⁷ Meanwhile, kinetic models are necessary to describe strain relaxation in terms of dislocation glide velocity and nucleation rates.

This section reviews the theoretical models necessary for controlling dislocations during lattice-mismatched epitaxy, leading to growth of high quality mismatched layers by means of compositionally graded buffers. A more extensive discussion of dislocation theory as it pertains to lattice-mismatched epitaxy can be obtained from Fitzgerald.¹⁸

2.2.1: Defect nucleation and strain relaxation

An epitaxial film with a lattice constant, a_f grown on a substrate with a lattice constant, a_s will experience a lattice mismatch, f given by Equation 2.1.

$$f = \frac{a_s - a_f}{a_f} \quad \text{Equation 2.1}$$

If the mismatch between the film and substrate is small (less than $\sim 2\%$), a layer below a critical thickness will grow in a state of pseudomorphic strain, taking on the in-plane lattice constant of the substrate as shown in Figure 2.2a. The out-of-plane lattice constant also deforms according to the Poisson ratio of the film, leading to a tetragonal distortion of the epitaxial deposit. As the thickness of the layer increases, it will accumulate elastic strain energy until a critical thickness, h_c . At this thickness, the layer undergoes plastic deformation where accumulated strain energy is released by formation of a misfit dislocation at the interface, as illustrated in Figure 2.2b for the case of $a_f > a_s$.

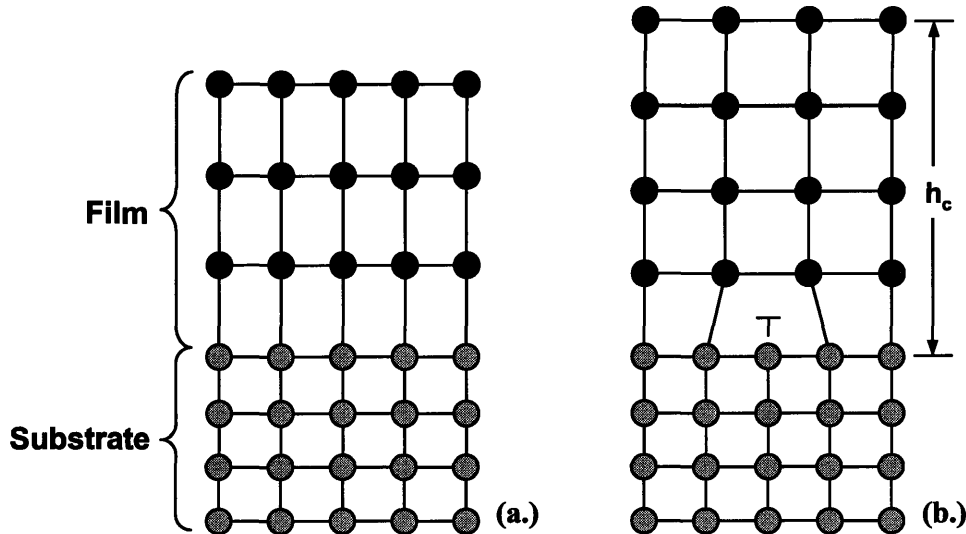


Figure 2.2: Illustration of a lattice-mismatched layer grown below (a.) and above (b.) the critical thickness. Supercritical layers relax through formation of misfit dislocations at the interface.

Misfit dislocations form at the expense of introducing line energy to the crystal. However, above the critical thickness it is energetically favorable for the dislocation to form in exchange for the strain energy released by the layer during relaxation. This can be expressed mathematically by minimizing the total energy of the system, leading to the classic Matthews-Blakeslee equation given by:

$$h_c = \frac{\bar{\mu}(1 - \nu \cos^2 \alpha)(b/b_{\text{eff}})[\ln(h_c/b) + 1]}{2Mf} \quad \text{Equation 2.2}$$

where $\bar{\mu}$ is the average shear modulus of the interface, ν is the Poisson ratio of the film, α is the angle between the dislocation line direction and the Burgers vector, b is the magnitude of the Burgers vector, b_{eff} is the magnitude of the edge component of the Burgers vector lying in the interface plane and M is biaxial modulus of the film.

Figure 2.3 illustrates the geometric aspects of dislocation formation in a lattice mismatched layer. Most semiconductors have either the diamond-cubic or zinc-blende crystal structure, which share the $a/2 \langle 110 \rangle \{111\}$ slip system. Equivalently, glissile dislocations have a Burgers vector of $a/2 \langle 110 \rangle$ and glide on $\{111\}$ slip planes. The line direction of the misfit dislocation, ξ is defined by the intersection of the $\{111\}$ slip planes with the mismatched interface. Therefore, strain relaxation of a lattice-mismatched layer grown on a (001) surface is characterized by formation of misfit dislocations along the mutually perpendicular $[\bar{1}10]$ and $[110]$ directions.

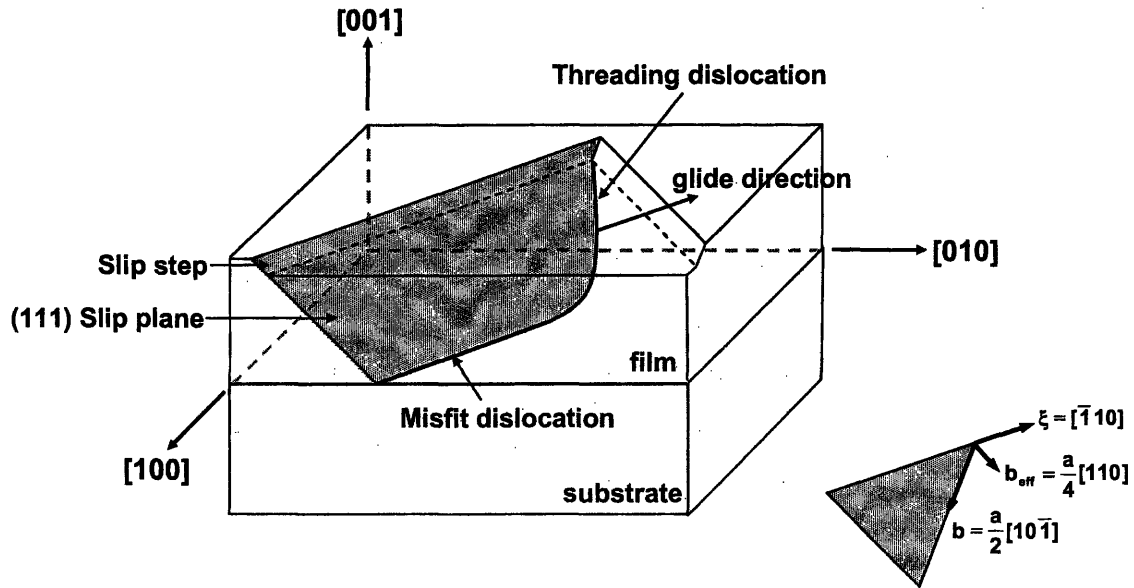


Figure 2.3: Illustration of misfit and threading dislocations within a lattice-mismatched film grown beyond the critical thickness. Dislocations cannot terminate within a perfect crystal; therefore misfit dislocations exit the film at the surface in the form of a threading dislocation. The Figure illustrates a case for compressive strain relaxation ($a_f > a_s$) in the lattice-mismatched layer as explained in the text.

The significance of b_{eff} stems from the fact that only the edge component of a dislocation is capable of relaxing strain. Strain relief is most efficient for dislocations that are in perfect edge orientation with a Burgers vector that is in the plane of the mismatched film. However, these dislocations are sessile and cannot glide to relax strain. Therefore glissile 60° dislocations are favored in low-mismatch ($f < 2\%$) systems where the misfit has both edge and screw components and a Burgers vector that is oriented 60° relative to the line direction, as shown in the inset of Figure 2.3. Consequently, the magnitude of the effective Burgers vector for lattice mismatch strain relief of diamond cubic semiconductors is $b/2$ or equivalently, $a/4 \langle 110 \rangle$.

The cross product between the line direction and the effective Burgers vector specifies the orientation of the extra half-plane of atoms relative to the mismatched interface. Consequently the direction of b_{eff} relative to ξ also denotes the orientation of strain relief. The Burgers vector portrayed in Figure 2.3 corresponds to relaxation of a

compressively strained layer since $\xi \times \mathbf{b}_{\text{eff}}$ points toward the substrate, matching the atomistic illustration used in Figure 2.2b.

No dislocation can terminate within a perfect crystal. Therefore, misfit dislocations exit the film at the surface in the form of a threading segment as indicated in Figure 2.3. Threading dislocations contribute to performance degradation and failure for devices fabricated on the surface. While misfit dislocations are thermodynamically necessary for strain relaxation, their threading counterpart only exists as a vehicle for introducing misfits to the interface. Fundamentally, zero threading dislocations are necessary to relax a strained layer, leading to continuing efforts toward reducing their density in heteroepitaxial layers.

Figure 2.4 is a mechanistic depiction of strain relaxation in supercritical layers. When a layer exceeds the critical thickness, threading dislocations will glide leaving behind strain-relaxing misfits at the interface. Matthews, et al.^{17,19} balanced the Peach-Kohler force, F_e on the threading segment with the line tension, F_ℓ of the misfit segment, yielding an alternate derivation of the critical thickness. This view of strain relaxation leads to kinetic models for dislocation formation in mismatched layers. Since misfits are formed by glide of threading dislocations, the rate at which strain is relieved is directly related to the number of available threading dislocations and their glide velocity. Consequently, the number of threading dislocations necessary for strain relaxation of a mismatched layer is reduced by minimizing the lattice strain and maximizing the dislocation glide velocity- a concept that is exploited during growth of compositionally graded buffers.

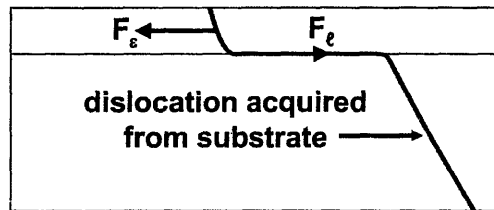


Figure 2.4: Misfit formation by threading dislocation glide. Above the critical thickness, the Peach-Kohler force, F_e , on the threading dislocation exceeds the line tension, F_ℓ , of the misfit dislocation thus resulting in glide of the threading segment.

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The misfit dislocation density, ρ_m is related to plastic strain relaxation, δ by Equation 2.3. Unlike thread density, which is expressed as the number of threading dislocations intersecting the surface per unit area [cm^{-2}], the misfit density is defined as the total misfit *length* per unit area, yielding units of cm^{-1} .

$$\delta = \frac{b_{\text{eff}} \rho_m}{2} \quad \text{Equation 2.3}$$

A fully relaxed ($\delta=f$) GaAs layer grown on Si requires a misfit dislocation density of $4.1 \times 10^6 \text{ cm}^{-1}$ or equivalently, an average misfit spacing of 2.4 nm. Formation of a misfit array with this density will invariably require rampant nucleation of threading dislocations, precluding growth of GaAs layers directly on Si for subsequent device integration. Such a small dislocation spacing will also lead to numerous dislocation blocking events²⁰, which will require further nucleation of threads. The compositionally graded buffer, discussed in the following section, provides a solution to this problem by relaxing large lattice strains across many low-mismatch interfaces.

2.2.2: Compositionally graded buffers

Many technologically important semiconductors have a large lattice constant mismatch with Si as previously seen in Figure 1.1. The dislocation nucleation rate in these systems can be managed by relaxing their large strains incrementally with a compositionally graded buffer.⁹ Graded buffers provide an epitaxial approach for fabricating an arbitrary lattice constant on a Si wafer, thereby creating a virtual substrate for subsequent material integration. Using the graded buffer approach, a large lattice mismatch is diluted over many low-mismatch interfaces thereby reducing the strain relaxation across any one layer. Furthermore, a single threading dislocation can be used to extend misfit dislocations in multiple grading steps as illustrated in Figure 2.5.

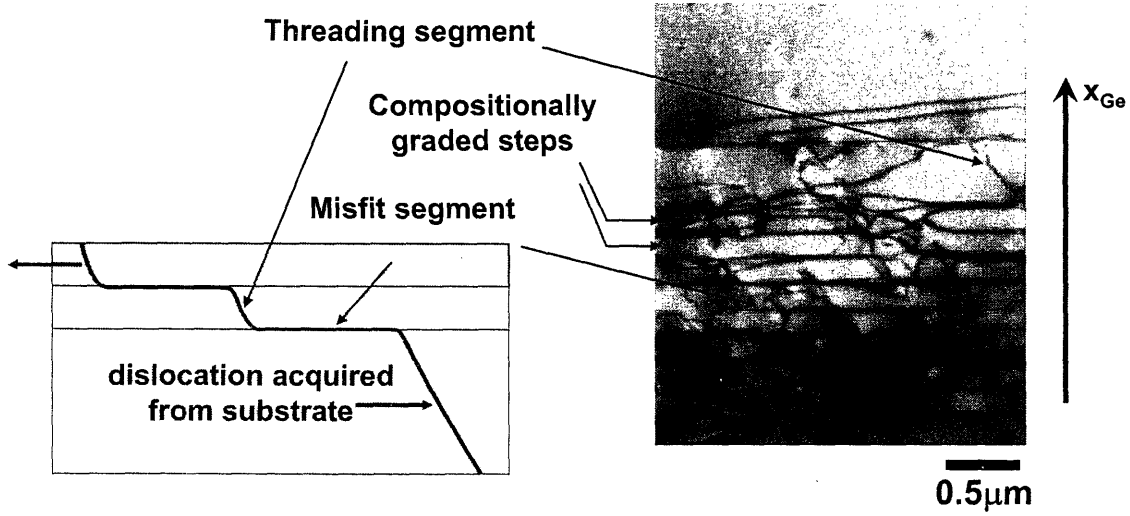


Figure 2.5: Illustration of strain relaxation in compositionally graded buffers. Large lattice strains are relaxed across many low-mismatch interfaces thereby limiting the nucleation rate of threading dislocations. Meanwhile, a single threading dislocation can be re-used in multiple grading steps. The accompanying TEM micrograph illustrates this process in a $\text{Si}_{1-x}\text{Ge}_x$ graded buffer.

The equilibrium threading dislocation density, ρ_t in a graded buffer is established by a balance between strain relaxation and thread nucleation. At equilibrium, the strain relaxation rate is proportional to the density of mobile threads multiplied by their glide velocity. In this glide-limited regime, the steady-state value of ρ_t is given by Equation 2.4²¹, where R_g is the growth rate [$\mu\text{m}/\text{min}$], R_{gr} is the grading rate [$\%\text{Ge}/\mu\text{m}$], E_{glide} is the activation energy for dislocation glide, B and m are constants where m is typically ~ 1 and ϵ_{eff} is the effective stress in the heteroepitaxial layer that drives dislocation glide.

$$\rho_t = \frac{2R_g R_{gr} \exp\left(\frac{E_{\text{glide}}}{k_B T}\right)}{bBY^m \epsilon_{\text{eff}}^m} \quad \text{Equation 2.4}$$

According to Equation 2.4, the threading dislocation density of compositionally graded buffers is reduced by maximizing the growth temperature and minimizing the growth and grading rates. The activation energy is always smaller for dislocation glide

than for nucleation. Consequently, the thread density during growth of compositionally graded buffers is generally glide-limited meaning that threading dislocations nucleate only when their glide velocity is insufficient to relax the strain rate imposed by the product of R_g and R_{gr} . However, at very high growth temperatures it is possible to enter a nucleation dominated regime where the thread density *increases* with temperature. This is particularly evident in materials with low yield strengths such as Ge which require an optimization of growth temperature to yield high quality virtual substrates.²²

According to Equation 2.4, ρ_t is independent of the total amount of strain relaxed within the graded buffer. In practice, threading dislocations can encounter obstacles inhibiting glide and thus rendering them ineffective for relaxing strain. This necessitates nucleation of a new threading dislocation leading to an increase in ρ_t . One blocking mechanism that is common in graded buffers is illustrated in Figure 2.6.¹⁰ Strain fields from misfit dislocations cause local depressions in growth rate, leading to the crosshatched surface morphology characteristic of compositionally graded buffers. The surface undulations combine with the underlying strain field to block dislocations gliding in the perpendicular direction. This blocking mechanism can be removed by planarizing the surface, allowing the trapped dislocations to glide and potentially annihilate in subsequently grown layers.²³

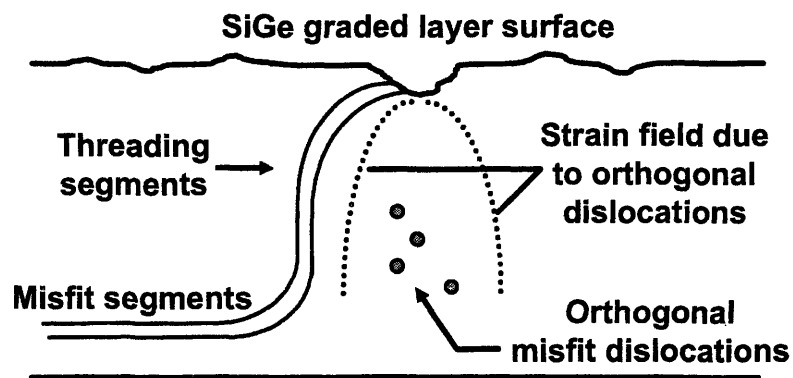


Figure 2.6: Illustration of a dislocation blocking mechanism in compositionally graded buffers. Local undulations of the surface coincide with the strain field generated by underlying misfit dislocations, posing an obstacle to dislocation glide. Figure adapted from Samavedam.¹⁰

Compositionally graded buffers have been used to provide low defect densities in mismatched systems such as SiGe/Si⁹⁻¹¹, InGaAs/GaAs¹² and InGaP/GaP¹³. The SiGe system is particularly useful for III-V integration with Si for two fundamental reasons. First, the SiGe system is fully miscible as evidenced by the phase diagram given in Figure 2.7.²⁴ This allows growth of thermodynamically stable Si_{1-x}Ge_x layers of arbitrary lattice constant ranging between that of Si and Ge. Secondly, the lattice mismatch between GaAs and Ge is only 0.07%. Consequently, SiGe buffer layers graded to Ge provide a platform for subsequent GaAs growth, thus bridging the gap between Si and III-V integration. The low defect density of these virtual substrates has enabled fabrication of compound semiconductor lasers^{25,26} and an optical circuit²⁷ utilizing an LED emitter, waveguide and detector all monolithically integrated on a Si wafer.

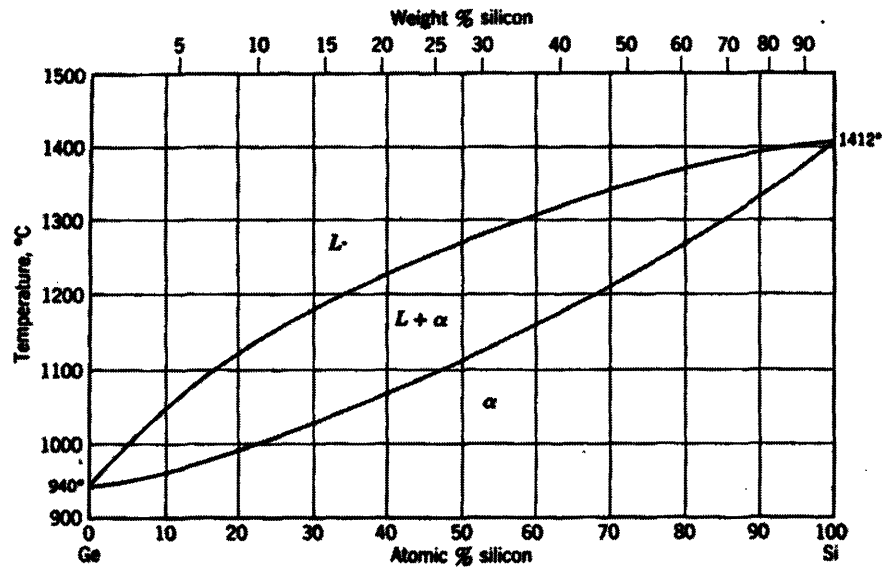


Figure 2.7: SiGe phase diagram showing complete solid solution miscibility of Si_{1-x}Ge_x alloys. Figure adapted from Gandhi.²⁴

2.3: Wafer bonding and layer transfer

Despite its ability to produce lattice-mismatched epitaxy of unprecedented quality²⁸, the virtual substrate requires growth of a thick graded buffer to maintain low strain levels throughout the structure. Ge virtual substrates typically require more than 10 μm of epitaxial growth to maintain high film quality. Such thick layers complicate

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subsequent device integration with the underlying Si substrate since the device levels are not coplanar and must be processed across a deep step.

Another approach that eliminates the need for thick epi-layers involves transferal of the mismatched film from one wafer to another by means of direct wafer bonding. In this technique, two wafers are used for fabricating a thin, lattice-mismatched layer directly on Si. The first wafer, denoted here as the donor wafer is brought into contact with a second Si handle wafer. Upon annealing, strong covalent bonds are formed across the bond interface. Finally, a thin layer of material is transferred from the donor to the handle by means of a variety of layer separation techniques which will be discussed in Section 2.4.3. This technique is similar to the procedure used in the hybrid integration approach, except that during direct wafer bonding the mismatched layer is transferred to the Si substrate prior to device fabrication. Whereas the hybrid method involves transferal of completed circuits or discrete devices, monolithically integrated layers are co-processed with the Si devices on the same wafer.

The wafer bonding approach does not eliminate the misfit dislocations at the film/substrate interface which are geometrically necessary to accommodate lattice mismatch between the bonded pair, as indicated in Figure 2.8a. However, the misfits that are introduced during wafer bonding form without dislocation glide. In contrast, an epitaxial layer is fully coherent to the substrate prior to strain relaxation. Upon reaching the critical thickness, misfits are formed by dislocation glide, necessitating threading dislocations as indicated in Figure 2.8b. Consequently, highly mismatched layers can be transferred to a Si wafer without the threat of creating dislocations in the device layer.

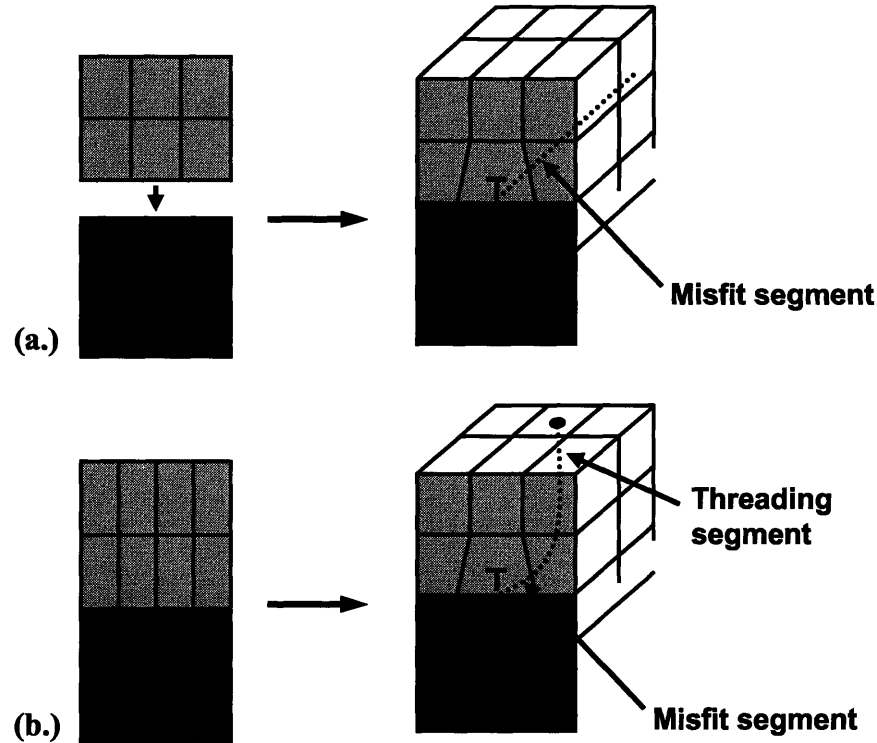


Figure 2.8: Two mechanisms for the formation of misfit dislocations at the interface of lattice-mismatched crystals. (a.) During wafer bonding, the misfits are created by short atomic displacements to accommodate the geometry of the interface. (b.) Epitaxial layers are initially coherent with the substrate and relax by dislocation glide. Using wafer bonding, lattice-mismatched layers can be placed directly on a Si substrate without nucleation of deleterious threading dislocations.

2.3.1: Bulk wafer bonding

Both GaAs and Ge have a coefficient of thermal expansion (CTE) that is larger than that of Si. As a result, bonding of bulk wafers is limited by the thermal stress that arises during high temperature annealing of bonded wafer pairs. The thermal strain energy scales linearly with the wafer thickness. For elastically deformed materials, the strain energy of two thermally mismatched wafers of thickness h_1 and h_2 can be determined from the stress profile, $\sigma(z)$ and biaxial moduli of the bonded pair using Equation 2.5, where the subscripts 1 and 2 denote the first and the second wafer, respectively.

Chapter 2: Integration of lattice-mismatched semiconductors

$$E_{el} = \int_0^{h_1} \frac{\sigma_1^2(z)}{2 M_1} dz + \int_{-h_2}^0 \frac{\sigma_2^2(z)}{2 M_2} dz \quad \text{Equation 2.5}$$

The stress profile in each wafer is determined using bending mechanics, leading to Equations 2.6a and 2.6b.

$$\sigma_1(z) = \frac{1}{\rho} \left[\left(z - \frac{h_1}{2} \right) M_{si} - \frac{2}{h_1 (h_1 + h_2)} \left(\frac{M_1 h_1^3 + M_2 h_2^3}{12} \right) \right] \quad \text{Equation 2.6a}$$

$$\sigma_2(z) = \frac{1}{\rho} \left[\left(z - \frac{h_2}{2} \right) M_{2s} - \frac{2}{h_2 (h_1 + h_2)} \left(\frac{M_2 h_2^3 + M_1 h_1^3}{12} \right) \right] \quad \text{Equation 2.6b}$$

where ρ is the radius of curvature of the bonded pair, given by:

$$\rho = \frac{(M_2 h_2^3 + M_1 h_1^3)(M_2 h_2 + M_1 h_1) + 3 M_2 M_1 h_2 h_1 (h_2 + h_1)^2}{6 M_2 M_1 h_2 h_1 (h_2 + h_1)(\alpha_1 - \alpha_2) \Delta T} \quad \text{Equation 2.7}$$

where α is the coefficient of thermal expansion and ΔT is the temperature change $T - T_0$, where T_0 is the reference temperature at which the wafer pair is stress-free. The biaxial modulus for cubic materials with a (001) substrate orientation is given in terms of the stiffness parameters by Equation 2.8.

$$M^{(001)} = C_{11} + C_{12} - \frac{2 C_{12}^2}{C_{11}} \quad \text{Equation 2.8}$$

	Si	Ge	GaAs
C_{11} [N/m ²]	1.66×10^{11}	1.24×10^{11}	1.19×10^{11}
C_{12} [N/m ²]	6.39×10^{10}	4.13×10^{10}	5.38×10^{10}
C_{44} [N/m ²]	7.96×10^{10}	6.83×10^{10}	5.99×10^{10}
α [K ⁻¹]	2.59×10^{-6}	5.8×10^{-6}	6.86×10^{-6}
$\gamma^{(001)}$ [J/m ²]	1.34	--	0.91

Table 2.1: Material parameters relevant to thermo-elastic energy calculations for bulk wafer bonding.

Applying the material parameters given in Table 2.1, the elastic strain energy between two 525 μm thick Si and GaAs wafers, bonded at 700 °C is 320 J/m². The bonded pair will be thermodynamically stable unless the strain energy exceeds the energy of adhesion given by:

$$E_{\text{ad}} = \gamma_{\text{Si}} + \gamma_{\text{GaAs}} - \gamma_{\text{int}} \quad \text{Equation 2.9}$$

where γ_{Si} and γ_{GaAs} are the unreconstructed surface energies of Si and GaAs, respectively and γ_{int} is the energy of the interface. Assuming a negligible interface energy, i.e. a perfect bond, the highest possible bond energy between (001) Si and GaAs surfaces is 2.25 J/m². Upon comparison of this result to the thermal strain energy of the bonded pair, $E_{\text{el}} \gg E_{\text{ad}}$ which leads to the experimentally verified conclusion that high temperature bonding of bulk Si and GaAs wafers is unfeasible.²⁹

The thermal strain energy arising during wafer bonding can be reduced by decreasing the thickness of the bonded materials. For example, the strain energy of a 100 nm GaAs layer bonded to a Si substrate at 700 °C is only 50 mJ/m². This value is significantly lower than the combined surface energy of Si and GaAs, yielding a mechanically stable bond interface. Transferal of thermally mismatched layers is

complicated by the large strain energies generated during the bonding of bulk wafers. This problem can be solved by replacing the vehicle for layer transfer with a virtual substrate, wherein the thermal mismatch is effectively confined to a thin transfer layer.

2.3.2: Virtual substrate bonding

Stress arising between thermally mismatched wafers can be managed by minimizing the annealing temperature of the bonded pair. The layer exfoliation process, which will be described in Section 2.4.3 allows separation of the thermally mismatched layer from the donor wafer at low temperature and has been demonstrated for transferal of GaAs³⁰ or Ge³¹ layers to Si from bulk GaAs and Ge wafers, respectively. A more serious limitation to bulk wafer bonding is the wafer size mismatch between Si and III-V or Ge wafers. Films transferred from III-V-sized wafers are limited to use in trailing-edge fabrication facilities using small diameter Si wafers. Although a limited supply of Ge bulk wafers is available in large diameters, their current production volume is too small to sustain the CMOS industry.

Fortunately, the advantages of virtual substrates and proven layer transfer processes via wafer bonding can be combined. By transferring epitaxial layers from virtual substrates, Ge or III-V films can be fabricated in close proximity to a Si substrate while taking advantage of the full diameter of modern Si wafers. This approach also eliminates the thermal stress that arises during bulk wafer-bonding since the bulk composition of both donor and handle wafers is Si. An illustration of the virtual substrate wafer bonding scheme is given in Figure 2.9.

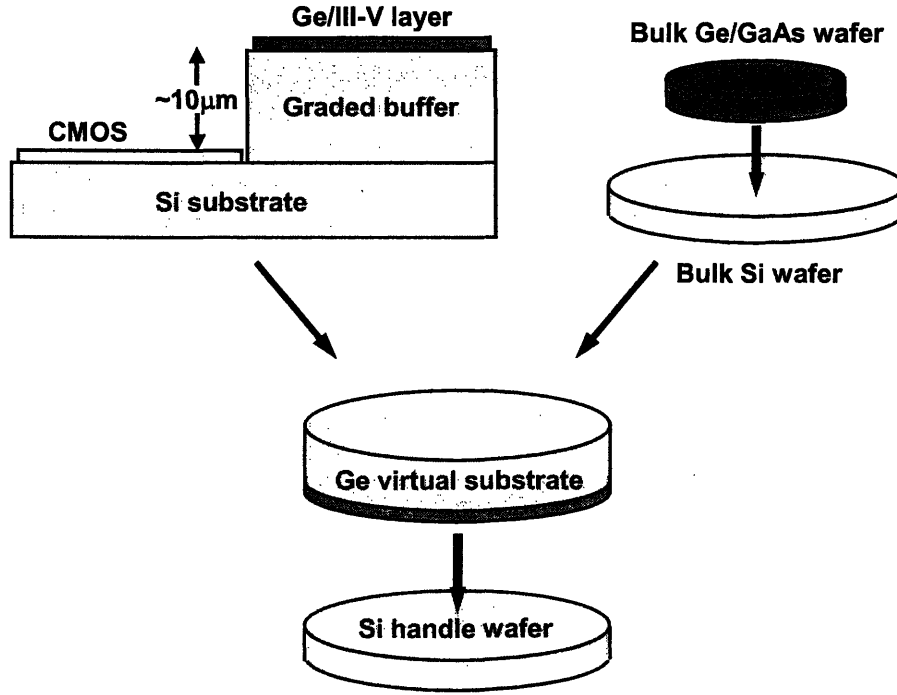


Figure 2.9: Schematic showing a material integration technique utilizing wafer bonding of virtual substrates. Layer transfer from virtual substrates enables formation of lattice-mismatched semiconductors directly on a silicon wafer thereby eliminating the wafer size and thermal mismatch constraints of bulk wafer bonding.

The benefits of virtual substrate bonding are somewhat offset by the surface roughness and wafer bow that forms during growth of relaxed graded buffers. The bondability of bowed wafers can be evaluated using gap-closing theories which are derived by considering the energy balance between substrate deformation and the surface energy released during bonding of wafers with flatness non-uniformities.³² Assuming complete surface contact, wafers having a flatness non-uniformity, h , over a lateral extension R will bond if the condition given by Equations 2.10a or 2.10b is met, where h_w is the wafer thickness and $Y' = Y/(1 - \nu^2)$.

$$h < \frac{R^2}{\sqrt{\frac{2 Y' h_w^3}{3 E_{ad}}}} \quad R > 2 h_w \quad \text{Equation 2.10a}$$

$$h < \frac{18}{5} \sqrt{\frac{R E_{ad}}{Y'}} \quad R < 2h_w \quad \text{Equation 2.10b}$$

In practice, 100 mm Si wafers with 525 μm thickness can tolerate bow of up to 25 μm during room temperature bonding, where a surface energy release of 0.15 mJ/m^2 is typical for van der Waals attraction between two hydrophobic Si surfaces. This number is in good agreement with the gap-closing model. However, growth of Ge virtual substrates can generate a wafer curvature of as high as 35 μm measured from center to edge of a 100 mm wafer. Such a large deflection cannot be as easily accommodated during wafer bonding.

A simple solution to the wafer bow problem is growth of the graded layers on double-side polished wafers using a hot-walled CVD reactor. In this way, deposition occurs equally on both sides of the substrate, eliminating any net stress which would lead to wafer bow if deposition were to occur on only one side. The experimental details for growth of flat virtual substrates are provided in Chapter 3.

A more imposing problem with virtual substrate bonding is the inherently large surface roughness of compositionally graded buffers. Gap closing models do not consider microscopic undulations of a wafer surface, which have a strong effect on the actual contact area at the bond interface. An alternate model that correlates surface roughness to the bondability of two wafers has been proposed by Gui *et al.*³³ in terms of a dimensionless adhesion parameter θ :

$$\theta = \frac{Y^*}{E_{ad}} \sqrt{\frac{\sigma^3}{\omega}} \quad \text{Equation 2.11}$$

where $Y^* = \frac{4}{3}[Y/(1-\nu^2)]$, σ is the standard deviation of the surface height (the RMS roughness) and ω is the average periodicity of the surface asperities. When $\theta \geq 12$, the

contact area approaches zero and spontaneous room temperature bonding is not observed. When $\theta \leq 1$, full contact is obtained and a complete release of surface energy results in spontaneous bonding of the wafer surfaces.

An RMS surface roughness of less than 0.8 nm measured on $10 \times 10 \mu\text{m}$ scale can be tolerated for room temperature bonding of hydrophilic Si surfaces, although bond propagation occurs slowly and only with application of pressure.³³ Spontaneous room-temperature bonding requires an RMS roughness of ~ 0.1 nm. Ge virtual substrates have a surface roughness of 10-15 nm on $25 \times 25 \mu\text{m}$ scale- much too high for spontaneous bonding. Consequently, in order to reduce their high as-grown surface roughness, virtual substrates require a chemo-mechanical polishing (CMP) step prior to wafer bonding. The process of planarization poses a particular problem for Ge virtual substrates, as discussed in detail in Chapter 4.

2.4.3: Layer separation techniques

The most common methods used for separating a film from a wafer-bonded substrate are mechanical grind with etch-back and layer exfoliation. The grind and etch-back technique, illustrated in Figure 2.10a, requires a chemical stop layer to selectively etch away the donor wafer. After wafer bonding, most of the donor is first removed by mechanical grinding to a thickness of $\sim 50 \mu\text{m}$. The remainder of the substrate is then selectively etched away, stopping on an etch-stop layer incorporated near the surface of the donor wafer. Heavily doped p-type Si has good etch stopping behavior in potassium hydroxide (KOH) and tetramethylammonium hydroxide (TMAH) solutions, a property utilized in the fabrication of the first silicon-on-insulator (SOI) substrates.³⁴ Similar selectivity is observed between Si and $\text{Si}_{1-x}\text{Ge}_x$ with $x > 0.2$ ^{35,36}, allowing transferal of $\text{Si}_{0.75}\text{Ge}_{0.25}$ layers from virtual substrates using the relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ cap as a natural etch-stop.^{36,37} However, there is no known etch that exhibits such selectivity to pure Ge across the entire $\text{Si}_{1-x}\text{Ge}_x$ composition range. Accordingly, the grind and etch-back approach is not practical for transferal of relaxed Ge layers from virtual substrates.

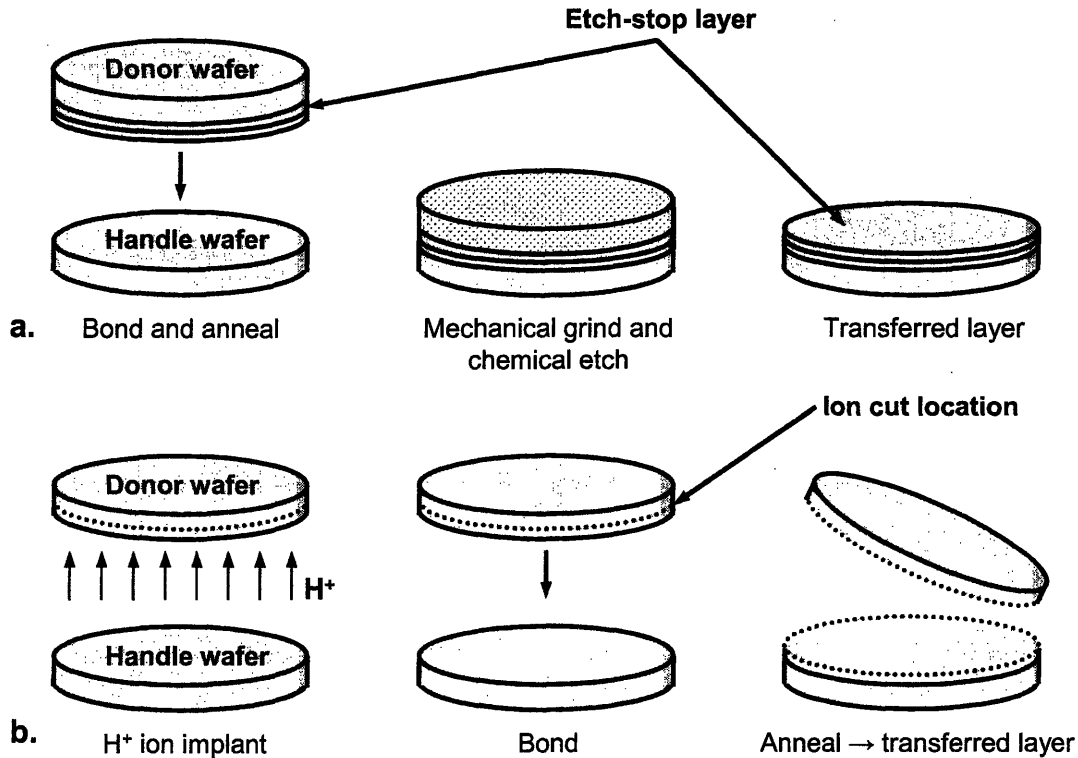


Figure 2.10: Schematic of two techniques for separating a layer from a bonded donor wafer. (a.) Grind and etch-back requires destructive removal of the donor wafer by mechanical grinding followed by selective chemical etching. (b.) Layer exfoliation requires H⁺ or He⁺ ion implantation of the donor wafer prior to bonding. Separation of the surface layer occurs during annealing of the bonded pair resulting in layer transfer to the handle wafer.

A more recent approach for layer separation involves H⁺ or He⁺ ion implantation into the donor wafer prior to bonding, as illustrated in Figure 2.10b. This process is generally termed “layer exfoliation” and commercially known as Smartcut™.³⁸ Layer exfoliation relies on the limited solubility of H₂ or He in a solid, causing the implanted ions to condense into gas-filled voids below the wafer surface. Upon wafer bonding and annealing, these voids coalesce into cracks causing separation of the surface layer from the donor wafer resulting in layer transfer to the handle wafer.

Layer exfoliation is highly versatile for integration of virtually any material with Si and is currently the leading method for the fabrication of SOI wafers. However, it involves a costly implantation step and causes surface damage in the transferred layer. The material damage is typically removed with a subsequent polishing step. However, as

shown in Chapter 4, CMP techniques are difficult to implement with thin Ge films. Consequently, Chapter 5 presents a novel method of removing such exfoliation damage with selective etching.

2.5: Conclusion

Of the numerous methods for integrating lattice-mismatched semiconductors with silicon, only wafer bonding of virtual substrates enables coplanar integration using large diameter substrates. This approach provides the ideal marriage of epitaxial growth on large diameter substrates and wafer bonding for integration of mismatched layers directly on silicon. However, in order to apply this technique virtual substrates must be planarized prior to bonding. The epitaxial growth and characterization techniques that will be used throughout the thesis are now presented in Chapter 3, while Chapter 4 will address the particular issue of virtual substrate planarization.

Chapter 3: SiGe heteroepitaxy by ultra-high vacuum chemical vapor deposition

This chapter gives an overview of the experimental techniques that are used throughout the thesis. The first three sections provide a description of ultra-high vacuum chemical vapor deposition (UHVCVD) and how it pertains to growth of the high-quality compositionally graded SiGe buffers used in this study. The chapter concludes with a description of the most important techniques which were necessary for characterizing film morphology, strain and layer composition in this work.

3.1: Introduction

The process of layer transfer by wafer bonding requires a large number of donor wafers for experimentation with process variations. This requires a technique which allows growth of thick epitaxial layers on numerous substrates. Furthermore, the advanced donor structures described in Chapters 5 and 6 will also require metastable growth of highly strained SiGe layers.

Several growth techniques are available for SiGe heteroepitaxy including molecular beam epitaxy (MBE) and chemical vapor deposition (CVD). While MBE is useful for low temperature growth of metastable layers, it is not particularly suited for growth of relaxed graded buffers which require thick layers grown at high temperature. Furthermore, MBE is not suitable for high throughput growth of these structures. UHVCVD combines the batch process capability of a hot-walled CVD system with the ability to grow metastably-strained layers at low temperature. All of the donor substrates that were used in this thesis were prepared using the UHVCVD growth technique which is now described.

3.2: Overview of UHVCVD

UHVCVD was developed in the mid 1980s to address the issue of low temperature CVD growth of epitaxial layers.³⁹ CVD is reaction limited at low temperature, yielding growth rates on the order of 0.1 Å/sec for Si at 550 °C and Ge at 350 °C using SiH₄ and GeH₄ precursor gases, respectively. In order to grow epitaxial layers with minimal incorporation of impurities at these low growth rates, the background contamination of the system must be minimized. The UHVCVD technique uses a turbo-pumped quartz tube furnace evacuated to a background pressure of 10⁻⁹ Torr with background oxygen levels typically <10⁻¹⁰ Torr. The low impurity background negates the need for a carrier gas for delivery of gaseous reactants to the wafer surface. Instead, the precursor gases are injected directly into the turbo-pumped system yielding growth pressures ranging between 1-25 mTorr.

A schematic of the UHVCVD reactor used in the present study is given in Figure 3.1. The load-locked system consists of a hot-walled quartz reaction chamber. The reaction chamber and load-lock are evacuated by turbo molecular pumps backed by a common foreline which is maintained at 10^{-3} Torr by roots and rotary-vane pumps. The reaction chamber is baked out to a background pressure of 10^{-9} Torr.

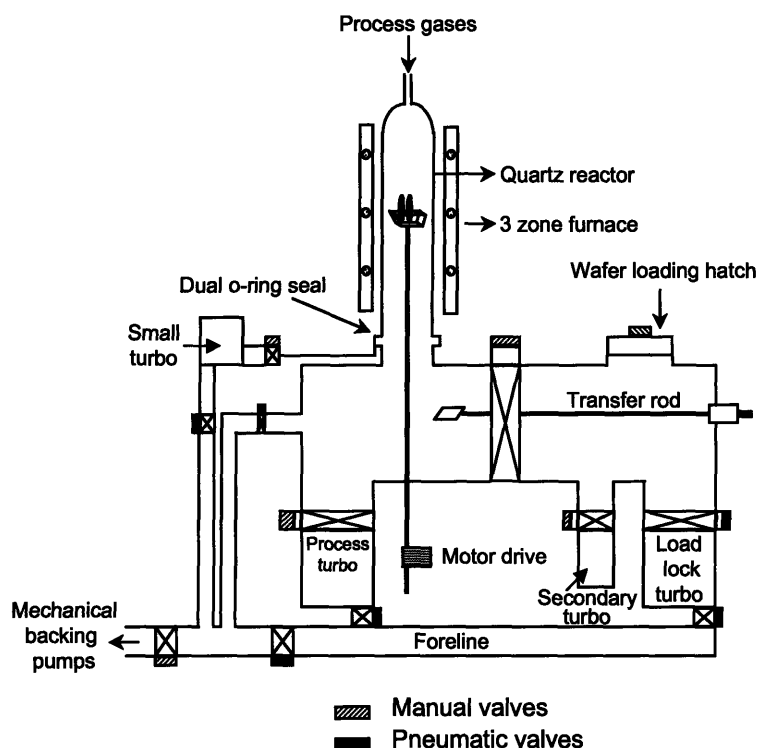


Figure 3.1: Schematic of UHVCVD system used for SiGe epitaxy. Figure adapted from Samavedam.⁴⁰

An added benefit of UHVCVD stems from its hot-walled reaction chamber which allows simultaneous growth on multiple substrates. The design of the system used in this work allows growth on as many as ten, 150 mm (6") substrates using a vertically oriented wafer boat. Furthermore, reasonable deposition rates (~ 10 Å/sec) can be obtained at the elevated temperatures that are required for growth of relaxed graded buffers. This makes

the system particularly convenient for the growth of the graded buffers as a batch process.

3.3 Growth of Ge/Si_{1-x}Ge_x/Si compositionally graded buffers

All of the wafer bonding work presented in this thesis made use of Si_{1-x}Ge_x compositionally graded buffers. These virtual substrates were prepared using double-side polished (DSP) Si wafers cut 6° from the (001) orientation toward the nearest {111} plane. Offcut substrates are necessary to suppress anti-phase boundary (APB) formation in GaAs layers grown on Ge substrates.^{6,41} In this work, only transfer of Ge films from virtual substrates is investigated. However layer transfer from virtual substrates is adaptable to other semiconductors such as III-Vs, and future work will involve growth of APB-free GaAs on Ge for III-V transfer.

Since film growth occurs on both sides of the wafer in a hot-walled CVD system, use of DSP substrates is a simple yet effective way to reduce wafer curvature induced by thermal stress. The deflection of a Ge virtual substrate grown on a single-side polished wafer is >35 μm across a wafer diameter of 100 mm. As noted in Chapter 2, minimizing wafer bow is essential for efficient mating of the wafer surfaces during bonding. DSP wafers reduce the wafer deflection to <15 μm allowing easy conformation between seed and handle substrates.

Prior to epitaxial growth, the wafers were cleaned in a standard piranha bath consisting of 3 H₂SO₄: 1 H₂O₂ for 10 minutes followed by deionized (DI) water rinsing. The piranha clean renders the wafers strongly hydrophilic with a residual chemical oxide layer that is subsequently removed with a diluted (10:1) HF etch for 1min. This procedure renders the surface highly hydrophobic and consequently, the wafer does not require final DI water rising. Furthermore, the HF etch terminates the surface with hydrogen bonds, protecting it against the formation of SiO₂. Despite exposure to ambient concentrations of oxygen and water vapor, the surface passivation is sufficiently stable to allow loading of oxide-free silicon wafers into the loadlock of the reactor.⁴²

According to Equation 2.4, the threading dislocation density of compositionally graded buffers is exponentially dependent on growth temperature. High growth temperatures are necessary to maximize the relaxation kinetics in graded buffers leading to low threading dislocation density. However, the relatively low cracking temperature of the GeH_4 precursor gas leads to nucleation of solid particles in the gas stream resulting in poor surface morphology of the deposited film. High growth temperatures that avoid gas phase nucleation⁴³ can be obtained using a GeCl_4 precursor for Ge.⁴⁴

Fortunately, the activation energy for dislocation glide decreases with Ge composition in the graded layers. Consequently, the growth temperature can be reduced when increasing the Ge content in the growing film thereby suppressing gas phase nucleation without sacrificing strain relaxation kinetics. Using SiH_4 and GeH_4 precursors, the growth temperature of the graded buffer was maintained between $0.7T_m$ and $0.8T_m$ as shown in Figure 3.2, where T_m is the absolute melting temperature of the growing $\text{Si}_{1-x}\text{Ge}_x$ film.

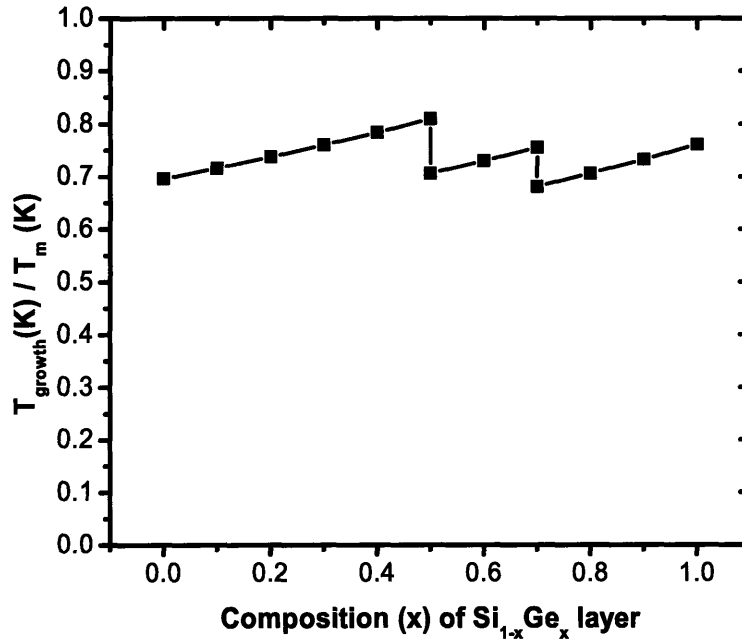


Figure 3.2: Growth temperatures used for growing $\text{Si}_{1-x}\text{Ge}_x$ compositionally graded buffers via UHVCVD using SiH_4 and GeH_4 precursor gases.

The surface morphology of the graded buffer evolves during growth as adatom diffusion is influenced by surface strain fields generated by the orthogonal arrays of misfit dislocations within the graded buffer.¹⁰ This can cause deep surface crosshatch which leads to dislocation pile-ups and increased threading dislocation density. To maintain low pile-up density and minimize the surface roughness of the final layer, Ge virtual substrates were grown in two intervals.²³ The compositionally graded buffer was grown by increasing the Ge fraction in 200 nm, 2% Ge intervals to an initial composition of $\text{Si}_{0.5}\text{Ge}_{0.5}$. The wafers were then removed from the reactor and planarized using a chemo-mechanical polishing (CMP) step to remove the surface crosshatch. Planarization of virtual substrates, as it pertains to wafer bonding, will be discussed in detail in the next chapter. After planarization, the virtual substrates were subjected to two consecutive piranha/HF cleaning procedures, as described earlier. The first piranha/HF etch was preformed to remove any residual CMP slurry particles from the $\text{Si}_{0.5}\text{Ge}_{0.5}$ surface, while the second was done to prepare the wafers for growth. The H-passivated wafers were then re-inserted into the UHVCVD system for continued grading from $\text{Si}_{0.5}\text{Ge}_{0.5}$ to Ge.

3.4: Material characterization

After film processing steps such as epitaxial growth, CMP and ion implantation, the wafers were subjected to various material characterization procedures to ascertain material quality, wafer bondability and film microstructure. This section provides a background of the major material characterization techniques that are used in this thesis.

3.4.1: Transmission electron microscopy

Transmission electron microscopy (TEM) provides information pertaining to film and defect morphology by passing a monochromatic electron beam through a thin sample foil. The interaction of the beam and a crystalline foil produces multiple diffracted beams that can be used to create an image of the internal structure of the sample. The variation in the intensity of these diffracted beams leads to diffraction contrast that is

Chapter 3: SiGe heteroepitaxy by UHV CVD

particularly useful for imaging features that cause distortion in the sample lattice. Diffraction contrast was used to image the dislocations in the graded buffer micrograph shown previously in Figure 2.5. High-resolution TEM (HRTEM) images can also be created by imaging the interference patterns arising between diffracted and forward scattered electrons enabling resolution of individual lattice planes. TEM is used extensively in Chapter 6 for deciphering the H-gettering behavior of tensilely strained SiGe layers.

TEM requires some sample preparation to provide an electron-transparent foil. Preparation of foils from bulk samples, such as a semiconductor wafer, involves mechanical grinding and/or dimpling to a thickness of 10-20 μm followed by Ar^+ ion milling until perforation. Cross-sectional samples were made by joining two pieces of the wafer face-to-face with epoxy. The sample is then thinned by mechanical grinding and ion milling of the glued interface. Two microscopes were employed in this thesis: a JEOL 2000FX and a JEOL 2010, both operating at 200 keV with a lanthanum hexaboride (LaB_6) cathode.

3.4.2: Atomic force microscopy

Atomic force microscopy (AFM) is a useful diagnostic tool for characterizing the surface morphology of deposited films. The underlying mechanism for forming an AFM image is a silicon cantilever mounted on a piezoelectric transducer. The tip of the cantilever has a sharp point which is rastered across the sample surface. A constant force is maintained between the cantilever and the sample surface by application of a voltage to the piezoelectric transducer thus generating a digital image of the surface topography. There are two main types of AFM employed in the characterization of semiconductor surfaces: contact-mode and tapping mode. Tapping mode utilizes a vibrating cantilever to minimize electrostatic interaction between the tip and the sample surface. All of the atomic-force micrographs presented in this thesis utilized tapping-mode imaging.

In addition to providing a qualitative description of surface morphology, AFM is useful for measuring the root-mean-square (RMS) roughness to ascertain the bondability

of a wafer surface. The RMS roughness is defined as the standard deviation of the surface topography relative to an average value. The bondability of a wafer has a stronger dependence on surface undulations on the small scale compared to long scale waviness. Therefore, meaningful surface roughness values must be accompanied by the AFM scan area from which the RMS value was calculated.

The AFM used in this thesis is a Digital Instruments Dimension 3000 equipped with a Nanoscope IIIa controller. The microscope is capable of a $118 \times 118 \mu\text{m}$ scan area with a maximum nominal vertical scan range of $6 \mu\text{m}$, making it ideal for imaging graded buffer surfaces which typically have a crosshatch period of $2\text{-}5 \mu\text{m}$ and a vertical range of 250 nm . Larger, macroscopic features are best observed using an optical microscope equipped with Nomarski contrast capabilities.

3.4.3: Nomarski contrast microscopy

Nomarski microscopy or the differential interference contrast (DIC) technique was originally developed for rendering contrast in transparent specimens. Conventional optical microscopy relies on variations in surface reflectivity to create contrast, requiring biological samples to be dyed prior to imaging. The reflectivity of a blanket semiconductor film also has little variation, yielding poor contrast using standard optical techniques. The DIC microscope utilizes a beam-shearing prism to produce an illumination source containing two optical paths which are slightly out of phase with each other. Contrast is produced by optical path gradients caused by variations of refractive index or topography of the sample. This technique greatly improves image contrast, particularly in samples that contain steep optical path gradients such as those present with crosshatched surfaces. These elements combine to make Nomarski microscopy ideal for imaging etch-pits for dislocation density measurements, as described in the next section.

3.4.4: Etch-pit density measurement

Measurement of defect density in semiconductor layers is perhaps the most misunderstood aspect of epitaxial layer characterization. The literature is plagued with inaccurate dislocation density measurements mainly due to incorrect application of characterization techniques.

A common and dependable method for determining dislocation density in lattice-mismatched layers is direct imaging of the defects with plan-view TEM (PVTEM). Unfortunately, the maximum imageable area using electron microscopy is typically smaller than $10 \times 10 \mu\text{m}$. Assuming a magnification of $10,000\times$, a sample containing 10^5 cm^{-2} threading dislocations will require hundreds of TEM micrographs to yield a statistically meaningful threading dislocation density (ρ_t) value.

An alternative method for revealing dislocations is selective etching of the sample and subsequent etch-pit density (EPD) measurements. Chemical etching reveals dislocations by selectively attacking their highly strained cores. EPD measurements are suitable for samples containing low ($<10^6 \text{ cm}^{-2}$) threading dislocations since the etch-pits can be imaged over a large surface area using DIC microscopy, as shown for a Ge virtual substrate in Figure 3.3. The diameter of the etch-pit is much larger than the diameter of the corresponding dislocation. Consequently EPD is not applicable to samples containing large defect densities where overlap of the etch pits seriously underestimates ρ_t values.

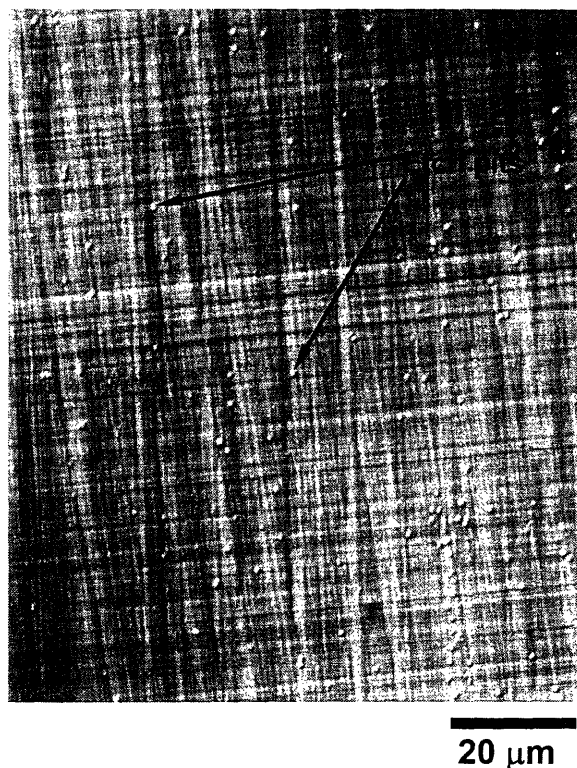


Figure 3.3: DIC micrograph of Ge virtual substrate after selective etching, revealing a threading dislocation density of $\sim 10^6 \text{ cm}^{-2}$.

Dislocations in Ge were revealed by etching in a 5:10:11 solution of hydrofluoric acid (HF), nitric acid (HNO_3) and acetic acid (CH_3COOH) with 30 mg of dissolved iodine (I_2).⁴⁵ This mixture yields an etch rate of $\sim 300 \text{ nm/sec}$, requiring a quick ($\sim 1 \text{ sec}$) dip and a Ge cap thickness of $>1 \mu\text{m}$ to prevent etching of dislocations in the underlying graded buffer. EPD was used in conjunction with PVTEM to monitor the quality of the epitaxial material used in this thesis.

3.4.5: Secondary ion mass spectroscopy

Secondary ion mass spectroscopy (SIMS) is a method for characterizing the composition of deposited layers by sputtering the surface with a focused ion beam. The secondary ions ejected from the sample are collected with a mass spectrometer, yielding a composition depth profile of the structure. The accuracy of SIMS data depends primarily

Chapter 3: SiGe heteroepitaxy by UHVCVD

on the quality of the standards used to calibrate the relative sensitivity factor (RSF), which are obtained by performing a SIMS experiment on a sample of known composition. Furthermore, SIMS samples a finite volume below the sputtered surface resulting in ion mixing artifacts for layer thicknesses on the order of the sampling depth. Ion mixing artifacts can be limited by minimizing the energy of the incident ion beam.

Chapter 5 applies SIMS analysis to characterize the thermal stability of epitaxial structures by monitoring interdiffusion between SiGe and Ge layers. SIMS is also used in Chapter 6 to characterize the hydrogen profiles of Ge donor substrates after H^+ ion implantation.

3.4.6: Raman spectroscopy

Raman spectroscopy is the collection of inelastically scattered light from a specimen surface. The Raman shift is defined as the energy shift of the incident light by the energies of molecular vibrations within the sample. The Raman shifts due to Si-Si and Si-Ge vibrations have been recently applied for measuring the strain and composition of strained Si and SiGe epitaxial layers.⁴⁶⁻⁴⁸ While x-ray measurements provide the most direct measurement of strain, this technique generally requires a minimum layer thickness of ~20 nm to generate sufficient diffracted signal for detection. Since light absorption occurs only at the surface of the analyzed sample, Raman spectroscopy has the capability of characterizing layers with thickness <10 nm by proper selection of the wavelength of the incident light. This technique is used in Chapter 6 for characterizing strain relaxation of tensilely strained SiGe layers after H^+ ion implantation.

3.5 Conclusion

The remainder of the thesis attacks the challenges of Ge layer transfer from compositionally graded buffers. UHVCVD enables growth of the large quantity of virtual substrates that will be needed for these experiments. Furthermore, the low growth

temperatures that are possible in an ultra-high vacuum environment are necessary for growth of highly strained SiGe layers for etch-stop and H-gettering structures, which are motivated in Chapters 5 and 6, respectively.

Chapter 4: Planarization of SiGe virtual substrates

Virtual substrate wafer bonding is severely limited by the crosshatched surface morphology that evolves during growth of strain-relaxed compositionally graded buffers. In order to maximize the contact area between two bonded wafers, the roughness must be reduced while also preserving the crystal quality of the polished surface. A proven method for planarizing Si and SiO₂ layers is the process of chemo-mechanical planarization (CMP). This chapter will apply the method of CMP for preparation of Ge virtual substrates for wafer bonding.

4.1: Introduction

As its name implies, chemo-mechanical planarization utilizes chemical and mechanical components for material removal during the polishing of a wafer surface. CMP was originally developed for preparing atomically flat semiconductor surfaces which enables fabrication of sub-micron device features in modern integrated circuits. Since the early 1990s, CMP has also found applications in back-end metallization where dielectric and metal films are planarized for interconnection of devices in very-large scale integrated (VLSI) circuits.^{49,50} As mentioned in Chapter 2, Ge virtual substrates have a 25×25 μm RMS surface roughness of 10-15 nm which must be reduced to <0.5 nm prior to wafer bonding. Consequently, the focus of this chapter is front-end planarization of virtual substrates for subsequent wafer bonding.

4.2: CMP tool description

A Strasbaugh 6EC laboratory polisher was used to planarize the virtual substrates used in this thesis. A schematic of this research tool is provided in Figure 4.1. The setup consists of a 22” polishing table with a Rohm and Haas IC1000 layered CMP pad. The pad is conditioned with a diamond-grit disc that lifts the nap of the CMP pad during wafer polishing, thereby maintaining a constant material removal rate. Cabot’s Semisperse-25 fumed silica slurry, diluted (1:1) with deionized water was used for polishing Si, SiGe and SiO₂ films as will be discussed in subsequent sections.

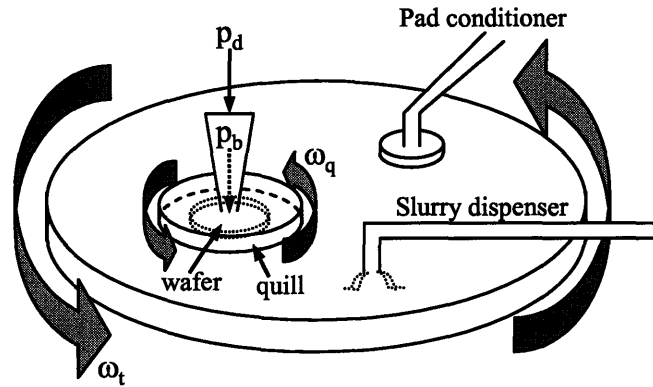


Figure 4.1: Schematic of the Strasbaugh 6EC laboratory polishing system. The wafer is held by the quill which rotates in the same direction as the polishing table. The pad conditioner lifts the nap of the CMP pad thereby maintaining a constant material removal rate.

Typical polishing parameters for the Strasbaugh 6EC are given in Table 4.1. The wafer is held in place in the quill that rotates in the same direction as the polishing table. The downforce (p_d) is applied by a hydraulic piston to the entire wafer carrier, thereby setting the contact pressure between the wafer and the CMP pad. Meanwhile, the backpressure (p_b) is a pneumatic pressure applied to the back of the wafer to maintain polish rate uniformity across the wafer surface.

Parameter	Value
Table speed (ω_t)	35 rpm
Quill speed (ω_q)	15 rpm
Downforce (p_d)	3-5 psi
Backpressure (p_b)	1-3 psi
Slurry flow	100-150 mL/min

Table 4.1: Typical parameters used with the Strasbaugh 6EC CMP tool.

4.3: Direct chemo-mechanical planarization of SiGe virtual substrates

The goal of CMP is to provide wafer scale planarization of a substrate while minimizing the damage incurred to the surface. CMP achieves this by supplying a chemical component which provides a self-limiting reaction with the wafer surface. The reaction byproduct is subsequently removed with the mechanical component comprised of abrasive particles suspended in the polishing slurry. Since these particles do not come in direct contact with the substrate, defect-free surfaces can be prepared in this manner.

Atomically flat silicon surfaces are traditionally prepared using a slurry consisting of a colloidal suspension of SiO₂ particles in potassium hydroxide (KOH), commercially known as Syton[®].⁵¹ Upon exposure to the alkaline electrolyte, the silicon surface oxidizes according to³⁵:



The resulting hydroxyl groups weaken the Si-Si back-bonds that link the Si(OH)₂ species to the bulk crystal, making them susceptible to further attack by (OH)⁻ ions that are generated by H₂O dissociation.^{52,53} The resulting silicic acid species, Si(OH)₄, are mechanically removed by interaction of the slurry particles with the wafer surface. This exposes a clean Si layer which is followed by subsequent passivation with hydrogen which renders the surface hydrophobic.^{54,55} Increasing the [OH⁻] concentration in the slurry increases the electrochemical potential of Equation 4.1, thereby increasing the size of the silicic acid species removed from the wafer surface.⁵⁴ As a result, the material removal rate (MRR) for Si CMP processes increases with pH until a value of ~11 at which point a stable hydrophilic oxide forms to passivate the surface causing a concomitant reduction of the MRR.⁵⁶

CMP of Si_{1-x}Ge_x layers is complicated by a reduction of the MRR when using KOH-based slurries. As mentioned earlier, the progression of Si surface oxidation is facilitated by a weakening of Si-Si back-bonds as a result of strong polarization by hydroxyl groups.

The large atomic number of Ge provides an electronic screen against polarization, rendering $\text{Si}_{1-x}\text{Ge}_x$ alloys less susceptible to this oxidation mechanism³⁵, thereby slowing down the polishing rate. Figure 4.2 is a plot showing the decline of the MRR vs. Ge fraction in $\text{Si}_{1-x}\text{Ge}_x$ films using the Cabot Semisperse-25 slurry.

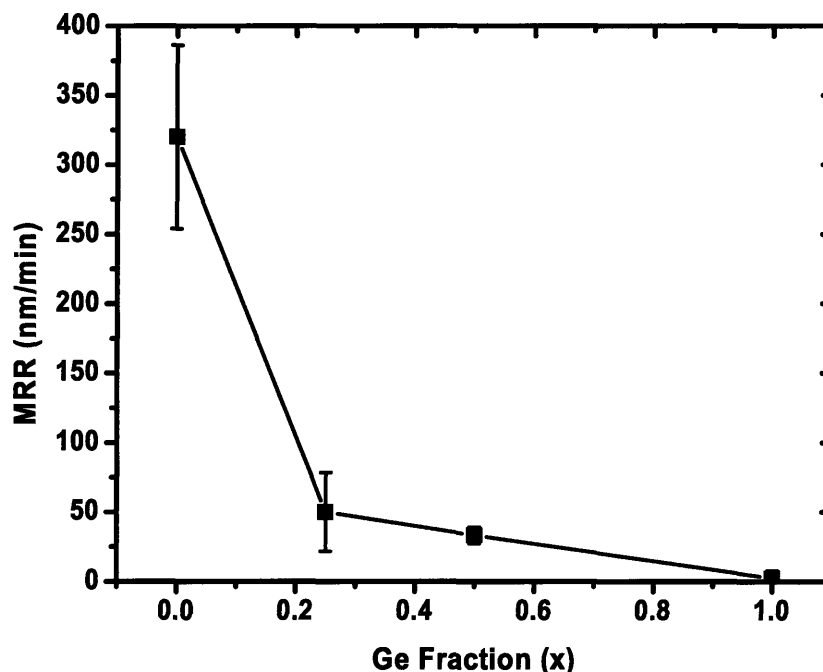


Figure 4.2: $\text{Si}_{1-x}\text{Ge}_x$ material removal rate (MRR) versus Ge content using the KOH-based Semisperse-25 slurry. The reduction of the polish rate is caused by the lower susceptibility of Ge oxidation by $(\text{OH})^-$ free radicals.

In practice, $\text{Si}_{1-x}\text{Ge}_x$ layers can be CMPed using KOH-based slurries up to a Ge fraction of ~ 0.5 . For Ge layers, the MRR drops to < 3 nm/min, rendering this technique highly ineffective for Ge virtual substrate planarization. To complicate matters, anisotropic etching leads to surface pitting without complete removal of the crosshatch roughness, as indicated in the atomic force micrograph shown in Figure 4.3b. The density of these surface pits is $\sim 5 \times 10^5 \text{ cm}^{-2}$ which roughly corresponds to the sample's threading dislocation density, suggesting that they are caused by etching of dislocations intersecting the Ge surface. Addition of hydrogen peroxide (H_2O_2) to the slurry can

significantly increase the MRR through aggressive oxidation of the surface.^{57,58}

However, GeO_2 is soluble in water and is removed from the wafer surface without mechanical aid, leading to poor selectivity to the asperities of the wafer surface.

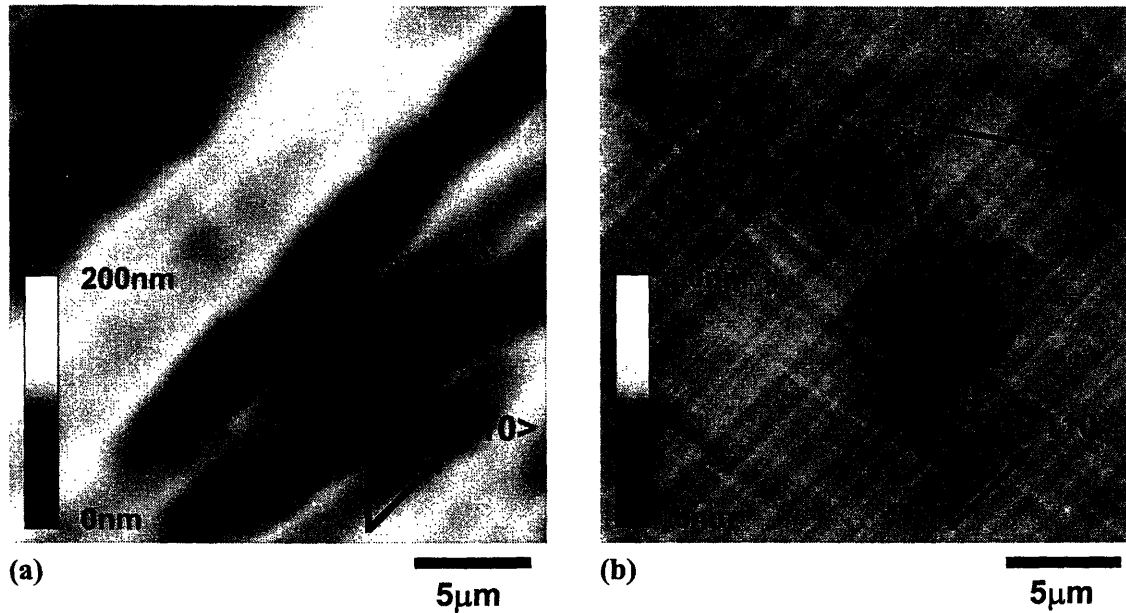


Figure 4.3: Atomic force micrographs of a Ge virtual substrate before (a.) and after (b.) 30 min of CMP using Semisperse-25 slurry. The surface pits result from selective etching of threading dislocations intersecting the wafer surface.

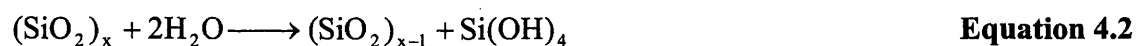
Bulk Ge CMP techniques cannot be easily applied to Ge virtual substrates since these generally have a strong chemical component consisting of H_2O_2 , requiring the removal of 10s of μm of material to achieve the required surface roughness. Unfortunately, the thickness of the uniform composition cap of a virtual substrate is limited by thermal stress and generally does not exceed $\sim 2 \mu\text{m}$. Furthermore, the inhomogeneous strain fields emanating from the underlying misfit arrays cause further roughening during chemical etching. As a result, a CMP process having a dominant chemical component will increase the crosshatch roughness and cannot be used for virtual substrate planarization.

4.4: Planarization of Ge virtual substrates using CMP layers

The difficulty in planarizing thin virtual Ge layers stems from the fact that current Ge CMP techniques feature a strong chemical component and high material removal rates. The source of this problem is the solubility of GeO₂ in water which leads to poor surface passivation in aqueous polishing slurries. These challenges should be addressed for integration schemes utilizing compositionally graded buffers, however an immediate solution consists of indirect planarization of virtual substrates using a CMP layer. The CMP layer can comprise SiO₂ or Si layers which are deposited onto the virtual substrate prior to planarization. CMP techniques for these materials benefit from decades of development. Furthermore, oxide and Si layers are also useful for protecting the underlying Ge during post-CMP and pre-bonding cleaning procedures. Both oxide and epitaxial Si CMP films were investigated for preparation of Ge virtual substrates prior to bonding. The results of these experiments are now presented.

4.4.1: Oxide CMP layers

Aside from Si, SiO₂ (oxide) CMP is the most mature of all CMP processes and can be easily applied to the planarization of virtual substrates for wafer bonding applications. An oxide CMP layer is also the obvious choice for preparation of Ge-on-insulator, where the CMP layer will also serve as the buried insulator after layer transfer. The material removal rate of an oxide layer is exceedingly sensitive to applied pressure, resulting in excellent selectivity to the surface asperities of the wafer. As a result, oxide surfaces can be planarized with minimal removal of material. The chemical component of oxide CMP is water, which serves to weaken the oxide bonds by the hydrolysis reaction given by Equation 4.2.



This reaction is accelerated by the compressive stress induced by the interaction of the CMP pad and slurry particles with the wafer surface.⁵⁹ The silicic acid reaction products are subsequently removed by the additional mechanical action of slurry particles. As a result, both chemical and mechanical removal mechanisms are enhanced at surface asperities leading to high planarization efficiency. In practice, a 15 nm RMS surface roughness can be reduced to bondable values (~ 0.5 nm) while removing <200 nm of oxide thickness.

A significant drawback to oxide CMP layers stems from the fact that they must be densified at high temperature prior to wafer bonding. Oxide layers are generally grown using low pressure or plasma-enhanced CVD which produce films containing significant amounts of residual hydrogen.⁶⁰ This hydrogen can lead to pockets of trapped gas at the bonded interface during wafer bond annealing. As a result, oxide CMP layers require a densification step which could be a problem for processes with a limited thermal budget.

Figure 4.4 contains an atomic force micrograph of a Ge virtual substrate planarized using a deposited low temperature oxide (LTO) CMP layer that was densified at 650 °C for 3 hours prior to polishing. The initial thickness of the oxide was 750 nm of which 500 nm was removed by CMP, yielding an RMS roughness of 0.26 nm as measured on a 1×1 μm scale and a value of 0.39 nm on a 10×10 μm scale.

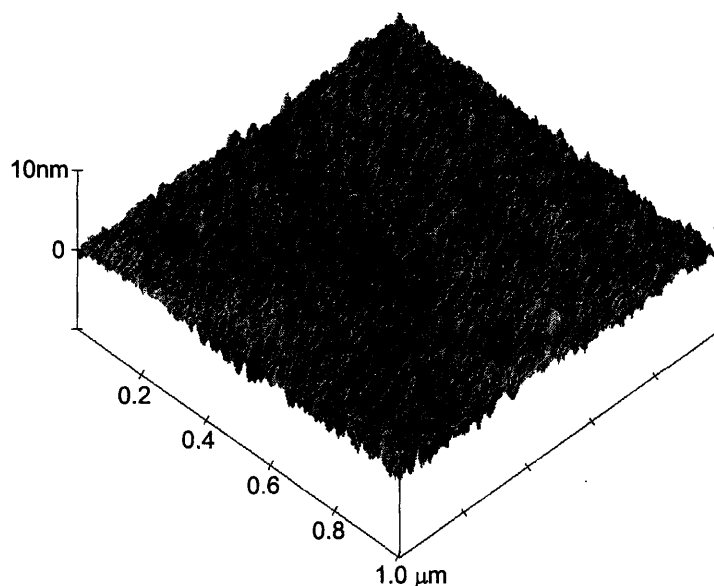


Figure 4.4: Atomic force micrograph of a Ge virtual substrate after planarization using an LTO CMP layer. The RMS surface roughness of this scan is 0.26 nm, a value which is acceptable for direct wafer bonding.

4.4.2: Epitaxial Si CMP layers

Epitaxial Si films grown by UHVCVD were also investigated as possible CMP layers for Ge planarization. The lattice constant mismatch between Si and Ge causes extensive nucleation of defects in the Si, as evidence by the XTEM micrograph shown in Figure 4.5. However, the dense misfit array at the Si/Ge interface provides an effective barrier to dislocation propagation to the underlying Ge. Consequently, the quality of the Ge device layer is not affected by the heavily dislocated CMP layer. The Si layer will become buried after layer transfer where its defect content will be inconsequential to devices fabricated on the Ge surface. Also, a highly defective region in the vicinity of the device layer could be useful from the standpoint of impurity gettering, thus providing an additional beneficial aspect to the structure.

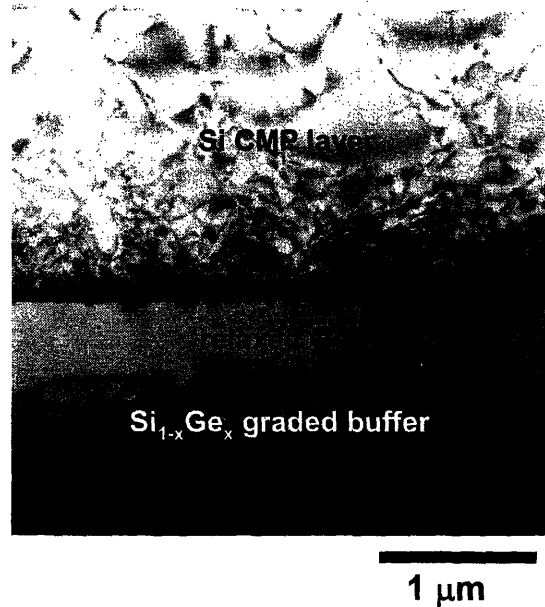


Figure 4.5: XTEM micrograph showing a highly defective epitaxial Si CMP layer grown directly on Ge. The Si layer can be readily CMPed to a surface roughness suitable for wafer bonding. The large density of defects is confined to the Si layer and does affect the quality of the underlying relaxed Ge.

Using the planarization process described in Section 4.2, large material removal rates were observed for Si CMP layers leading to poor selectivity to the surface asperities of the wafer. Consequently, a 2 μm thick Si layer was required to reduce the RMS surface roughness to <0.5 nm. After removal of 1 μm of film, the 1×1 μm surface roughness was reduced to 0.25 nm as measured from the atomic force micrograph shown in Figure 4.6. Epitaxial Si provided excellent 1×1 μm planarity; however the associated 10×10 μm AFM scans yielded roughness values of 0.89 nm, signifying relatively poor long scale planarity.

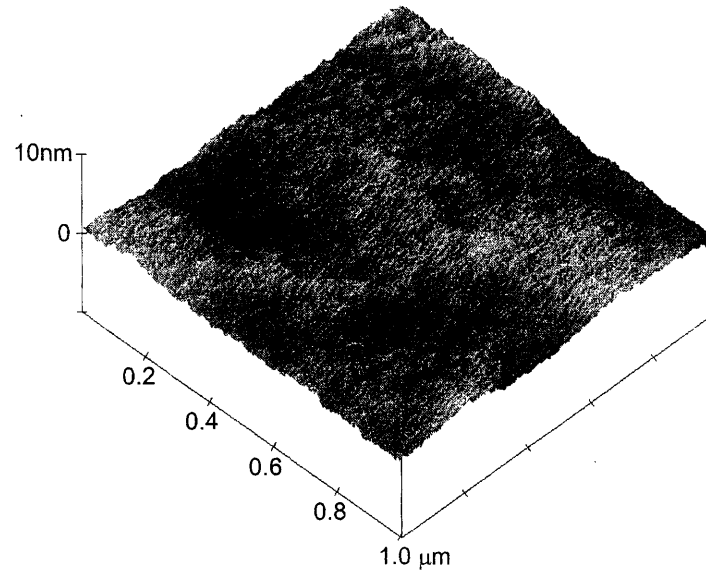


Figure 4.6: Atomic force micrograph of a Ge virtual substrate after planarization using an epitaxial Si CMP layer. Analogous to the LTO case, the $1 \times 1 \mu\text{m}$ RMS roughness is reduced to 0.25 nm allowing subsequent wafer bonding.

Both oxide and epitaxial Si CMP layers enabled planarization of virtual substrates for wafer bonding. Figure 4.7 is a XTEM micrograph showing a Ge layer transferred using an epitaxial Si CMP layer and the hydrogen-induced exfoliation process. In this particular instance the donor wafer (comprised of a Ge virtual substrate planarized via an epitaxial Si layer) was implanted with H_2^+ ions to a dose of $4 \times 10^{16} \text{ cm}^{-2}$ at 200 keV prior to hydrophilic wafer bonding. The micrograph shows extensive implantation damage in the relaxed Ge film which is partly due to the limited penetration depth of the implanted ions into the Ge layer. Exfoliation damage occurs near the end of the implanted ion trajectory. Therefore a reduction of the CMP layer thickness will allow deeper ion penetration to the Ge and provide a larger thickness of undamaged Ge. Nevertheless, some surface damage is inevitable with layer exfoliation and must be removed prior to device fabrication.

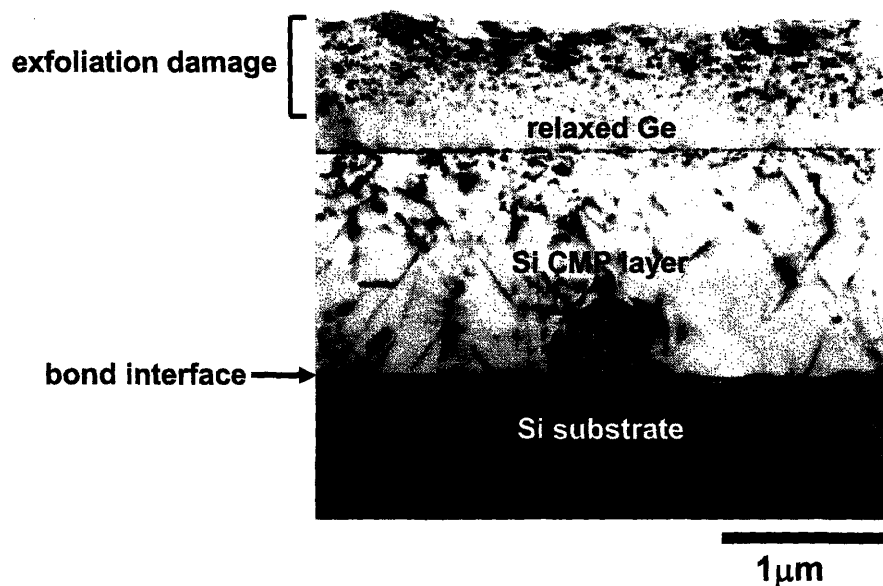


Figure 4.7: XTEM micrograph showing relaxed Ge on Si transferred from a virtual substrate. The Si CMP layer is buried between the Si wafer and transferred Ge film. Ge layer separation was induced using the exfoliation technique, resulting in significant damage near the surface of the transferred Ge layer.

The damaged layer induced by layer splitting is typically removed with a CMP step during the commercial fabrication of SOI wafers. However traditional CMP is difficult to implement with thin Ge films. This problem has been recently addressed with low temperature growth of Ge after layer exfoliation, which was shown to reduce the roughness of the damaged surface.³¹ However, the damaged region has a finite depth, as evidenced in Figure 4.7 and can only be eliminated by physical removal with polishing or chemical etching. Chapter 5 is dedicated to solving this problem with selective chemical etching, thus providing the final element necessary for coplanar integration of Ge with large diameter Si substrates.

4.5: Conclusion

Direct planarization of Ge is restricted by the solubility of GeO_2 in water which yields poor surface passivation in aqueous slurry chemistries. As a result, the strong

Chapter 4: Planarization of SiGe virtual substrates

chemical component of existing Ge CMP techniques precludes their use with thin Ge layers, including compositionally graded buffers. Therefore, the challenge of planarizing Ge virtual substrates was addressed with a deposited CMP layer prior to wafer planarization. Epitaxial Si and oxide layers can be readily CMPed using KOH-based slurries, enabling wafer bonding and Ge layer transfer. A remaining issue is the removal of the surface damage of exfoliated Ge films, which is addressed in the next chapter.

Chapter 5: Etch-stop structures for Ge layer transfer

This chapter addresses the problem of exfoliation damage in Ge layers by combining hydrogen ion implantation with selective chemical etching, thereby eliminating the need for CMP of the transferred layer. After motivating this approach, the etch selectivity and thermal stability of etch-stop structures containing tensilely strained SiGe layers are investigated. Furthermore, a method for initiating homoepitaxial growth on a Ge surface is described for applications involving transferal of epitaxial layers from bulk Ge wafers. Finally, layer transfer and selective etching is demonstrated for transferal of relaxed Ge-on-insulator (GOI) layers from virtual substrates onto large-diameter Si substrates.

5.1: Introduction

The layer exfoliation process is best suited for Ge transfer from virtual substrates since the grind and etch-back approach is hindered by the lack of a chemical etch that exhibits selectivity to SiGe across the entire $\text{Si}_{1-x}\text{Ge}_x$ composition range. However, this process induces damage in the transferred film that must be removed prior to device fabrication. Exfoliation damage can be readily removed with a CMP step in SOI⁶¹ or SiGe-on-insulator (SGOI)⁶² wafers prepared in this manner. However, as seen in Chapter 4, current CMP techniques are difficult to implement with thin Ge films. To solve this problem, a novel process was developed for removal of exfoliation damage from exfoliated Ge layers using a chemical etch and a strained SiGe etch-stop layer. The process flow for fabrication of relaxed GOI layer using this approach is illustrated in Figure 5.1.

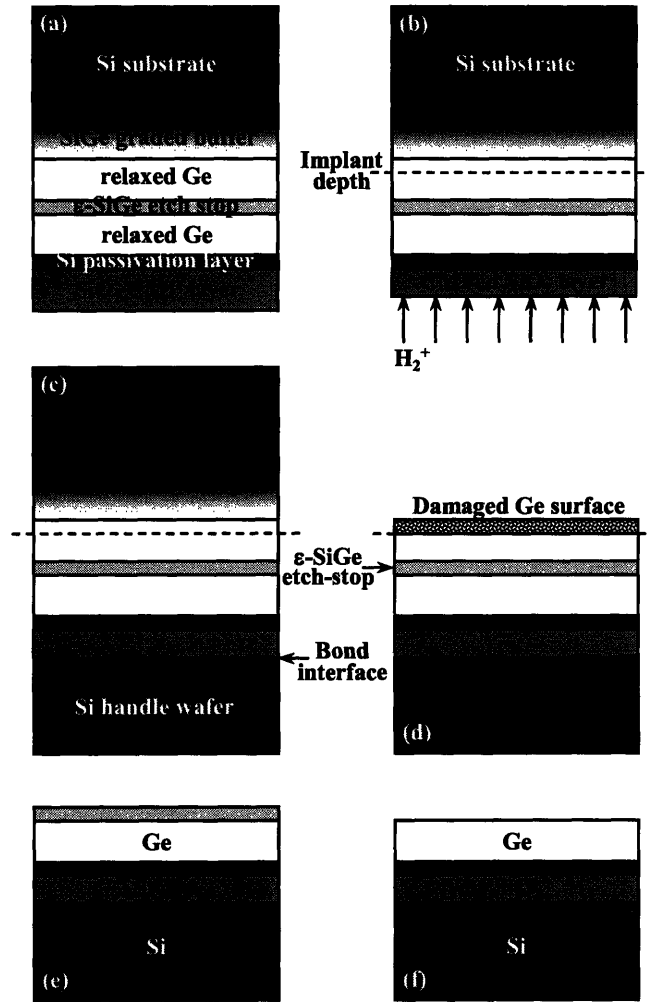


Figure 5.1: Outline of a Ge transfer process utilizing an etch-stop layer for CMP-less removal of exfoliation damage. (a) Preparation of the virtual substrate: growth of Ge transfer structure on a relaxed Ge/Si_{1-x}Ge_x/Si virtual substrate using UHVCVD followed by SiO₂ CMP layer deposition using LPCVD. (b) Planarization of the SiO₂ CMP layer and subsequent H₂⁺ implantation. (c) Wafer bonding the virtual substrate to a Si handle wafer. (d) Anneal and exfoliation of the SiO₂, Ge and Si_{0.4}Ge_{0.6} etch-stop layers to a Si handle wafer. (e) Selective etching in H₂O₂ to remove the damaged Ge surface. (f) Selective CMP to remove the Si_{0.4}Ge_{0.6} etch-stop layer.

The etch-stop structure consists of a tensilely strained SiGe layer grown upon a relaxed Ge virtual substrate, followed by a relaxed Ge transfer layer. The epitaxial structure may also be capped with a thin Si or Si₃N₄ layer to protect the Ge surface from

subsequent chemical processing steps. The process depicted in the Figure utilizes a deposited oxide CMP layer for planarization of the substrate surface prior to bonding (Figure 5.1(a)). After CMP of the oxide layer, the substrate is implanted with H^+ or H_2^+ ions with sufficient energy to induce splitting below the depth of the etch-stop (Figure 5.1(b)). The surface is then activated via wet chemical cleaning using the RCA cleaning procedure summarized in Table 5.1. The donor structure is bonded to a Si handle wafer (Figure 5.1(c)). Next, the bonded wafer pair is annealed at low temperature ($\sim 200^\circ C$) to strengthen the bond, followed by layer exfoliation at $400^\circ C$, transferring the oxide, Ge and etch-stop layer to the Si handle (Figure 5.1(d)). Finally, the transferred film structure is selectively etched to remove the damaged Ge surface, stopping on the buried SiGe layer (Figure 5.1(e)). Finally, a selective CMP step could be applied to remove the remaining etch stop layer (Figure 5.1(f)).

Step	Chemistry/Procedure	Time
1	5 H_2O : 1 NH_4OH : 1 H_2O_2 ($80^\circ C$)	10 min
2	DI H_2O rinse	3 min
3	50 H_2O : 1 HF	15 sec
4	DI H_2O rinse	3 min
3	6 H_2O : 1 HCl: 1 H_2O_2 ($80^\circ C$)	15 min
6	DI H_2O rinse	3 min
7	Spin-rinse dry	10 min

Table 5.1: RCA cleaning procedure used for hydrophobic surface activation prior to wafer bonding.

5.2: Ge/SiGe etch selectivity

The chemical behavior of a $Si_{1-x}Ge_x$ alloy can be correlated to its band gap energy (E_g). Figure 5.2 is a plot of E_g as a function of Ge fraction, x .⁶³ The rapid decrease of the band gap at $x \geq 0.8$ corresponds to the crossover from the Si-like Δ minimum to the Ge-like L minimum in the conduction band. This trend influences the electronic and

chemical properties of $\text{Si}_{1-x}\text{Ge}_x$ alloys which are “Si-like” for $x < 0.8$ and “Ge-like” for $x > 0.8$. As a result, the highest etch selectivity of Ge layers to $\text{Si}_{1-x}\text{Ge}_x$ is expected for alloys with $x < 0.8$.

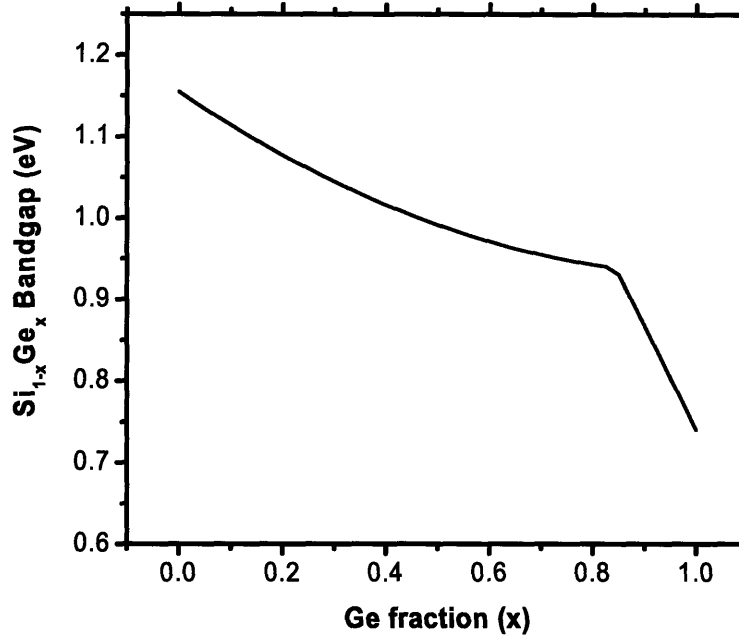


Figure 5.2: Plot of $\text{Si}_{1-x}\text{Ge}_x$ band gap as a function of Ge fraction.

The etch selectivity of Ge to relaxed $\text{Si}_{1-x}\text{Ge}_x$ alloys was characterized for two different etch chemistries. The first is a mixture of 1.7 g KI: 60 mg I_2 : 100 mL H_2O , while the second consists of (30%) hydrogen peroxide (H_2O_2). Figure 5.3 is a plot of the etch rate and selectivity vs. Ge fraction for both etches carried out at room temperature. As expected, the selectivity in both chemistries increases rapidly when the Ge fraction of the alloy is decreased below 0.8. Despite its higher selectivity compared to hydrogen peroxide, the iodine etch is incompatible with general CMOS processing due to its potassium component. Peroxide also has the added benefit of a significantly higher etch rate, making it a better choice for removing thick layers.

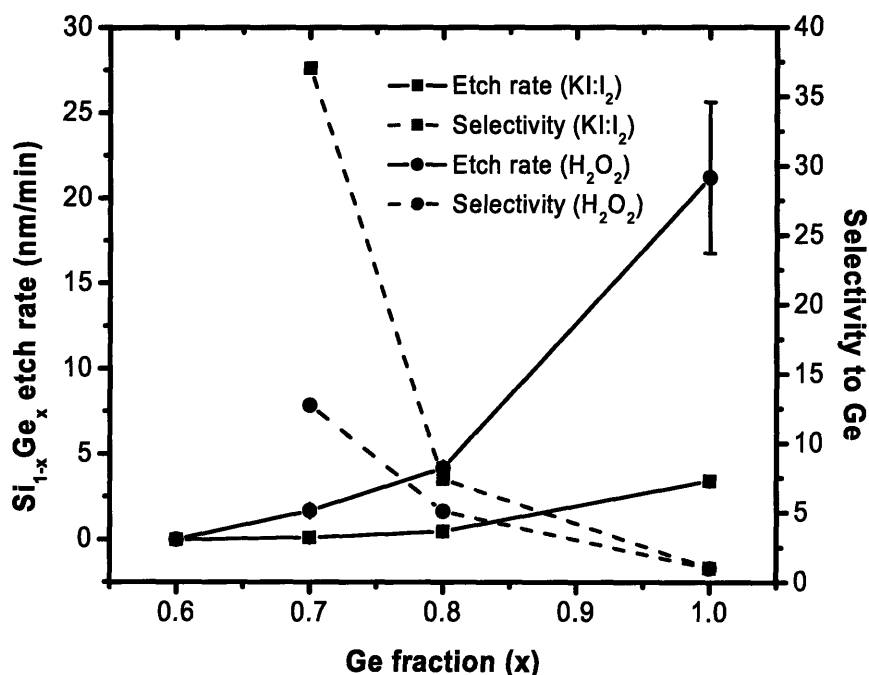


Figure 5.3: $\text{Si}_{1-x}\text{Ge}_x$ etch rates and selectivity to Ge in $\text{KI}:\text{I}_2$ and H_2O_2 etch chemistries. As expected, the selectivity of both etches to Ge increases rapidly when the Ge fraction is reduced below 0.8.

The etch selectivity is generally reduced for strained $\text{Si}_{1-x}\text{Ge}_x$ relative to relaxed layers of the same composition. However, for the case of tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ on Ge, a Ge thickness of 1 μm can be etched in hydrogen peroxide long before a 6 nm thick strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer is breached (yielding a selectivity >150). In order to preserve this processing window with a $\text{Si}_{0.3}\text{Ge}_{0.7}$ stop layer, an etch-stop thickness of 80 nm is required. The critical thickness for a $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer buried in Ge is 11 nm, while that of a $\text{Si}_{0.3}\text{Ge}_{0.7}$ layer is 17 nm making an 80 nm layer difficult to grow without significant relaxation and/or defect nucleation. Furthermore, a $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop is expected to tolerate higher processing temperatures compared to a layer containing a larger Ge fraction. In light of these observations, a composition of $\text{Si}_{0.4}\text{Ge}_{0.6}$ was chosen for the etch-stop in the layer transfer structure.

5.3: Growth of $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop structures

The growth temperature of highly strained SiGe layers must be minimized to prevent undulation of the surface.⁶⁴ Furthermore, while the critical thickness of a buried $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer is 11 nm, layers of this thickness must initially be grown under metastable strain as a surface layer to prevent nucleation of dislocations. UHVCVD is the ideal tool for low temperature growth of such highly strained lattice-mismatched layers. This section describes the growth and defect characterization of $\text{Si}_{0.4}\text{Ge}_{0.6}$ /Ge etch-stop structures grown on virtual substrates and bulk Ge wafers. The etch-stop layers are characterized for defect density and morphology as a function of their thickness. A second subsection is devoted to the initiation of homoepitaxial growth of Ge for layer transfer from bulk Ge wafers or CMPed Ge virtual substrates.

5.3.1: Etch-stop structures on virtual substrates

A cross-sectional TEM micrograph of the as-grown Ge virtual substrate incorporating the etch-stop layer is shown in Figure 5.4. The transfer structure was grown on a relaxed Ge virtual substrate and is comprised of a strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop buried by a relaxed Ge transfer layer. The structure is capped with a thin (7 nm) Si passivation layer to protect the Ge during subsequent processing. Both the etch-stop and Ge transfer layers were grown at 450 °C using SiH_4 and GeH_4 precursor gases. The Si cap was grown by flowing SiH_4 while simultaneously raising the growth temperature from 450 to 550 °C. This procedure was used to initiate growth of the highly mismatched layer while minimizing strain-induced surface undulations.⁶⁴

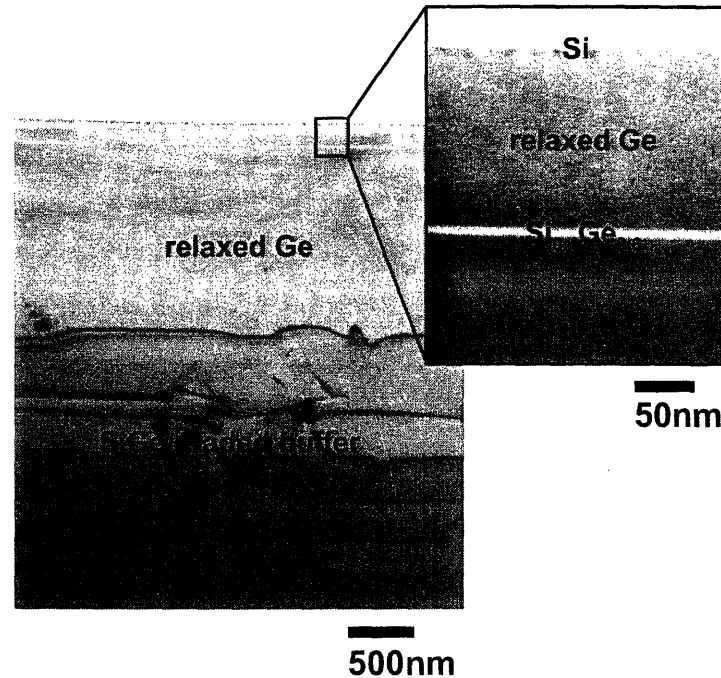


Figure 5.4: Cross-sectional TEM of a Ge transfer structure grown on a relaxed Ge/Si_{1-x}Ge_x/Si virtual substrate. The transfer structure consists of a strained Si_{0.4}Ge_{0.6} etch-stop layer followed by a relaxed Ge transfer layer. The entire structure is capped with a 7 nm Si passivation layer to protect the underlying Ge during subsequent chemical processing steps.

The critical thickness of a buried strained layer is doubled relative to a surface layer since layer relaxation requires misfit formation at two interfaces. Nevertheless, a small degree of relaxation can occur during growth, before the etch-stop is buried by the Ge transfer layer. Misfit dislocations at the Si_{0.4}Ge_{0.6}/Ge interface can be detected using PVTEM as shown in Figure 5.5 for 7.5 nm (a.) and 24 nm (b.) Si_{0.4}Ge_{0.6} layers.

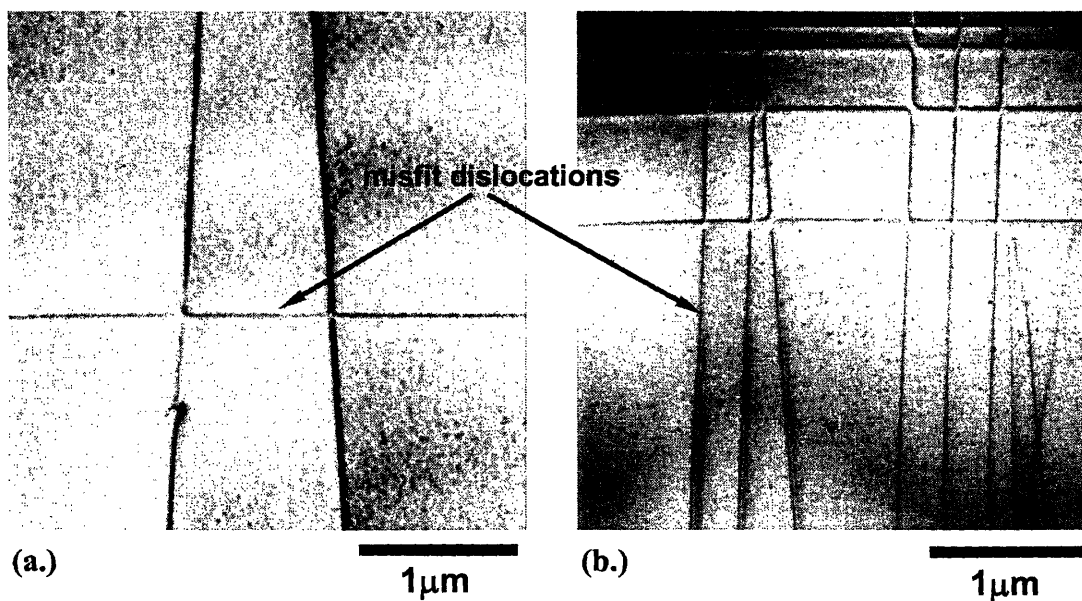


Figure 5.5: PVTEM micrographs showing misfit dislocations at a Si_{0.4}Ge_{0.6}/Ge interface. The misfit density is a function of Si_{0.4}Ge_{0.6} thickness for layers grown beyond the critical thickness. The thickness of the layer is 7.5 nm in (a.) and 24 nm in (b.).

Figure 5.6 is a plot of threading and misfit dislocation densities as a function of Si_{0.4}Ge_{0.6} layer thickness, as determined by PVTEM measurements. As expected, the misfit density increases with thickness for layers greater than the critical thickness of a surface layer (~6 nm). However, relaxation of the Si_{0.4}Ge_{0.6} layer occurs by glide of existing threading dislocations as evidenced by the insensitivity of the thread density to layer thickness in this range.

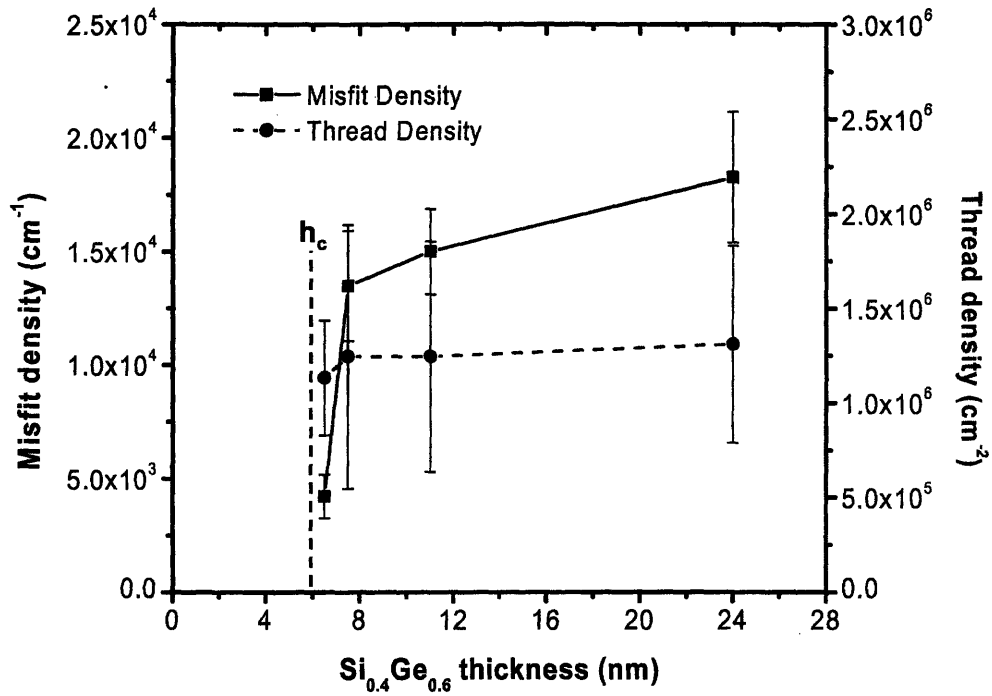


Figure 5.6: Misfit and thread density measurements for Ge transfer structures containing a strained Si_{0.4}Ge_{0.6} etch-stop layer. The Si_{0.4}Ge_{0.6} layer begins to relax when grown beyond the critical thickness (h_c) of a surface layer, resulting in a monotonic increase in misfit density. The constant thread density indicates that strain relaxation occurs without nucleation of new threading dislocations.

Increasing the etch-stop thickness significantly beyond 25 nm causes observable dissociation of threading dislocations into two Shockley partials. In the SiGe system, the commonly occurring 60° dislocation undergoes dissociation under tensile strain, resulting in 30° and 90° partial dislocations separated by an intrinsic stacking fault. This dissociation reaction proceeds as Equation 5.1, where the $a/6[1\bar{1}2]$ partial is a 90° edge and the $a/6[\bar{2}11]$ is a 30° mixed dislocation.

$$\frac{a}{2}[\bar{1}01] \rightarrow \frac{a}{6}[1\bar{1}2] + \frac{a}{6}[\bar{2}11]$$

Equation 5.1

The stacking fault is a planar defect; therefore it is readily detectable using XTEM when present in moderate densities. Figure 5.7 is a XTEM micrograph showing a number of such stacking faults within a 50 nm thick $\text{Si}_{0.4}\text{Ge}_{0.6}$ strained layer grown on Ge. A much larger density of these defects is present in the Si cap due to the larger lattice strain between Si and Ge. Stacking faults pose obstacles to dislocation propagation in strained layers and must be avoided to prevent nucleation of new threads. Even so, $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop layers deposited at 450 °C can be grown 2-3 times their critical thickness without a detectable increase in thread density, as seen in Figure 5.6.

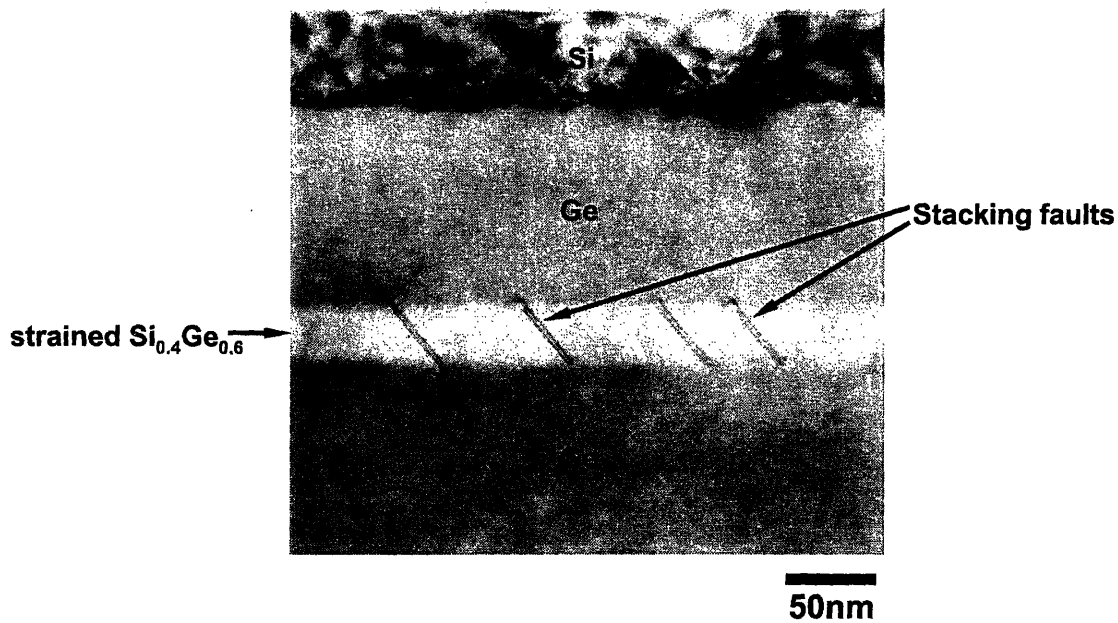


Figure 5.7: XTEM micrograph showing stacking faults in the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer due to dislocation dissociation under tensile strain.

5.3.2: Epitaxy on bulk Ge substrates

All of the Ge transfer structures grown on virtual substrates were initiated on a $\text{Si}_{0.5}\text{Ge}_{0.5}$ surface prior to compositional grading to Ge. However, in some instances it is also necessary to initiate Ge epitaxy on a pure Ge surface. Some of these applications include layer transfer from bulk Ge wafers and growth of etch-stop structures on planarized virtual substrates when Ge CMP techniques reach maturity. Si and $\text{Si}_{0.5}\text{Ge}_{0.5}$ epitaxy can be readily initiated on an HF-dipped wafer without in-situ etching of the substrate prior to growth. In contrast, the high surface diffusivity of Ge adatoms renders initiation of homoepitaxial layers highly susceptible to surface contaminants which nucleate three-dimensional growth at elevated temperature. This section addresses this issue for initiation of planar Ge layers using UHVCVD.

Despite several improvements of Ge surface cleaning procedures⁶⁵⁻⁶⁸, initiation of homoepitaxial growth at high temperature invariably leads to Volmer-Weber (three-dimensional) film morphology. Figure 5.8 contains XTEM and AFM micrographs of a Ge transfer structure initiated on a bulk Ge wafer at 650 °C. After initiation, the temperature was dropped to 450 °C for growth of the $\epsilon\text{-Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ layers. The Si cap was grown by raising the furnace temperature in SiH_4 flow as described earlier.

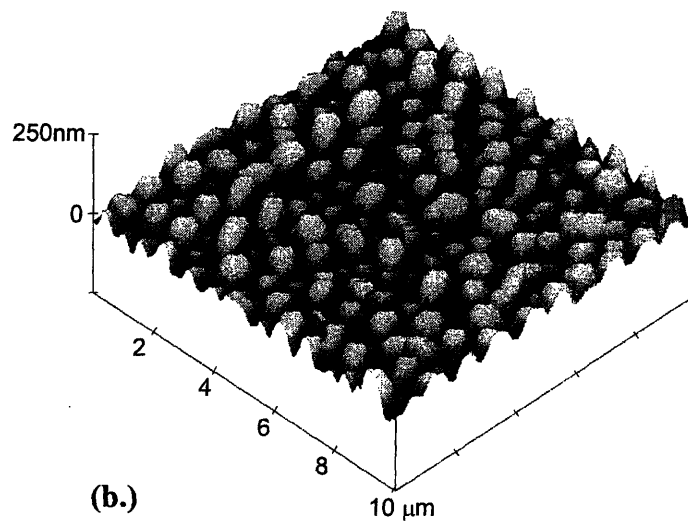
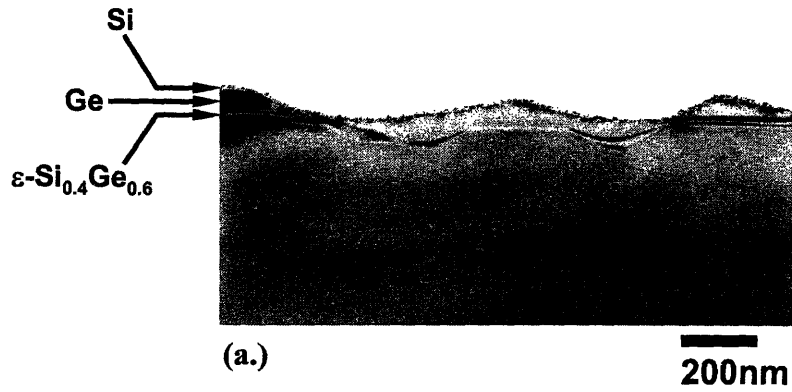


Figure 5.8: XTEM (a.) and (b.) atomic force micrographs of Ge growth initiation at 650 °C. The high surface diffusivity of Ge enables ad-atom interaction with surface contaminants leading to three-dimensional growth and islanding. The RMS surface roughness of the structure is 23 nm as measured from a 10×10 μm AFM scan.

The XTEM micrograph shown in Figure 5.8a illustrates local depressions of growth rate occurring at ~1 μm intervals. This pinch-off of film deposition causes significant roughening of the surface on the μm scale as further demonstrated by the atomic force micrograph given in Figure 5.8b. The total thickness of the deposited layers was 300 nm, leading to a surface RMS roughness of 23 nm as measured from a 10×10 μm AFM scan.

The source of surface roughening of homoepitaxial Ge is partly rooted in the instability of GeO₂ in water, acid and alkali solutions- the same attribute that impedes chemo-mechanical polishing of thin Ge films. Consequently, formation of a protective oxide layer by wet chemical treatment is difficult, leading to surface contamination prior to loading the wafer into the deposition tool. Furthermore, Ge homoepitaxy is particularly sensitive to carbon contamination of the surface. These problems are addressed by a number of methods for MBE-grown films, including *in-situ* sputtering of the surface and low temperature nucleation of a Ge seed layer. Ion sputtering imparts damage to the Ge surface which is not readily removed during subsequent annealing.⁶⁹ Other techniques require careful surface preparation by oxidation with UV-ozone⁷⁰ or chemical⁶⁸ treatments of the Ge wafer prior to insertion into the deposition system. The GeO₂ oxide layer is highly volatile and can be desorbed in UHV by annealing at ~425 °C.⁷¹ Despite subsequent desorption of these oxidized surfaces, some residual carbon contamination of the wafer surface remains^{65,66}, requiring low temperature initiation⁷⁰ of planar Ge layers. The ~350 °C cracking temperature of GeH₄ enables low temperature initiation of Ge, which was investigated for UHVCVD-grown layers in this work.

The substrates used in this study were bulk Ge wafers supplied by Umicore, specified as “epi-ready” for GaAs epitaxy. The wafers were n-doped with $1 \times 10^{17} \text{ cm}^{-3}$ phosphorus and oriented (001) with a 6° offcut toward the nearest {111} plane. Although these substrates are specified for direct insertion into the load chamber, some of them were cleaned prior to growth. This was done to evaluate the potential applicability of this process for planarized virtual substrates, which will require chemical etching prior to epitaxial regrowth. The Ge cleaning procedure that was used is outlined in Table 5.2.

Step	Chemistry/Procedure	Time
1	3 H ₂ O : 1 H ₂ O ₂	30 sec
2	DI H ₂ O rinse	2 min
3	10 H ₂ O : 1 HF	30 sec
4	DI H ₂ O rinse	2 min
5	3 H ₂ O : 1 H ₂ O ₂	30 sec
6	DI H ₂ O rinse	2 min
7	Spin-rinse dry	5 min

Table 5.2: Ge cleaning procedure used prior to epitaxial growth on bulk Ge wafers.

Prior to growth, the wafers were annealed in UHV at 600 °C for 10 minutes to desorb the native GeO₂ oxide. The furnace temperature was then dropped to 350 °C for growth of a 10 nm thick initiation layer. The growth rate of Ge with GeH₄ precursor gas is only 0.1 Å/sec at 350 °C, therefore this is the lowest practical temperature for initiation of Ge epitaxy in the present study. Furthermore, the low surface ad-atom mobility which leads to planar layers can eventually lead to the breakdown of epitaxial growth and a transition to amorphous deposition. Once growth of the initiation layer was completed, the furnace temperature was raised to 450 °C for subsequent growth of the etch-stop and transfer layers.

Figure 5.9 is a XTEM micrograph showing the planar morphology of the Ge transfer structure initiated on a bulk Ge wafer after receiving the cleaning procedure prior to growth. Both substrates (with and without pre-epi cleaning) yielded planar morphology in TEM observations, however some surface roughening of the cleaned wafer is evident from AFM measurements performed after growth. Figure 5.10 is a comparison of the corresponding surface morphologies. The 1×1 μm RMS roughness of the epi-ready Ge wafer prior to any processing was <2 Å. After growing 120 nm of epitaxial film, the surface roughness of this substrate increased to 2.3 Å. The surface roughness of the epitaxial layers was slightly higher on the cleaned wafer, yielding a value of 2.6 Å. These values were determined from an average of five 1×1 μm AFM

scans. The roughening of the etched substrate could probably be reduced by diluting the peroxide and HF etching chemistries. In any case, the roughness values of both samples are sufficiently low for direct wafer bonding.

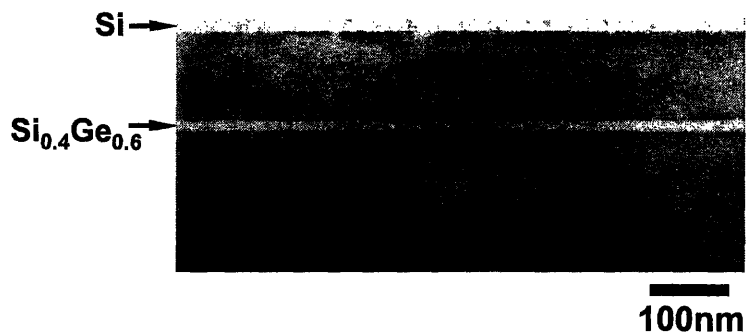


Figure 5.9: XTEM micrograph of a Ge transfer structure grown on bulk Ge after growing a 350 °C initiation layer. By initiating growth at low temperature, Ge ad-atom mobility is reduced yielding planar film morphology.

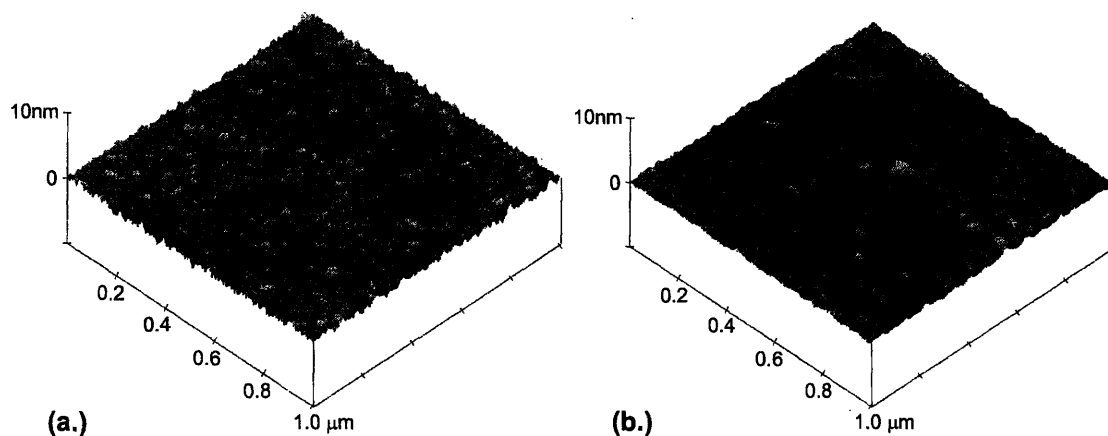


Figure 5.10: Atomic force micrographs showing surface morphology of epitaxial Ge transfer structures grown on epi-ready (a.) and peroxide/HF cleaned (b.) bulk Ge wafers. The cleaned wafer was slightly roughened by the chemical etching process; however both substrates have an RMS surface roughness of $<3 \text{ \AA}$, which is sufficiently low for wafer bonding.

5.4: Thermal stability of Ge transfer structures

Ge layer exfoliation temperatures are generally below 400 °C, where interdiffusion of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop layer is negligible. However, the thermal stability of Ge transfer structures becomes an important issue when utilizing oxide CMP layers. As discussed earlier in Chapter 4, LTO and PECVD oxides contain significant amounts of hydrogen and require densification at temperatures exceeding 600 °C prior to wafer bonding. Unfortunately, interdiffusion of the $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ structure becomes significant at these temperatures as evidenced in Figure 5.11. The thermal budget of Ge transfer processes are then limited by etch-stop interdiffusion during densification of deposited oxides. Figure 5.11 contains SIMS data showing the Si concentration profile of a Ge transfer structure after various annealing treatments. The SIMS data indicate an initial Si fraction of 0.3 in the $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop, however this is an ion-mixing^{72,73} artifact caused by the small thickness (11 nm) of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer.

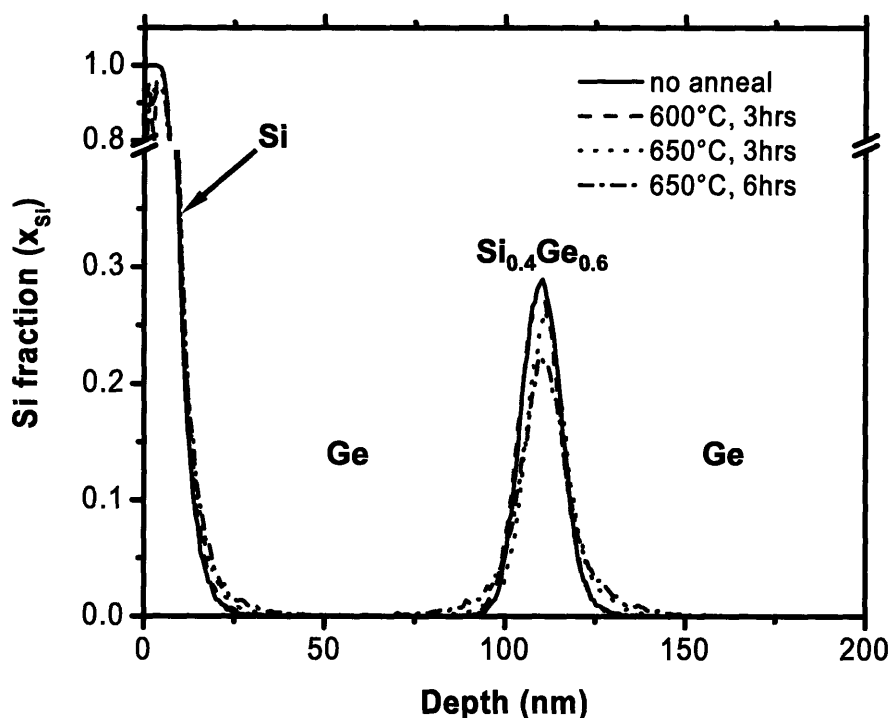


Figure 5.11: SIMS data showing interdiffusion of a $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ etch-stop structure after typical oxide densification anneals. Nominal layer compositions are indicated.

Chapter 5: Etch-stop structures for Ge layer transfer

The decrease in Si fraction is significant after 6 hours of annealing at 650 °C, however the $\text{Si}_{1-x}\text{Ge}_x$ layer loses its etch-stopping characteristics only after the composition of the layer drops below $x \sim 0.2$. This does not occur until after ~ 12 hours at 650 °C for an 11 nm thick $\text{Si}_{0.4}\text{Ge}_{0.6}$ film. Therefore, Ge transfer structures incorporating a $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop are thermally stable relative to minimal LTO densification time/temperature schedules. Other donor structures which have potential applications for transferal of strained GOI layers are not as fortunate.

Figure 5.12 contains interdiffusion data for a strained GOI donor structure grown upon a $\text{Si}_{0.5}\text{Ge}_{0.5}$ virtual substrate. This structure was designed for transferal of a compressively strained Ge film from a $\text{Si}_{0.5}\text{Ge}_{0.5}$ virtual substrate using the layer exfoliation process. In this example, the exfoliation damage is induced in the rightmost $\text{Si}_{0.5}\text{Ge}_{0.5}$ layer which can be readily removed using CMP. The $\text{Si}_{0.2}\text{Ge}_{0.8}$ layer provides an etch-stop layer for exfoliation damage removal using such a CMP process. The large composition mismatch between the strained Ge layer and $\text{Si}_{0.5}\text{Ge}_{0.5}$ substrates results in significant interdiffusion during 600 °C annealing, limiting the densification temperature for oxide CMP layers to ≤ 550 °C.

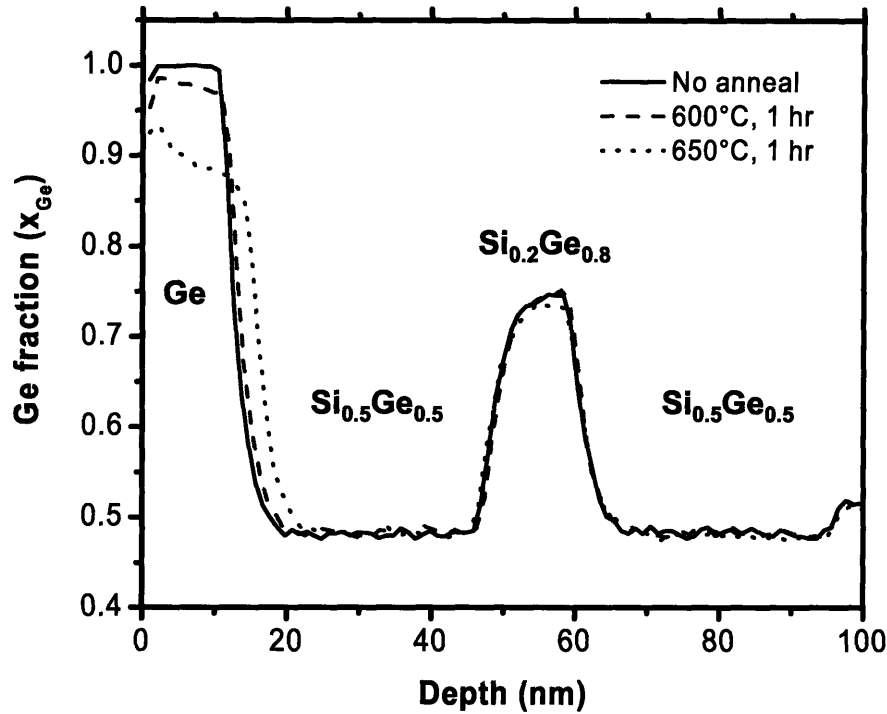


Figure 5.12: SIMS data characterizing the thermal stability of a strained GOI donor structure grown upon a Si_{0.5}Ge_{0.5} virtual substrate. Nominal layer compositions are indicated. The buried Si_{0.2}Ge_{0.8} serves as an etch-stop for subsequent removal of exfoliation damage using chemo-mechanical polishing.

Under-densified oxide layers have a tendency to produce surface bubbles in H-exfoliated layers, as will be demonstrated in the following section. Fortunately, the tolerance of strained GOI donor wafers to thermal treatments could be improved by cladding both sides of the strained Ge layer with tensilely strained Si using a triple-channel device structure.⁷⁴ In addition to increasing the thermal budget of device processing temperatures, these structures also provide carrier confinement to the high hole mobility Ge channel and a Si etch-stop for exfoliation damage removal. Layer transfer of triple channel structures was not attempted in this thesis; however these device layers may provide the optimal combination of thermal stability, etch-stopping behavior and device performance in epitaxially strained GOI structures.

5.5: Epitaxial Ge transfer from virtual substrates

Relaxed Ge containing a 6.5 nm strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop layer was transferred to Si from a virtual substrate after planarization with a LTO CMP layer. The CMP layer was densified at 650 °C for 3 hours prior to planarization. The donor wafers were then implanted with $4 \times 10^{16} \text{ cm}^{-2} [\text{H}_2^+]$ ions at 200 keV prior to hydrophilic wafer bonding to a Si handle wafer. The bond was strengthened by annealing at 250 °C for ~12 hours in N_2 ambient followed by an exfoliation anneal at 450 °C for 30 minutes. Finally, the transferred structure was etched in (30%) hydrogen peroxide to selectively remove the exfoliation damage from the transferred Ge.

Figure 5.13 show a cross-sectional TEM micrograph of the GOI structure after layer transfer and selective etching, demonstrating a process for epitaxial Ge layer transfer from a virtual substrate to a large diameter Si wafer. Using an etch-stop layer, the surface damage induced by the exfoliation process was removed using a selective etch rather than CMP. As a result, the GOI thickness is defined by the etch-stop layer, which is located at the surface of the structure in the micrograph. The selectivity of CMP to Si-rich SiGe alloys could then be used to remove the etch-stop relative to the underlying Ge device layer, leaving a damage-free Ge surface suitable for device fabrication.

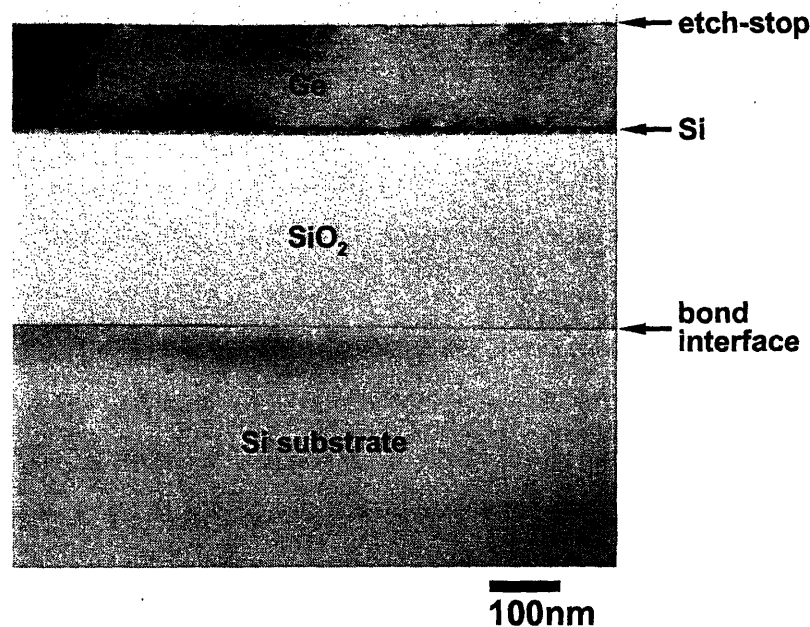


Figure 5.13: XTEM micrograph of an epitaxial GOI structure transferred from a relaxed Ge virtual substrate. A 7 nm thick Si passivation layer is buried between the LTO and Ge device layer. Selective CMP can be used to remove the remaining etch-stop layer at the surface.

The donor structures used in this thesis were capped with a thin Si passivation layer to protect the Ge surface during wafer processing. Because this layer is grown directly on Ge, it contains a high density of defects resulting from the 4% lattice mismatch between Si and Ge. After bonding and film transfer, the passivation layer is incorporated within the GOI structure between the Ge and LTO layers, as shown in Figure 5.13. Since the defects are confined to the Si film, they will have a negligible effect on devices fabricated on the Ge device layer. This is analogous to the virtual substrate where the large dislocation density in the graded region is inconsequential to devices fabricated on the uniform composition cap layer, where the defect density is low. In any case, for the purpose of surface passivation an alternate material such as Si_3N_4 instead of epitaxial Si could be used for GOI fabrication schemes if required for devices fabricated on these substrates.

The surface morphology of the transferred film before and after selective etching is shown in Figure 5.14. Figure 5.14a shows the surface of the as-transferred Ge film,

which is heavily damaged from the exfoliation process. The RMS surface roughness of the GOI at this stage of processing is >50 nm, measured over a 25×25 μm area. After selective etching in hydrogen peroxide, this value is reduced to <15 nm and the crosshatch pattern of the original virtual substrate is revealed as shown in Figure 5.14b. It is interesting to note that the re-emergence of the crosshatch is not caused by anisotropic etching but is a result of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop layer being grown on a crosshatched surface without prior planarization. When the peroxide etch reaches the etch-stop layer, an inverted version of the original crosshatch pattern is delineated in the transferred film. Upon comparison of the surface morphology of the original virtual substrate with that of the transferred and etched Ge layer, we note that the sharp peaks of Figure 4.3a correspond to the narrow valleys of Figure 5.14b. On the submicron length scale, the RMS surface roughness is reduced from 30 to 1.4 nm, as determined from an average of ten 1×1 μm AFM scans.

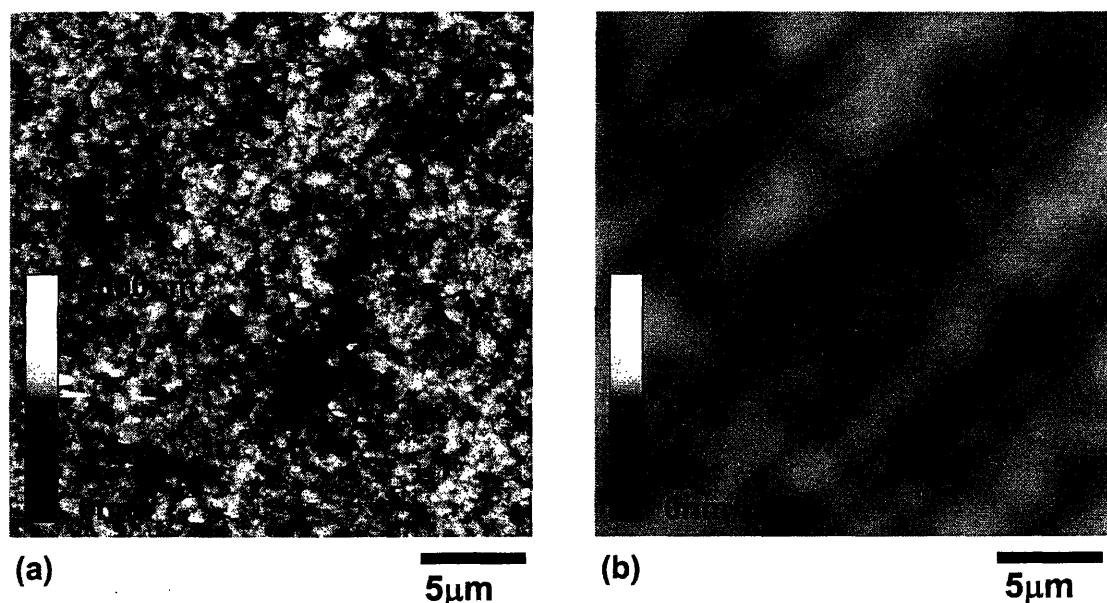


Figure 5.14: Atomic force micrographs showing the transferred Ge surface morphology before (a.) and after (b.) selective chemical etching. (a.) The Ge surface is highly damaged immediately after layer exfoliation, having an RMS roughness of 50 nm. (b.) After selective etching, the roughness is reduced to <15 nm and the crosshatch pattern of the original virtual substrate is revealed.

The exceedingly high surface roughness of the as-transferred film is partially attributed to the large stopping distance associated with the 200 keV energy used to implant the H_2^+ ions for layer transfer. This exfoliation-induced roughness is completely removed after selective etching; however lower ion energies could be used to minimize the surface damage. This would be particularly achievable if the thickness of the oxide CMP layer was reduced or removed entirely. The latter requires direct CMP of the Ge virtual substrate which, if done prior to growth of the etch-stop and Ge transfer layers, will result in a crosshatch-free surface morphology of the transferred Ge film. However, this optimal solution will not be possible until advanced planarization techniques for Ge virtual substrates become available.

It is worthwhile to mention a limitation to H-induced exfoliation of Ge layers. The tendency for H_2 effusion from ion implanted Ge can lead to trapped gas at the bond interface, leading to surface bubbles in the transferred layer. Figure 5.15 contains DIC optical micrographs of Si (5.15a) and Ge (5.15b) layers transferred with a CMPed LTO layer. In this instance, the LTO layer was densified at 550 °C for 3 hours prior to planarization. The inset of the Figure illustrates the structure in cross-sectional view. Both samples were implanted to a dose of $6 \times 10^{16} \text{ cm}^{-2} [H^+]$ at 100 keV. Layer transfer was induced by annealing the bonded wafers at 200 °C for 12 hours following a 30 min exfoliation anneal at 400 °C. The micrograph shown in Figure 5.15a was taken near the edge of the handle wafer, showing the boundary between the transferred Si/LTO layers and Si substrate (the transferred Si layer is otherwise featureless). However, the same process applied to Ge layer transfer yielded bubbles in the transferred layer as indicated in Figure 5.15b.

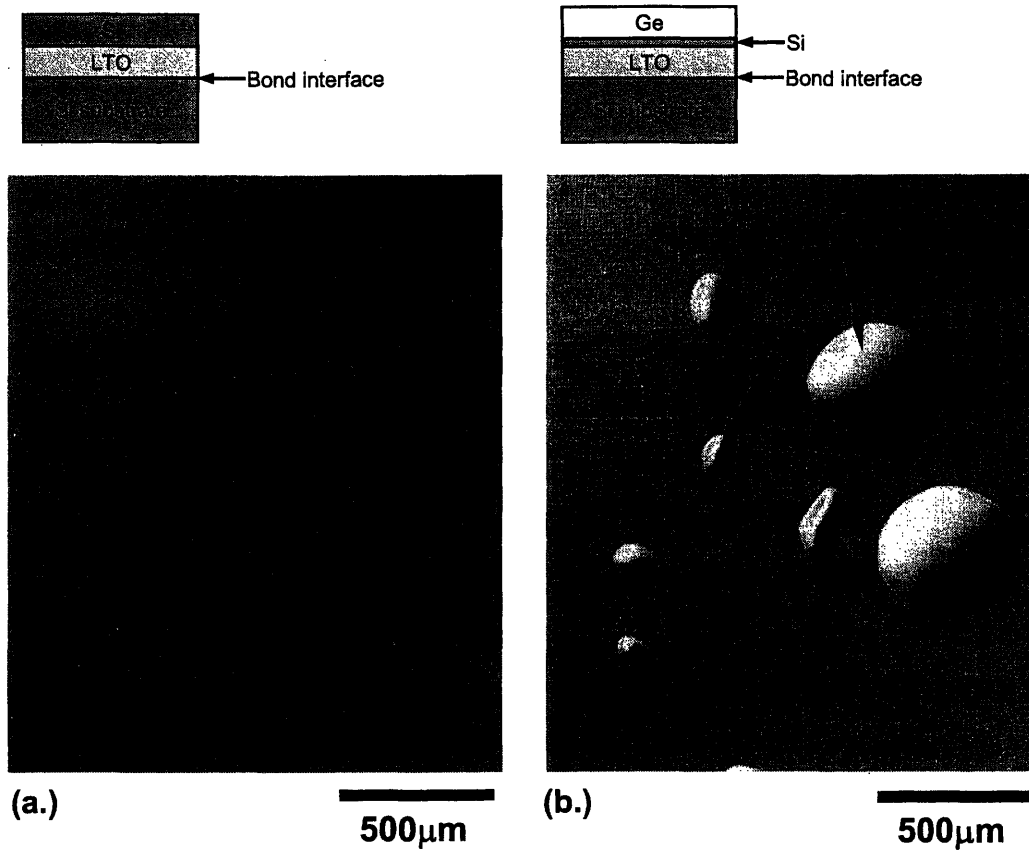


Figure 5.15: DIC optical micrographs of relaxed Si (a.) and Ge (b.) transferred to Si using an LTO CMP layer. The LTO layer was densified at 550 °C for 3 hours in both cases. Both samples were processed identically, however surface bubbles were observed only in the transferred Ge layer.

H₂ effusion from Ge begins at temperatures as low as 200 °C. Consequently, the surface bubbles are likely due to hydrogen gas trapped at the bond interface as illustrated in Figure 5.16. Notably, Ge layers transferred using LTO densified at 650 °C did not contain surface bubbles, indicating that denser oxide layers are less susceptible to this phenomenon.

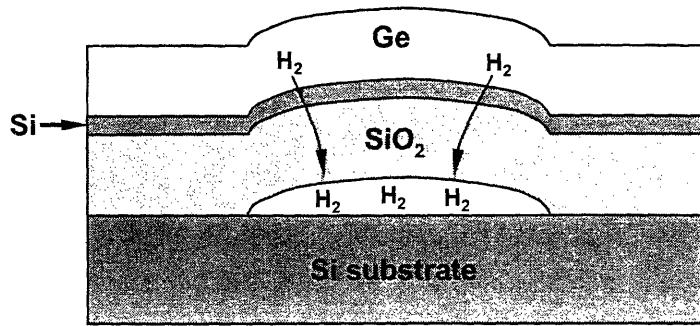


Figure 5.16: Schematic showing surface bubble formation in an exfoliated Ge layer by an accumulation of gaseous hydrogen at the bond interface.

5.6: Conclusion

Etch-stop structures in combination with virtual substrate wafer bonding enable transfer of lattice-mismatched layers to large diameter Si wafers. A demonstration of this technique was presented for epitaxial GOI by means of a $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop incorporated with a relaxed Ge virtual substrate. The use of an etch-stop layer avoids the need for CMP which is difficult to implement with thin Ge films. The etch-stop also serves to define the final thickness of the transferred film entirely by epitaxy, thereby allowing transfer of arbitrarily thin device layers.

Several aspects pertaining to H-exfoliation of Ge remain to be solved. First, some applications may require the bonding of bulk wafers which is limited by thermal mismatch constraints. Secondly, the tendency for H_2 effusion from Ge can lead to trapped gas at the bond interface, reducing the useable area of the transferred layers. Both of these issues can be alleviated by lowering the exfoliation temperature with an increase in hydrogen ion dose. The drawback to this approach is that it will increase the cost of the ion implantation step.

The next chapter presents a fortuitous side-effect of incorporating a tensilely strained layer within hydrogen-implanted donor structures. This has led to a novel solution for decreasing the annealing temperature and implantation dose necessary for layer transfer by utilizing the H-gettering effect of tensilely strained films.

Chapter 6: H-gettering structures for Ge layer transfer

In the previous chapter, a tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer was incorporated with the donor wafer to serve as an etch-stop. Coincidentally, tensile strain also attracts interstitial hydrogen and drives platelet nucleation, both of which are necessary to induce exfoliation of a hydrogen-implanted surface. This chapter describes layer exfoliation as a gettering phenomenon and investigates the use of gettering structures incorporating a tensilely-strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ film for layer transfer applications.

6.1: Introduction

H-induced layer exfoliation, commercially known as the Smartcut™ process, utilizes hydrogen or helium ion implantation to separate a thin layer of material from the surface of a semiconductor wafer.³⁸ Since its initial disclosure in 1995, Smartcut™ has become a central component for transfer of thin Si layers for fabrication of silicon-on-insulator (SOI) substrates. Layer exfoliation is also likely to become the preferred approach in generalized monolithic material integration schemes, owing to its versatility of use with a variety of materials as well as having the possibility of re-using the donor wafer, which is attractive for layers generated by epitaxy due to cost concerns. However, the Smartcut™ process has the disadvantage of requiring an expensive ion implantation step, the cost of which increases with ion dose. Furthermore, the bonding of thermally mismatched substrates such as bulk Si and Ge wafers typically has a limited thermal budget. Therefore, the layer splitting temperature must be minimized to prevent bond failure or wafer cracking from thermal stress.

In lieu of its drawbacks, the potential importance of layer exfoliation for advanced substrate technologies has encouraged researchers to devote the last decade to the study of the underlying mechanisms responsible for H-induced blistering and layer separation. Although radiation-induced surface blistering has been observed in metals for decades⁷⁵⁻⁷⁷, early investigations were dedicated to preventing their formation in radiation-exposed nuclear reactor components. During the Smartcut™ process, substrates are encouraged to blister after H⁺ or He⁺ ion implantation. Therefore studies have been geared toward improving the exfoliation efficiency of the implanted ions. The following section reviews the progress made in layer exfoliation technology from the perspective of H and He gettering in semiconductors, leading to H-gettering structures which is the focus of this chapter.

6.2: Layer exfoliation as a gettering process

The underlying basis for H-induced layer exfoliation is the inherently low solubility ($\sim 10^7 \text{ cm}^{-3}$) of hydrogen in semiconductors.⁷⁸ Ion implantation is therefore the only means of introducing the supersaturated amount ($\sim 10^{21} \text{ cm}^{-3}$) of H to the semiconductor wafer that is necessary to induce layer separation. Post-implant annealing causes the H_2 to come out of solution and segregate to subsurface voids. The enthalpy of solution released during this process provides the energy necessary to pressurize these voids with H_2 leading to blistering and eventual exfoliation of the wafer surface.

A limited solubility is not enough to achieve the process of exfoliation however, since hydrogen could simply diffuse out of the wafer during annealing. The implant has a second important feature in that it creates damage, composed of defects and traps, which can act as nucleation-sites for H_2 bubbles. Interstitially dissolved H_2 can find space in the lattice at these defects, thereby limiting the loss of hydrogen to surface effusion during post-implant annealing. Numerous studies⁷⁹⁻⁸⁵ and reviews^{86,87} are available on the subject of layer exfoliation. The following section provides an overview of the layer exfoliation process from the standpoint of gettering in semiconductors.

6.2.1: Overview of H-gettering in layer exfoliation processes

Gettering phenomena are based on the electronic interaction of an impurity species with the gettering medium. Insertion of a hydrogen molecule into an interstitial site of a crystal lattice causes a local distortion of the electron cloud, which increases the energy of the system. This interaction limits the solubility of hydrogen in semiconductors where the small amount of dissolved H_2 molecules is accommodated by the entropy of mixing. However, implanted hydrogen atoms can lower their energy by bonding to dangling valence states present at crystal defects or hetero-valent impurity species. These interactions manifest themselves by the passivation of dangling bonds at vacancies or interstitials or by the formation of hydrogen-dopant complexes. Hydrogen is also getterred by tensile strain where the dilation of the crystal lattice reduces the

electronic repulsion between the impurity species and the host crystal, causing a local enhancement of the solubility of interstitially-dissolved hydrogen relative to the unstrained state.

Most of the point defects generated by light ion implantation are vacancy-like in nature (i.e. consisting of vacancies and vacancy clusters) and are generated toward the end of the ion trajectory.⁸⁸ Thus, the damage distribution closely resembles that of the implanted species with the exception that the range of the implanted species is slightly larger than that of the damage profile. A vacancy provides gettering opportunities by creating four dangling bonds and a tensile strain field in the surrounding lattice, both of which provide a low energy site for hydrogen. These interactions result in seemingly anomalous diffusion behavior of the implanted hydrogen during subsequent annealing whereby the implanted hydrogen diffuses uphill, thus consolidating the original distribution.

A second gettering mechanism that is exclusive⁸⁹ to the presence of hydrogen are $\{111\}$ and $\{100\}$ planar defects called platelets which consist of internal semiconductor surfaces terminated with hydrogen.⁷⁶ An atomistic schematic of a platelet is shown in Figure 6.1, revealing a close resemblance of the lattice distortion generated by a platelet to that of a prismatic dislocation loop.

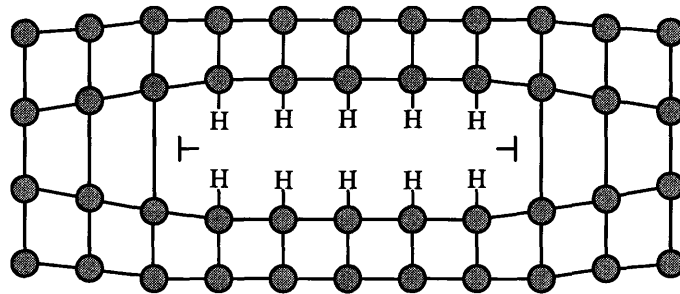


Figure 6.1: Atomistic schematic of a H-induced platelet in a crystalline semiconductor.

Once nucleated, platelets become the central gettering mechanism in hydrogen-induced layer exfoliation.⁸⁷ In practice, platelets are best observed edge-on in TEM

observations. A $\{111\}$ platelet in Ge is shown in the high-resolution TEM micrograph of Figure 6.2.

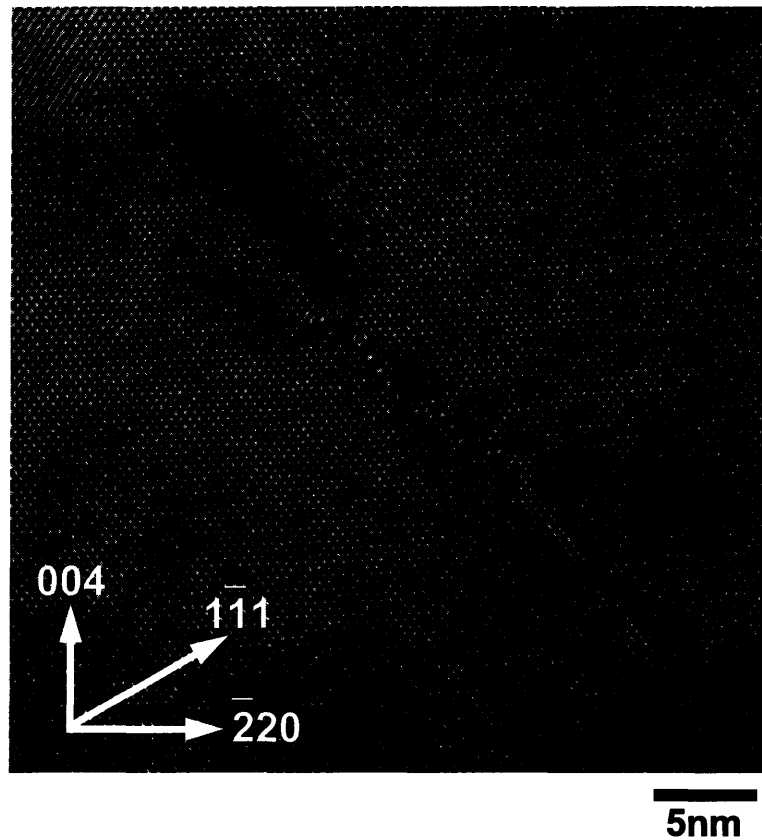


Figure 6.2: A HRTEM micrograph of a $\{111\}$ platelet viewed edge-on in a H^+ -implanted Ge crystal. The defect consists of H-terminated $\{111\}$ surfaces within the crystal. Platelets are efficient gettering sites for hydrogen and contribute to the H-induced layer exfoliation process.

Platelet microstructure and the mechanism of their formation have been the subject of numerous studies.^{79, 89-97} Platelets are unlike stacking faults in that they are not bound by a dislocation loop and do not exhibit stacking fault contrast in TEM observations.^{89, 92, 94, 96} Furthermore, the lattice displacement perpendicular to the platelet habit plane is generally smaller than that of an extrinsic stacking fault.⁹³

While the nucleation mechanism of platelets is still a debated subject; it is known that they can form without the aid of radiation damage. Platelets can be induced in bulk

silicon by a remote hydrogen plasma treatment.⁹² The platelet and H concentration introduced by remote plasma is insufficient to cause layer separation. Consequently, this technique is not employed for layer transfer processes. However, the fact that platelets can be induced without direct exposure of high energy ions leads to the general belief that they are not the result of point defect condensation as once thought.

As mentioned earlier, platelets are efficient gettering centers for hydrogen. As the platelet fills with gaseous H₂ molecules, it grows into a nanocavity pressurized with hydrogen gas. The pressure builds until it is finally released through blistering of the surface- an occurrence that is the basis for the exfoliation process. However, the extreme reactivity of atomic hydrogen causes a portion of the implanted atoms to also become occupied at the dangling bonds of vacancies, interstitials and other implantation-induced defects. This reduces the amount of hydrogen available for pressurizing internal cavities. In contrast, helium is unreactive and can effectively pressurize the H-stabilized platelet structures. As a result, the combined effect of hydrogen and helium has been shown to reduce the total dose necessary for layer exfoliation from $\sim 10^{17}$ to $\sim 2 \times 10^{16} \text{ cm}^{-2}$ compared to implantation of either H⁺ or He⁺ alone.⁹⁸

Another improvement to layer exfoliation is described by Tong, *et al.* who propose the co-implantation of boron with hydrogen in a process termed “smarter cut”.⁹⁹ It is well known that electronically active acceptor states getter hydrogen through formation of H complexes.¹⁰⁰ This effect is expected to prevent effusion of the implanted hydrogen and provide nucleation sites for platelet formation. Secondly, co-implantation of boron creates additional damage in the wafer, further increasing its hydrogen trapping efficiency. Although the exact mechanism by which layer exfoliation occurs in B and H co-implanted Si is somewhat ambiguous, the final result clearly indicates a significantly reduced thermal budget for layer transfer.

Yet another proposal by Usenko, *et al.* for H-gettering involves formation of a damaged region below the surface of the donor wafer by means of an inert gas or self ion implantation.¹⁰¹ The subsurface damage caused by the ion implant is then used to getter H introduced to the wafer through H plasma exposure. Unfortunately, this process is not

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suitable for applications with a limited thermal budget since the substrate must eventually be treated at high temperature to anneal out the implantation damage.

6.2.2: H gettering in tensilely strained Si_{0.4}Ge_{0.6}/Ge layers

Introduction of an interstitial hydrogen molecule into a crystal causes a local distortion of the surrounding lattice. The energy associated with this process is related to the specific volume of the impurity atom, $\bar{V}_{H_2^i}$, and the stress state in the vicinity of the occupied interstitial site. Accordingly, the solubility of interstitial hydrogen, $[H_2^i]$, is affected by hydrostatic pressure according to Equation 6.1.

$$[H_2^i](P) = [H_2^i]_0 \exp\left[\frac{-P\bar{V}_{H_2^i}}{k_B T}\right], \quad \text{Equation 6.1}$$

where $[H_2^i]_0$ is the hydrogen solubility in the absence of pressure, k_B is Boltzmann's constant and T is the absolute temperature. The applied pressure, P , can be expressed in terms of the principal stress components as $-(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})/3$. For a layer under biaxial strain, $\sigma_{xx} = \sigma_{yy}$ and $\sigma_{zz} = 0$. Therefore, in terms of the biaxial strain of the layer, ϵ , Equation 6.1 becomes

$$[H_2^i](\epsilon) = [H_2^i]_0 \exp\left[\frac{2\epsilon M \bar{V}_{H_2^i}}{3k_B T}\right], \quad \text{Equation 6.2}$$

where M is elastic modulus for a state of biaxial strain.

Since interstitial hydrogen creates a dilation of the surrounding lattice, $\bar{V}_{H_2^i}$ is always a positive value. Therefore tensile strain ($\epsilon > 0$) causes an exponential increase of H solubility while compression ($\epsilon < 0$) has the opposite effect.¹⁰²

It is a well known fact that interstitially-dissolved hydrogen accumulates at tensile stress concentrators such as crack tips leading to hydride-induced fracture.¹⁰³⁻¹⁰⁶ Hydrogen will also accumulate in the vicinity of a dislocation, forming a Cottrell¹⁰⁷ atmosphere by congregating in its tensile strained regions. Similarly, the author has discovered that tensile strained epitaxial layers also exhibit hydrogen gettering behavior. Figure 6.3a is a SIMS profile showing preferential accumulation of H at the tensile strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop layer discussed in the previous chapter. Figure 6.3b shows corresponding selective damage induced by the hydrogen to the etch-stop layer after H^+ implantation and subsequent annealing at 400 °C. The damage in the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer degrades its etch stopping performance- a problem that can be easily avoided by means of sufficiently deep implants. Conversely, compressively strained SiGe layers grown upon a Si substrate have previously been shown to *reject* implanted H in US patent #6,500,732.¹⁰⁸ This behavior is expected based on Equation 6.2.

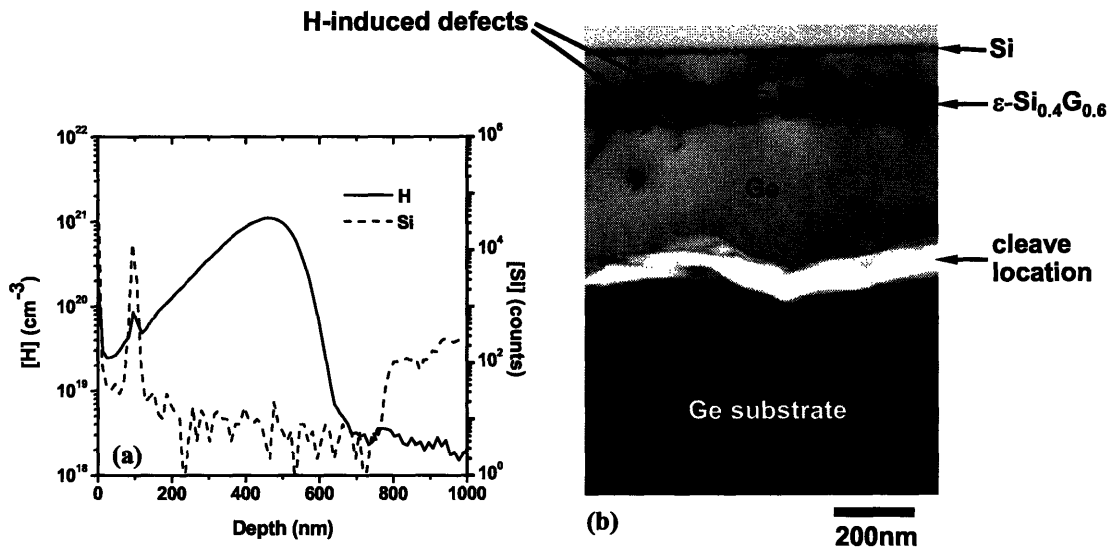


Figure 6.3: SIMS (a.) and XTEM (b.) data demonstrating an accumulation of H at the tensile strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop layer. The TEM micrograph illustrates hydrogen-induced defects in the etch-stop layer. The nature of these defects will be discussed in Section 6.4.

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In addition to enhancing the solubility of hydrogen, tensile strain drives precipitation of lattice-dilating platelet defects. In bulk materials, platelets occur with $\{111\}$ and $\{100\}$ orientations and cause a corresponding lattice dilation in the direction perpendicular to their habit plane. The strain energy released by a platelet with a dilation vector, $\vec{b}$, in the biaxial plane is given by Equation 6.3, where r is the radius of the platelet and σ is the stress in the layer.

$$\Delta W = -\pi r^2 \bar{\sigma} \cdot \vec{b}$$

Equation 6.3

As a result, the author has found that platelets habit the (100) and (010) planes in tensile layers to optimally relieve biaxial strain, as illustrated in Figure 6.4. Analogously, pure edge (90°) dislocations are most efficient for relieving mismatch stress in diamond cubic semiconductors. However, unlike 90° dislocations, $\{100\}$ platelets can nucleate easily as shall be seen experimentally in Section 6.4 and analytically in Chapter 7. These platelets provide additional gettering sites for hydrogen and H-decorated point defects. Consequently, preferential nucleation of platelets in tensilely strained layers provides a secondary gettering mechanism for hydrogen which can be exploited for layer exfoliation applications.

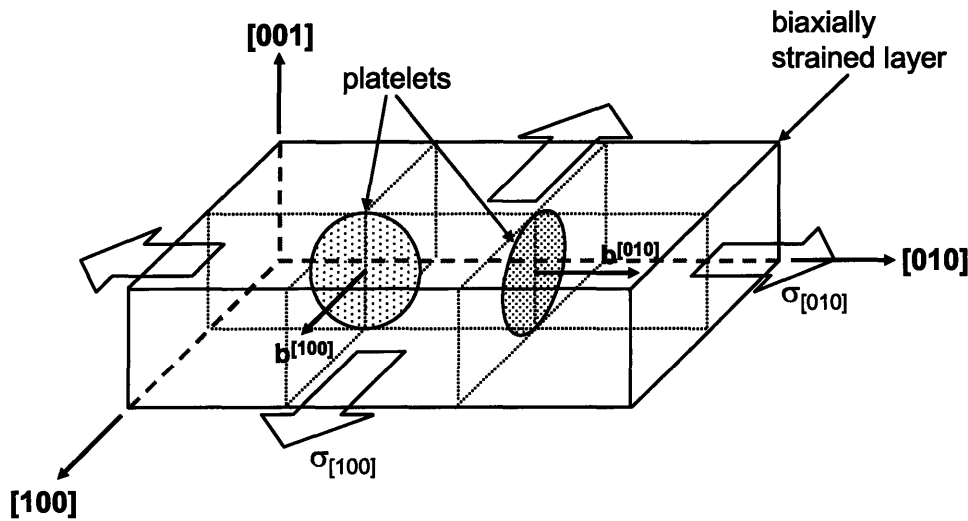


Figure 6.4: Schematic of (100) and (010) platelets within a biaxially strained layer.

The gettering mechanisms that are active in the various H^+ or He^+ induced exfoliation methods discussed in this section are summarized in Table 6.1.

Method	Gettering mechanism(s)	Advantages	Disadvantages
H^+ implantation	Platelets	Platelets are effective gettering sites for gaseous H_2	H trapped at dangling bonds do not contribute to pressurizing voids
He^+ implantation	Vacancy clusters	He is non-reactive, effective at pressurizing voids	No platelets, fewer gettering centers compared to H
H^+/He^+ co-implantation	Platelets	He is non-reactive, efficient at pressurizing H-induced platelets	Excess crystal damage caused by He^+ ions
H^+/B^+ co-implantation	Platelets, H-B complexes	Boron aids platelet nucleation and reduces H_2 effusion	Excess crystal damage caused by B^+ ions, H trapped at H-B complexes does not contribute to pressurizing voids
H^+ implantation of a tensilely strained layer (this work)	Tensile strain, preferentially nucleated platelets	H solubility and platelet nucleation exponentially enhanced with strain	Requires epitaxy

Table 6.1: Summary of various hydrogen or helium gettering mechanisms active during layer exfoliation applications.

6.2.3: Ge layer transfer structures incorporating a hydrogen-gettering layer

In the previous chapter, tensilely strained $Si_{0.4}Ge_{0.6}$ was used as an etch-stop for selective exfoliation damage removal. In addition, tensilely strained layers can be used to getter hydrogen and enhance layer exfoliation efficiency. Transfer structures can therefore include two tensilely strained $Si_{0.4}Ge_{0.6}$ layers: one to serve as an etch-stop and a second to getter hydrogen as shown in Figure 6.5.

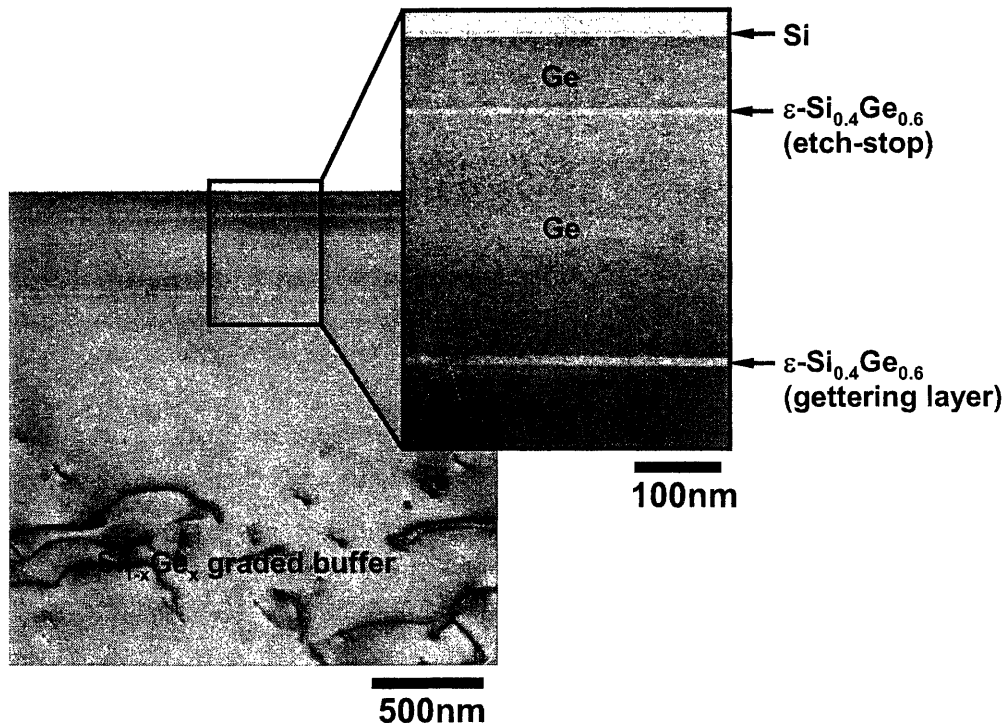


Figure 6.5: XTEM micrograph of a Ge donor structure incorporating a gettering layer. The topmost Si_{0.4}Ge_{0.6} layer is an etch-stop for exfoliation-induced damage removal. The second Si_{0.4}Ge_{0.6} layer provides a hydrogen gettering site for the exfoliation process.

The Ge transfer structure shown in Figure 6.5 consists of two tensilely strained Si_{0.4}Ge_{0.6} layers separated by a relaxed Ge spacer. The bottommost Si_{0.4}Ge_{0.6} layer serves to getter hydrogen and is therefore called the “gettering layer”, while the upper Si_{0.4}Ge_{0.6} layer provides an etch-stop for Smartcut™ damage removal. The uppermost relaxed Ge will serve as the device layer in the transferred film. The epitaxial structure is capped with a thin Si film that serves to protect the underlying Ge during subsequent processing steps, while also reducing the interfacial trap density of the back oxide-semiconductor interface in the final GOI structure.

Figure 6.6 is a process flow utilizing this described Ge transfer structure for fabrication of GOI substrates.

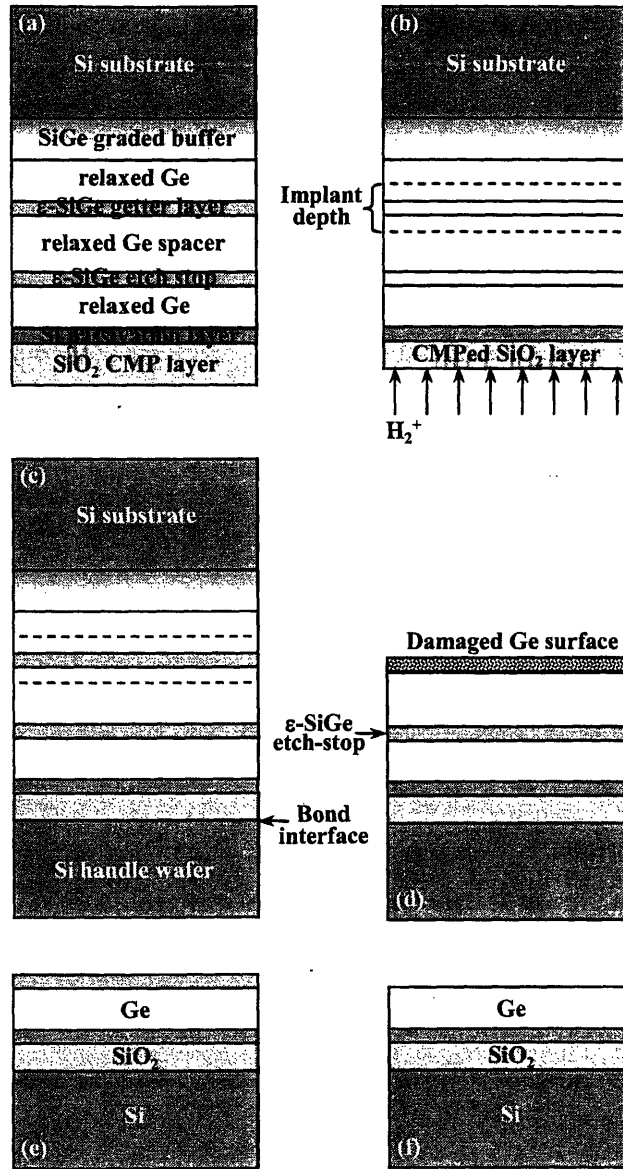


Figure 6.6: Germanium transfer process utilizing a H-gettering layer. (a) Preparation of the virtual substrate: growth of Ge transfer structure incorporating a tensilely strained Si_{0.4}Ge_{0.6} gettering layer on a relaxed Ge/Si_{1-x}Ge_x/Si virtual substrate followed by SiO₂ CMP layer deposition using LPCVD. (b) Planarization of the SiO₂ CMP layer and subsequent H₂⁺ implantation. (c) Wafer bonding the virtual substrate to a Si handle wafer. (d) Anneal and exfoliation of the SiO₂, Ge and Si_{0.4}Ge_{0.6} etch-stop layers to a Si handle wafer. (e) Selective etching in H₂O₂ to remove the damaged Ge surface. (f) Selective CMP to remove the Si_{0.4}Ge_{0.6} etch-stop layer.

6.3: Design of H-gettering experiments

Hydrogen gettering by tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layers was studied by characterizing H distribution, layer morphology and strain relaxation of H^+ -implanted samples as a function of annealing temperature. The following section provides the experimental details to the work presented in the remainder of the chapter.

6.3.1: TRIM simulations for projecting H and vacancy profiles

All of the Ge donor wafers were implanted with H^+ prior to characterization of their gettering behavior. SRIM-2003.20 simulation code¹⁰⁹ was used to model the ion trajectories and vacancy distribution generated during the implantation step. The vacancy distribution was modeled with the Kinchin-Pease formulation using a displacement energy value of 15 eV for Ge. This model does not account for recombination of vacancies and interstitials during irradiation and cannot be used to calculate the overall point defect concentration; however it is useful for estimating the damage distribution relative to the hydrogen profile. The simulation results were used to tailor the conditions necessary to generate the desired hydrogen profiles relative to the $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer. Figure 6.7 shows the simulation result for a 100 keV H^+ implantation into Ge. The corresponding H and vacancy profiles closely resemble experimental results published in the literature.⁸³

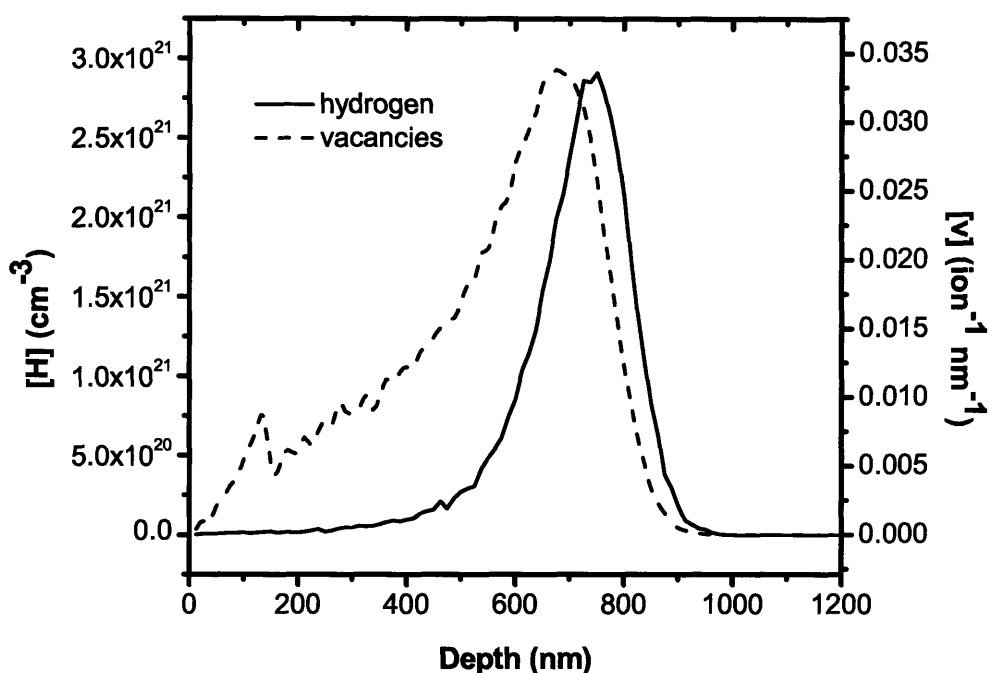


Figure 6.7: SRIM simulation of a 100 keV H^+ implantation into Ge. Most of the vacancies generated during light ion implantation are formed near the end of the ion trajectory, leading to a damage profile that closely resembles that of the implanted species.

According to Figure 6.7, the average range of the implanted ions is slightly larger than that of damage production. This is correlated with the fact that the ion-induced cut location does not coincide with the depth of highest hydrogen concentration.⁸⁰ Instead, the cleavage plane occurs at a shallower depth, corresponding to a region of higher damage production. The location of the cleavage plane relative to the implanted H distribution illustrates the role of implantation damage in layer exfoliation, as discussed in Section 6.2.

6.3.2: Experimental

Figure 6.8 contains schematics of the epitaxial structures along with the various implantation profiles studied in the H-gettering experiments. Sample 1 (Figure 6.8a)

contains a single, 11 nm thick tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer located ~ 100 nm below the wafer surface. Sample 2 (Figure 6.8b) contains two tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layers separated by a relaxed Ge spacer layer. The first strained layer is 6 nm thick and ~ 100 nm below the surface while the thickness of the second layer is identical to that of Sample 1, however it is buried ~ 450 nm below the surface. A XTEM micrograph of Sample 2 was previously shown in Figure 6.5. A third sample containing two thicker tensilely strained layers (11 and 24 nm, respectively) was also grown but will be omitted here for clarity. The experimental results for this sample (Sample 3) are given in Appendix A.

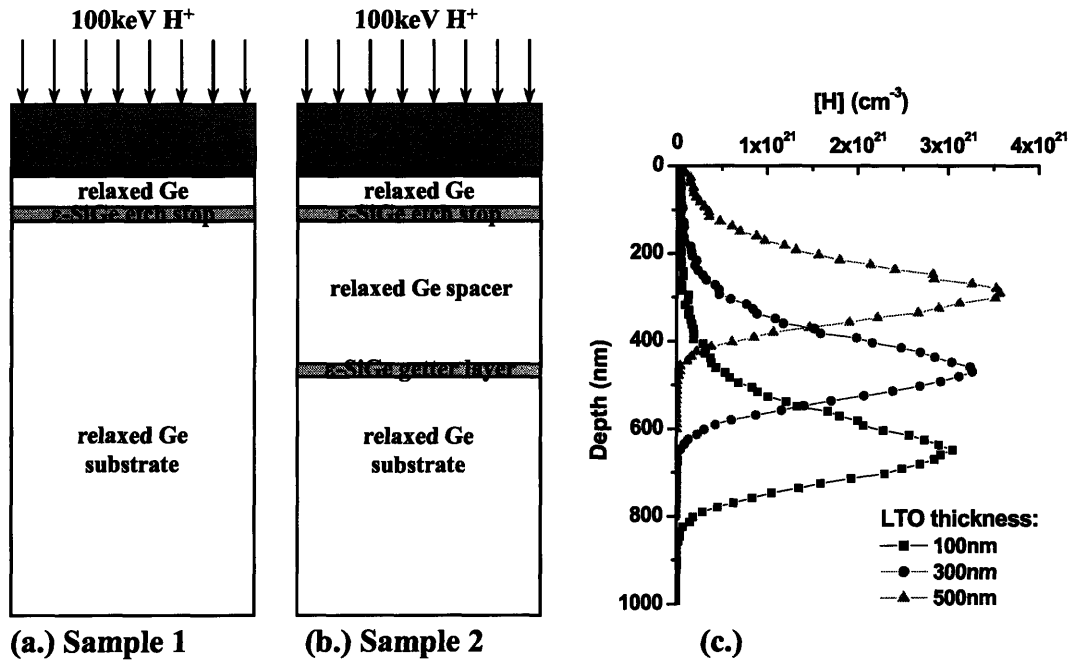


Figure 6.8: (a.)-(b.) Schematic of two epitaxial structures used in the gettering experiment. Sample 1 contains a single strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop layer near the surface. Sample 2 contains an additional $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer at a depth coinciding near the maximum implanted H concentration. The implant depth was varied by means of an LTO layer deposited prior to implantation with H^+ at 100 keV. (c.) H profiles simulated by TRIM are shown relative to the epi surface for various LTO thickness values.

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The samples used in this study were grown upon relaxed Ge virtual substrates using 150 mm (001) Si wafers, offcut 6° towards the nearest {111} plane. Using UHVCVD for the growth technique, the same considerations were made as for growing the etch-stop structures discussed in Chapter 5.

To investigate the effect of implant proximity to the strained layer, the implantation depth into the gettering structure was modulated by means of a low temperature oxide (LTO) layer deposited on the surface of the wafer prior to implantation. By utilizing this approach, multiple implant depths can be obtained in a single implantation lot. The oxide layer provides an additional benefit as a screen against ion channeling.

The LTO was deposited at 400 °C and subsequently densified at 600 °C for 1 hour prior to ion implantation. The implantation step was carried out in an Axcelis GDS ion implanter maintaining a beam current of 5.8-5.9 mA and a substrate temperature of <50 °C. Each sample was implanted with H⁺ at 100 keV and a dose of either $1 \times 10^{16} \text{ cm}^{-2}$ or $6 \times 10^{16} \text{ cm}^{-2} [\text{H}^+]$. The $1 \times 10^{16} \text{ cm}^{-2}$ dose did not yield surface blistering in these samples and will not be discussed in the following sections. Figure 6.8c is a summary of SRIM simulation results showing the nominal implantation depths for the three LTO thickness values used in the experiment.

After implantation, the LTO layer was stripped from the substrates by etching in hydrofluoric acid. The wafers were then cleaved into $\sim 1 \times 1$ cm pieces and annealed in a tube furnace under N₂ ambient prior to characterization for H redistribution, morphology and Si_{0.4}Ge_{0.6} layer relaxation. The results of these experiments are discussed in the next section.

6.4: H-gettering behavior of strained Si_{0.4}Ge_{0.6}/Ge structures

Hydrogen gettering by a buried layer of tensile SiGe manifests itself by an accumulation of H and nucleation of (100) and (010) platelets within the biaxially strained layer. This section characterizes the accumulation of H at the strained layer using secondary ion mass spectroscopy (SIMS). These results are then correlated to the

evolution of platelet microstructure which is characterized using transmission electron microscopy (TEM) and strain relaxation measurements using Raman spectroscopy.

6.4.1: Hydrogen redistribution during annealing

SIMS was used to analyze the hydrogen profiles in the samples after various thermal treatments. The sensitivity of the analysis to H in Ge was initially calibrated with a hydrogen forward scattering (HFS) experiment, also providing a verification of the implantation dose which was found to be $6.2 \times 10^{16} \text{ cm}^{-2} [\text{H}^+]$. Figure 6.9 shows annealed hydrogen profiles for Samples 1 (Figure 6.9a) and 2 (Figure 6.9b) after implantation at 100 keV to a nominal value of $6 \times 10^{16} \text{ cm}^{-2} [\text{H}^+]$ through a 300 nm LTO layer. The annealing time for both samples was 1 hour.

Sample 2 (Figure 6.9b) shows enhanced hydrogen concentration in the strained layers after annealing at 200 °C. The apparent decrease in total hydrogen content after annealing at temperatures ≥ 250 °C can be attributed in part to the dissolution of interstitial hydrogen, forming gaseous H_2 in internal cavities and platelets. The sensitivity of SIMS is reduced for hydrogen in a gaseous state leading to an apparent decrease of total hydrogen by SIMS analysis. The strain fields generated by the internal pressure of H_2 -filled cavities are readily observable by TEM, as described in Section 6.4.2. Accumulation of gaseous H_2 also increases the dilation vector of platelets which can be characterized by combining TEM data with strain relaxation measurements as discussed in Section 6.4.3.

Hydrogen gettering is also observable at the tensilely strained etch-stop layer near the surface of both samples. Excessive accumulation of hydrogen in this film can have a deleterious effect on etch-stop performance and should be avoided with a sufficiently deep implant, typically ~ 500 nm relative to the etch-stop layer.

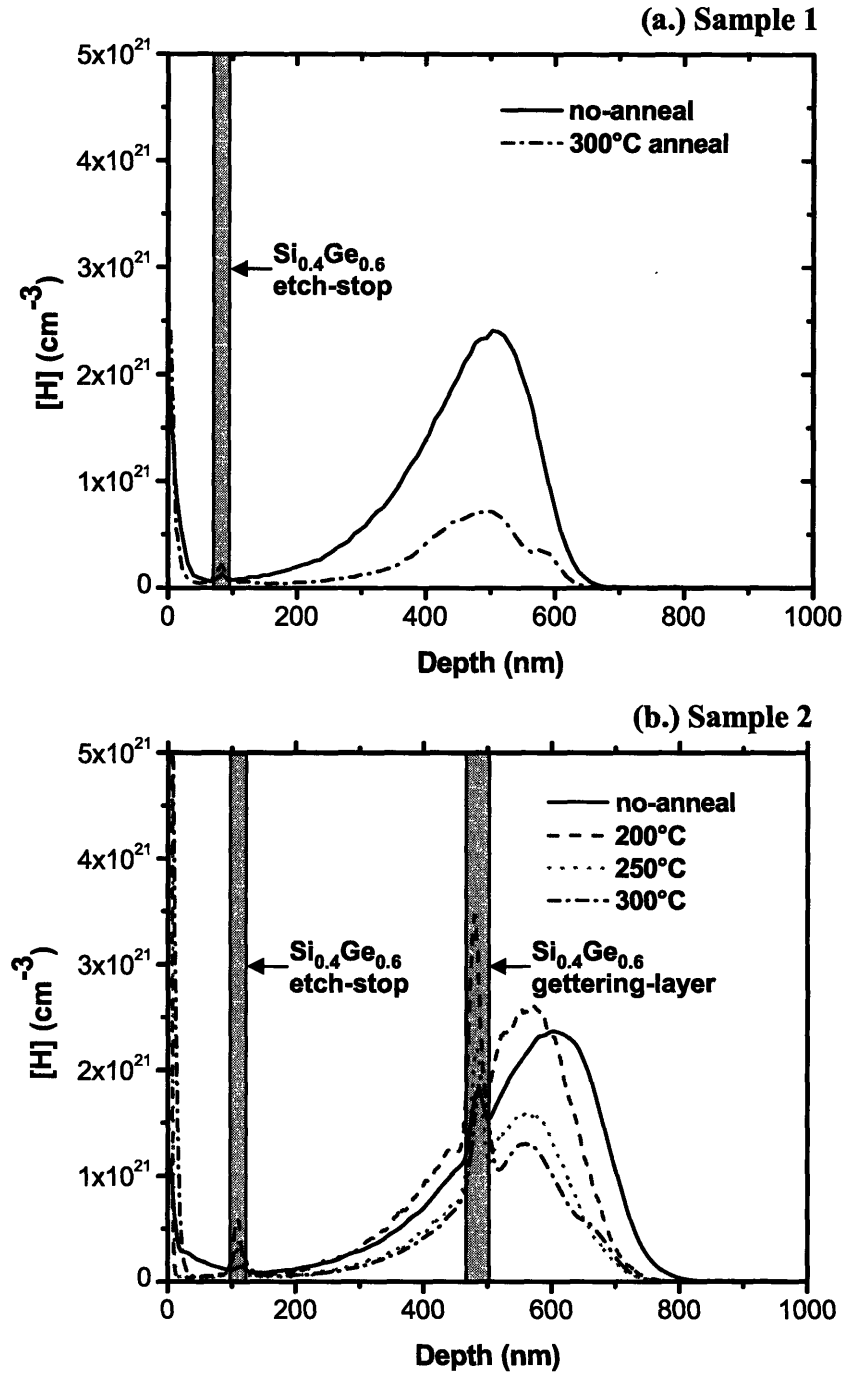


Figure 6.9: SIMS data showing the evolution of H distribution in Samples 1 (a.) and 2 (b.) during low temperature annealing for 1 hour. Sample 2 shows significant accumulation of hydrogen at the gettering layer after 200 °C annealing. The subsequent drop in hydrogen concentration is partially caused by the formation of gaseous H_2 at platelets by the dissolution of interstitial hydrogen. The shift of the H profile toward the surface and the shoulder at the tail end of the annealed profiles is the result of hydrogen gettering by radiation damage.

The series of H concentration profiles in Figure 6.9 also illustrates the effects of implantation damage on the redistribution of hydrogen during annealing. First, the annealed profiles are shifted toward the surface of the sample by ~50 nm, corresponding to the separation of the H and vacancy distributions simulated by TRIM in Figure 6.7. Furthermore, the 200 °C anneal (shown in Figure 6.9b) consolidates the implanted H, resulting in a sharper profile compared to the unannealed sample. The uphill diffusion of H that created this profile is caused by the preferential accumulation of H at the defects generated during the implantation step. In addition, the shoulder that appears near the end of the annealed profiles (at ~550 nm in Figure 6.9a and ~650 nm in Figure 6.9b) is the result of radiation damage cutoff that occurs when the accelerated ion nears the end of its trajectory. Point defect production ceases at this depth leading to a cutoff of hydrogen gettering by implantation damage.

6.4.2: Evolution of $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}(\text{H})$ microstructure during annealing

Accumulation of hydrogen is accompanied by platelet nucleation within the gettering layer. During annealing, the mechanism for platelet growth in tensilely strained layers changes from strain-limited to H_2 -diffusion limited, resulting in an evolution of platelet morphology. This section will use TEM to study these regimes in H^+ -implanted gettering structures as a function of annealing temperature.

All of the cross-sectional samples imaged in this section correspond to Sample 2 after implantation through a 300 nm LTO layer. The resulting hydrogen profile for this implant was shown in Figure 6.9b. The TEM foils were prepared by mechanical lapping of wafer sections to a thickness of 5-10 μm , followed by Ar^+ ion milling to electron transparency. The temperature was kept below 80 °C during the thinning steps to minimize hydrogen interaction with the sample.

The plan-view samples correspond to Sample 1 implanted through a 500 nm LTO layer, corresponding to the nominal H-profile shown in Figure 6.8c. The purpose for using a sample containing a single strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer near the surface for plan-view imaging was to facilitate sample preparation since the 100 nm thick Ge layer above the

strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ film is sufficiently thin to provide electron transparency. Prior to lapping and Ar^+ ion thinning, the Si passivation layer was selectively removed by chemical etching in KOH (at 60 °C).

Interaction of the gettering layer with the implanted hydrogen atoms begins during the implantation step. Figure 6.10 is a XTEM micrograph of the as-implanted gettering structure. Although this micrograph was taken prior to annealing, some *in-situ* heating of the substrate occurs during ion implantation, allowing a portion of the hydrogen to diffuse to the strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer. Furthermore, a narrow (~10 nm) zone denuded of implantation damage is observed on either side of the gettering layer.

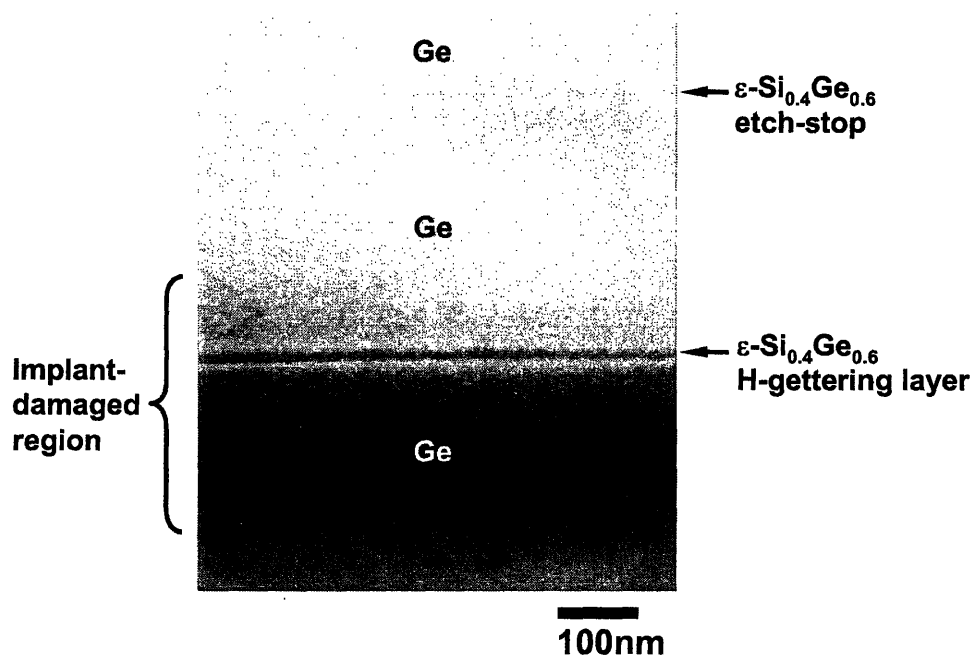


Figure 6.10: XTEM micrograph along the [110] pole showing a H^+ -implanted gettering structure prior to furnace annealing. A narrow (~10 nm) denuded zone is visible on either side of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer.

Upon annealing at 200 °C, the denuded zone becomes more clearly defined as seen in Figure 6.11a. In this example, the TEM foil was prepared along the [100]

direction, revealing preferential nucleation of (010) platelets within the gettering layer, as indicated in Figure 6.11b.

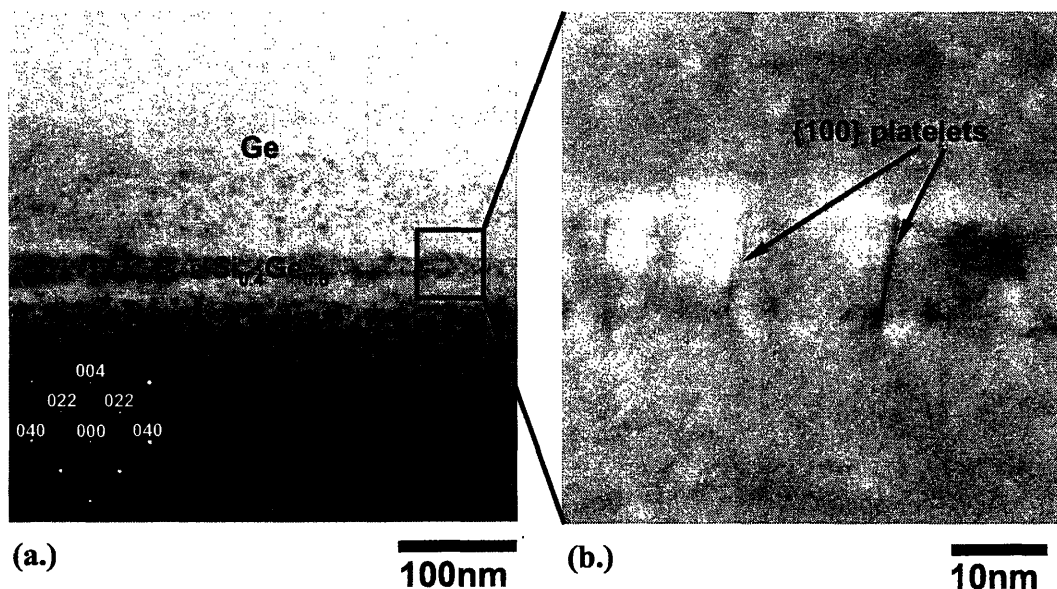


Figure 6.11: XTEM micrographs taken along the [100] pole showing a H⁺ implanted gettering layer after 200 °C annealing. The denuded zone on either side of the gettering layer is clearly defined after low temperature annealing. Preparation of the foil along [100] reveals preferential nucleation of (010) platelets within the strained Si_{0.4}Ge_{0.6} layer.

Figure 6.12 shows the plan-view microstructure at the Si_{0.4}Ge_{0.6} layer in relation to the sample foil. A thin (100 nm) Ge layer was left intact during sample preparation so that the Si_{0.4}Ge_{0.6} layer remains buried during plan-view imaging. At the thinnest region of the foil, only the platelet-free Ge layer is imaged, confirming preferential nucleation of platelets within the strained Si_{0.4}Ge_{0.6} layer. The platelets display Ashby-Brown¹¹⁰ contrast and appear as coffee-beans when $\bar{g} \cdot \bar{b} \neq 0$. Alternately, little or no contrast is seen in the platelets with habit plane parallel to $\bar{g}$ due to the $\bar{g} \cdot \bar{b} = 0$ invisibility criterion.

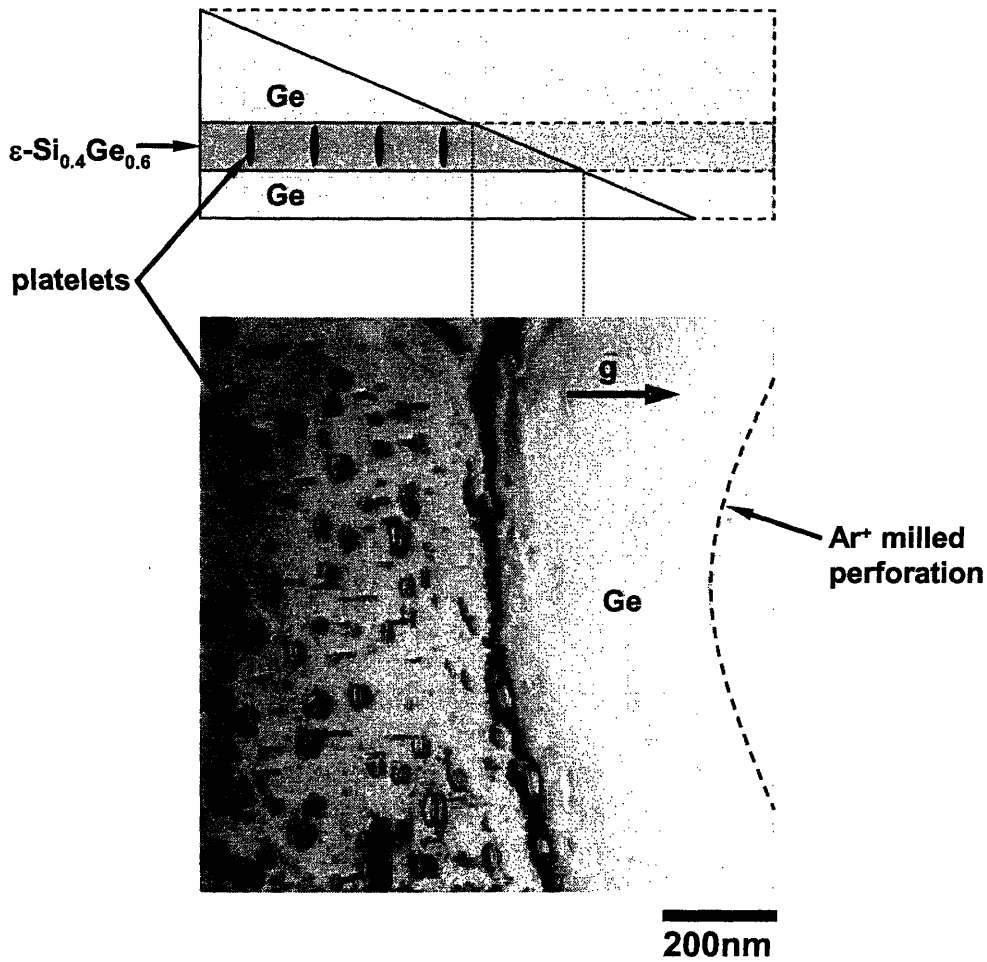


Figure 6.12: PVTEM micrograph of a H^+ -implanted gettering structure after 200 °C annealing shown in relation to a cross-sectional schematic of the sample foil. The thickness gradient caused by Ar^+ ion milling allows various depths of foil to be imaged simultaneously. In the thinnest region of the foil, only the Ge layer is imaged, confirming preferential nucleation of platelets within the $Si_{0.4}Ge_{0.6}$ layer.

The pair of PVTEM micrographs shown in Figure 6.13 illustrates how platelet morphology evolves during 200 °C annealing. The as-implanted sample (Figure 6.13a) contains $\sim 10^{11} \text{ cm}^{-2}$ platelets with a diameter $\leq 11 \text{ nm}$. Notably, 11 nm corresponds to the thickness of the $Si_{0.4}Ge_{0.6}$ layer implying that the diameter of the platelet is limited by the dimension of the host layer. Upon annealing at 200 °C, approximately 1/3 of these undergo a confined ripening process, growing in the [100] and [010] directions to an average length of $\sim 40 \text{ nm}$. A comparison of the plan-view and cross-sectional micrographs indicate that the platelets have an aspect ratio of ~ 4 after 200 °C annealing.

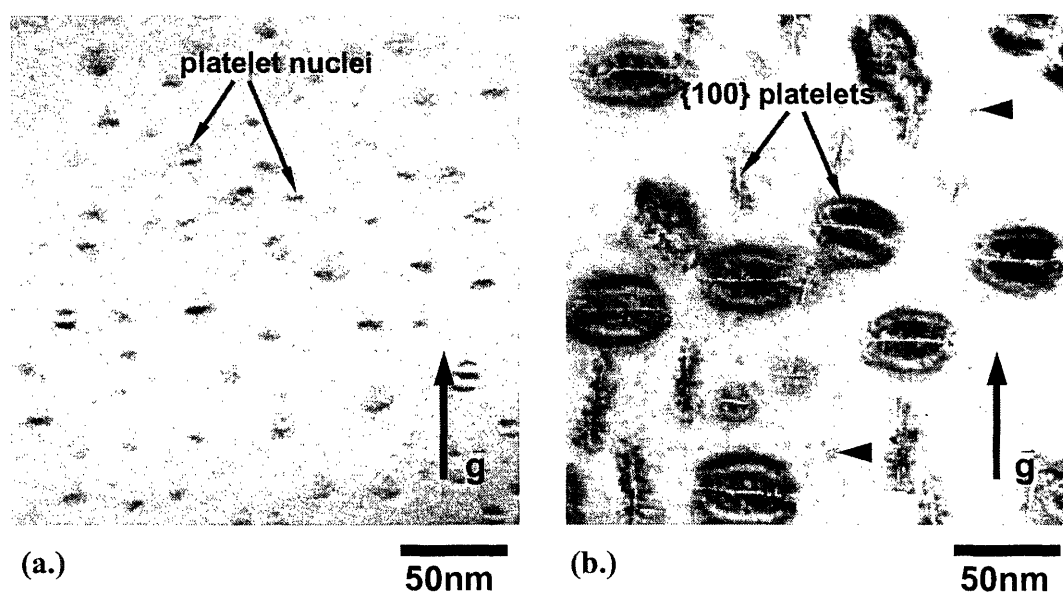


Figure 6.13: PVTEM of H^+ -implanted strained $Si_{0.4}Ge_{0.6}$ layer (a.) before and (b.) after 200 °C annealing. During the anneal, the platelets undergo competitive growth yielding a density of $3.4 \times 10^{10} \text{ cm}^{-2}$ platelets having an average in-plane length of 40 nm. Residual platelet nuclei remain in (b.), as indicated by the triangles.

The platelet nuclei remaining in the micrograph after annealing (indicated by the triangles in Figure 6.13b) indicate a competitive growth mechanism. There are two noteworthy observations that can be made upon comparison of the gettering layer microstructure in plan (Figure 6.13b) and cross-sectional (Figure 6.11a) views. First, the platelets are not associated with a denuded zone in the plane of the strained layer. This is in contrast to the corresponding cross-sectional view of Figure 6.11 which showed a ~ 10 nm denuded zone *out of the plane* of the gettering layer. Secondly, the hydrogen concentration profile (as determined from SIMS) of the gettering structure after 200 °C rules out hydrogen depletion from the denuded zone on either side of the gettering layer.

One possible explanation for these observations is that platelet growth involves an influx of point defects from the relaxed Ge layers neighboring the strained $Si_{0.4}Ge_{0.6}$ layer. Vacancy clusters are responsible for the implantation damage contrast seen in Figure 6.11a. Upon annealing, radiation-induced point defects either agglomerate to

form immobile defect clusters or diffuse into the adjacent gettering layer. However, the point defects that arrive at the gettering layer become immediately incorporated by the large density of growing platelets, requiring negligible lateral diffusion within the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer. Consequently, a zone denuded of radiation damage is observed only in the region out of the plane of the gettering layer.

This proposed mechanism is accompanied with one caveat in that a tensilely strained layer is expected to attract interstitials and repel vacancies. Point defect diffusion into a region of tensile strain requires a net increase of volume in order to be thermodynamically favorable. Since vacancies have a smaller atomic volume than an occupied lattice site, vacancy migration into a tensilely strained layer would appear unfavorable. However, the volume effect may be offset by the H-gettering capability of vacancies. Hydrogen is attracted to vacancies due to their highly reactive dangling bonds and also by the tensile strain field in the vicinity of the point defects, thus offsetting their negative volume of migration. Therefore, it is possible that vacancies serve as a vehicle for transporting H to the platelets in the gettering layer. Furthermore, as will be shown in the next section, platelets cause relaxation of the originally strained gettering layer, further reducing the thermodynamic force that drives vacancies away from regions of tensile strain.

Platelets in the gettering layer are accompanied by a large lattice distortion as evidenced by PVTEM observations in thicker regions of the sample foil. By increasing the thickness of the TEM foil, a larger extent of the strain field can be imaged. Figure 6.14 is a comparison of platelet image contrast produced in thin (~ 100 nm) and thicker (~ 300 nm) regions of the same TEM sample. Both images are reproduced on the same scale and were generated using identical contrast conditions. In thin regions of the foil, Ashby-Brown contrast causes the platelets to appear as coffee beans when their dilation vector is aligned with $\bar{g}$ (Figure 6.14a). In thicker regions of the foil (~ 300 nm), occasional platelet defects generate non-symmetric contrast as shown in Figure 6.14b. The density of these features is a strong function of foil thickness. In very thick (>300 nm) regions of the TEM foil, the overlapping strain fields of the numerous platelets prevent accurate calculation of their density. The micrograph in Figure 6.14b

was taken from a region of intermediate foil thickness (~300 nm), where only occasional platelets generate the extended strain contrast seen in the Figure. The contrast in this case is caused by the H₂ pressure within the defect, effectively behaving as a pressurized void. Therefore the variation of their density with thickness is likely caused by H₂ effusion from thin regions of the foil during sample preparation.

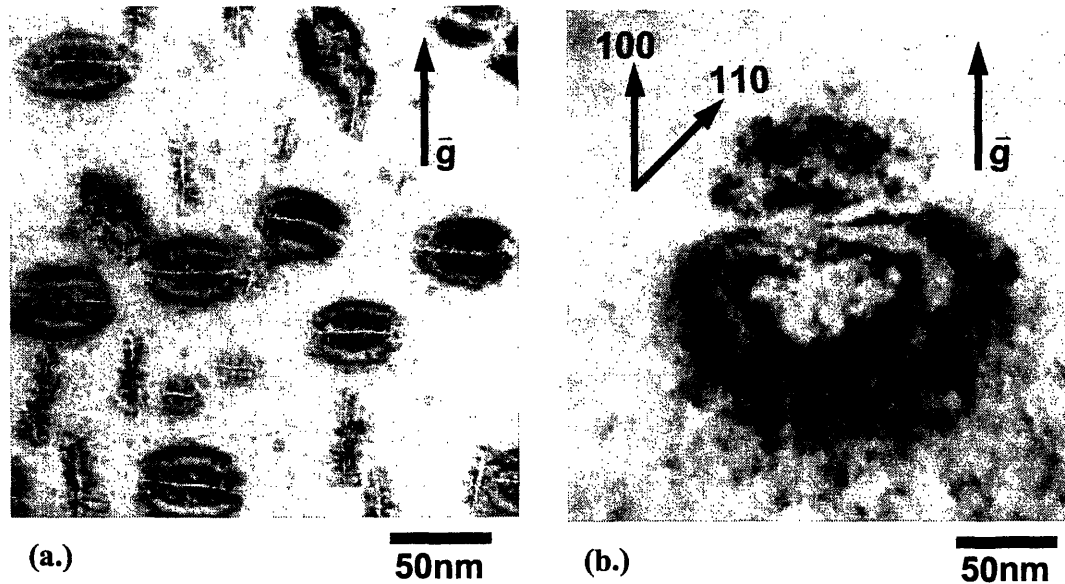


Figure 6.14: PVTEM images of platelets in Si_{0.4}Ge_{0.6}/Ge layer after H⁺ implantation and annealing at 200 °C in thin (a.) and thick (b.) regions of the sample foil. Strain contrast is the result of {100}-oriented platelets in the strained Si_{0.4}Ge_{0.6} layer.

The non-symmetrical strain contrast observed in Figure 6.14b is a result of a non-symmetric TEM sample, as illustrated in Figure 6.15. The platelets are confined to the Si_{0.4}Ge_{0.6} layer which is located near the bottom of the TEM foil. Since the foil is tilted for a given 2-beam condition, the projection of the strain field generated by the platelet extends further in the direction opposite the $\bar{g}$ -vector.

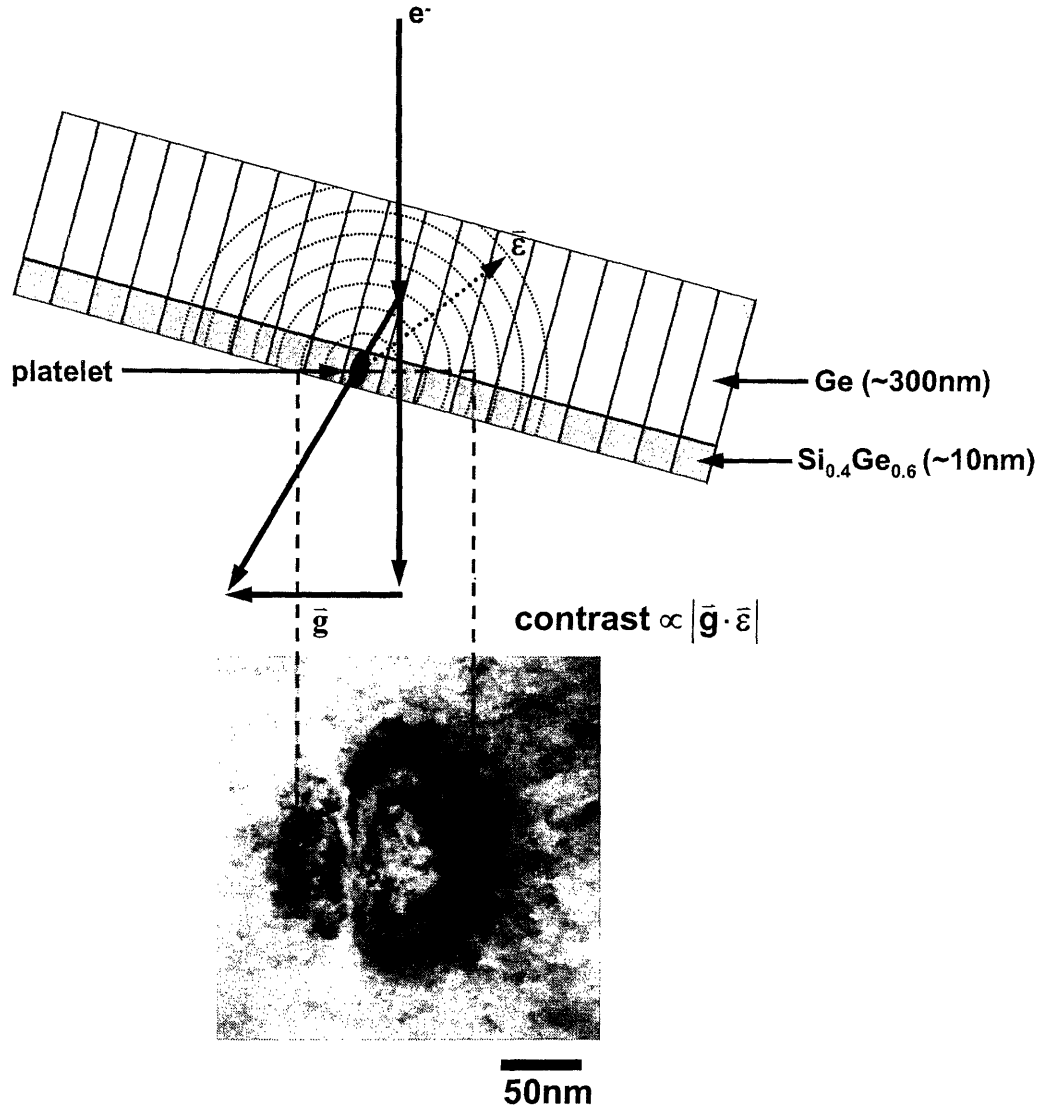


Figure 6.15: Schematic illustrating the origin of unsymmetrical strain contrast for H_2 -pressurized platelets imaged in a thick (~ 300 nm) region of a plan-view TEM sample.

A corresponding $\bar{g}$ -vector analysis of the H_2 -pressurized platelet shows that the strain field surrounding the defect is symmetrical in the plane of the TEM foil. The series of PVTEM images shown in Figure 6.16 were taken from the same location on the sample foil. The orientation of the strain fields emanating from the platelet defects varies systematically with $\bar{g}$ -vector orientation. Figure 6.16 also shows hydrogen gettering by misfit dislocations at the $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ interface.

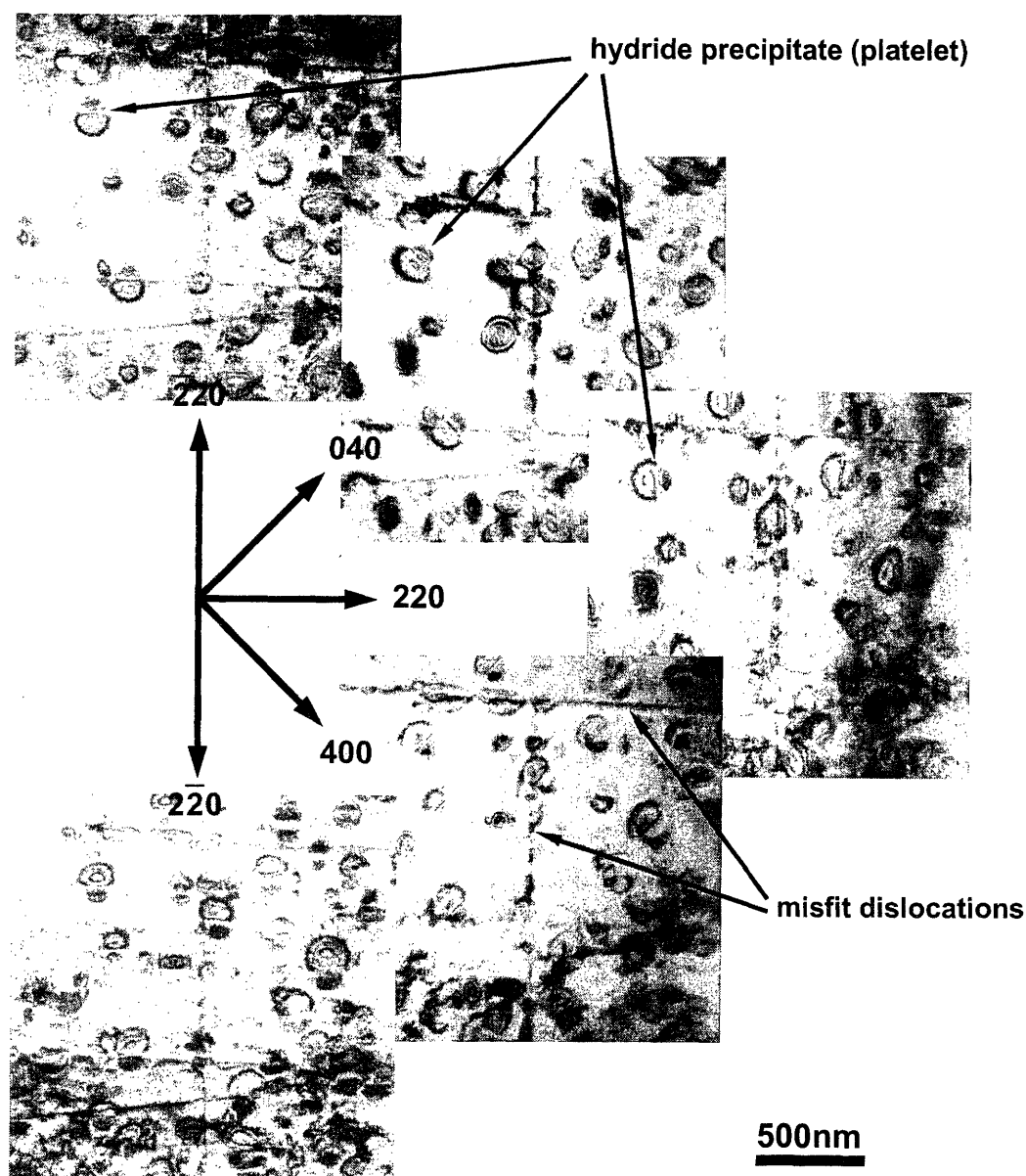


Figure 6.16: g-vector analysis of platelets in a thick (~300 nm) region of a plan-view TEM foil. The symmetry of the strain contrast generated by the platelets varies with g-vector orientation.

Dislocations provide an additional sink for hydrogen which manifests itself through strain contrast along the misfit segments extending along $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ interface. These misfit dislocations arise as a result of slight relaxation during epitaxial growth of the strained layer beyond its critical thickness, as discussed in Chapter 5. Accumulation

of hydrogen at misfit dislocations is clearly evidenced after 300 °C annealing in Figure 6.17. The average spacing between misfits ($\sim 1 \mu\text{m}$) is much larger than the average platelet spacing of $\sim 50 \text{ nm}$. Therefore misfit dislocations are only a minor contributor to the overall gettering behavior of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer.

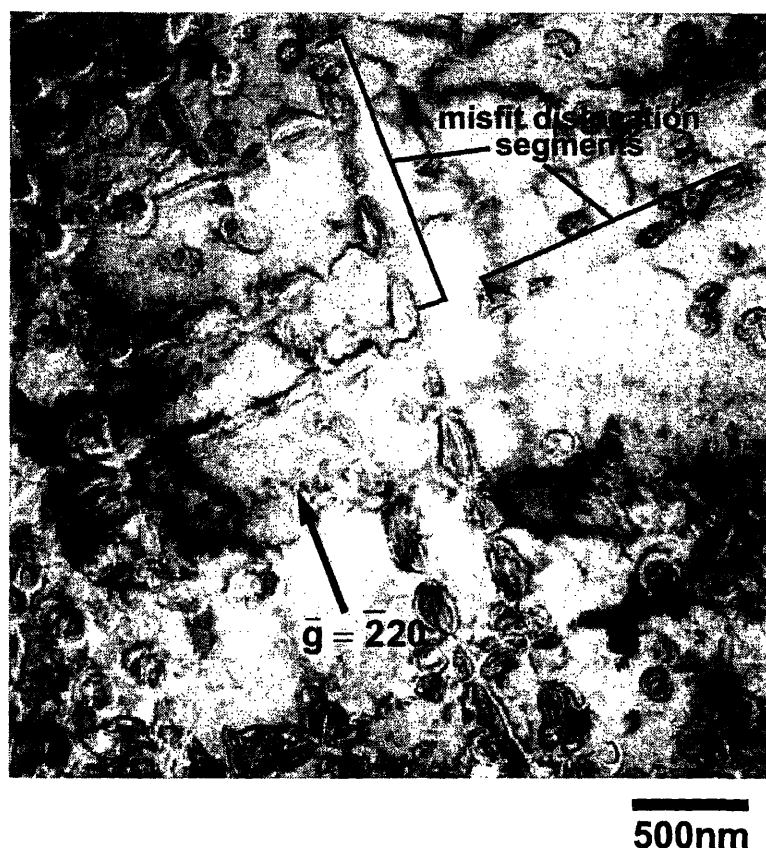


Figure 6.17: PVTEM micrograph of a strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer showing H-gettering by misfit dislocations. The sample was annealed at 300 °C for 1 hour.

The platelets were observed to grow confined to the biaxially strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer during 200 °C annealing, propagating only in the directions parallel to the film plane. However, increasing the temperature to 250 °C causes the platelets to grow in the [001] direction and extend into the relaxed Ge cladding as shown in Figure 6.18a. Meanwhile, PVTEM analysis showed that growth in the [100] and [010] directions is discontinued. Orientation of the TEM sample along the [110] direction also reveals

extension of {111} platelets from the strained $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ interface as illustrated in Figure 6.18b.

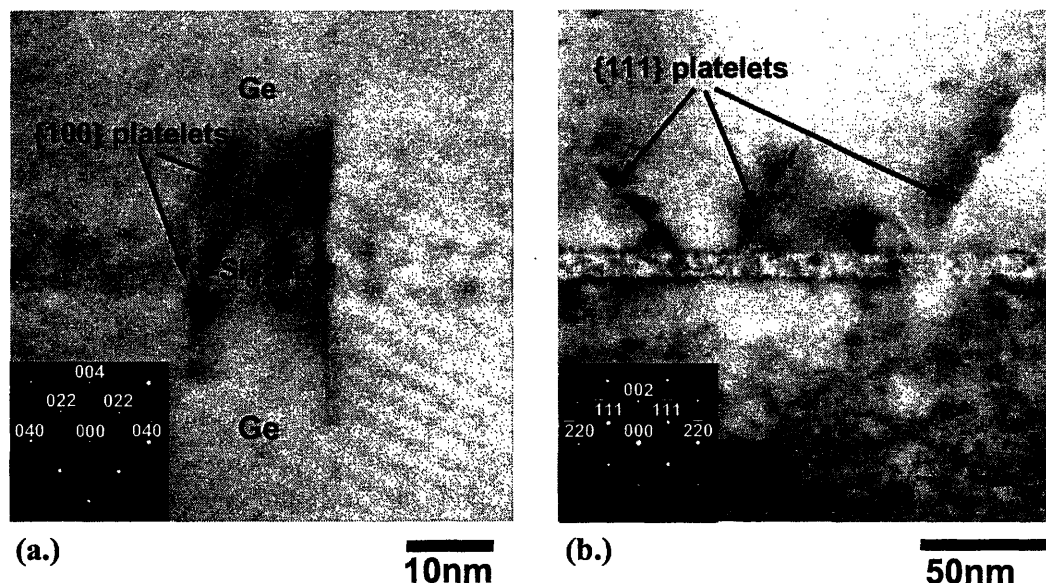


Figure 6.18: XTEM micrographs of H^+ -implanted $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer after 250 °C annealing along (a.) [100] and (b.) [110] pole directions. After 250 °C annealing, {100} platelets propagate beyond the tensilely strained layer in (a.) and new {111} platelets are nucleated as revealed in (b.).

Upon annealing the gettering structure at 300 °C, nanocrack nucleation is observed in the vicinity of the strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer, as shown in Figure 6.19. Although the majority of the platelets are oriented perpendicular to the sample surface, their spacing (50 nm) is smaller than the distance to the sample surface (500 nm), causing the crack to propagate between points within the gettering layer. An annealing temperature of 300 °C also corresponds to the onset of surface blistering of samples containing gettering layers which will be discussed in Section 6.5.

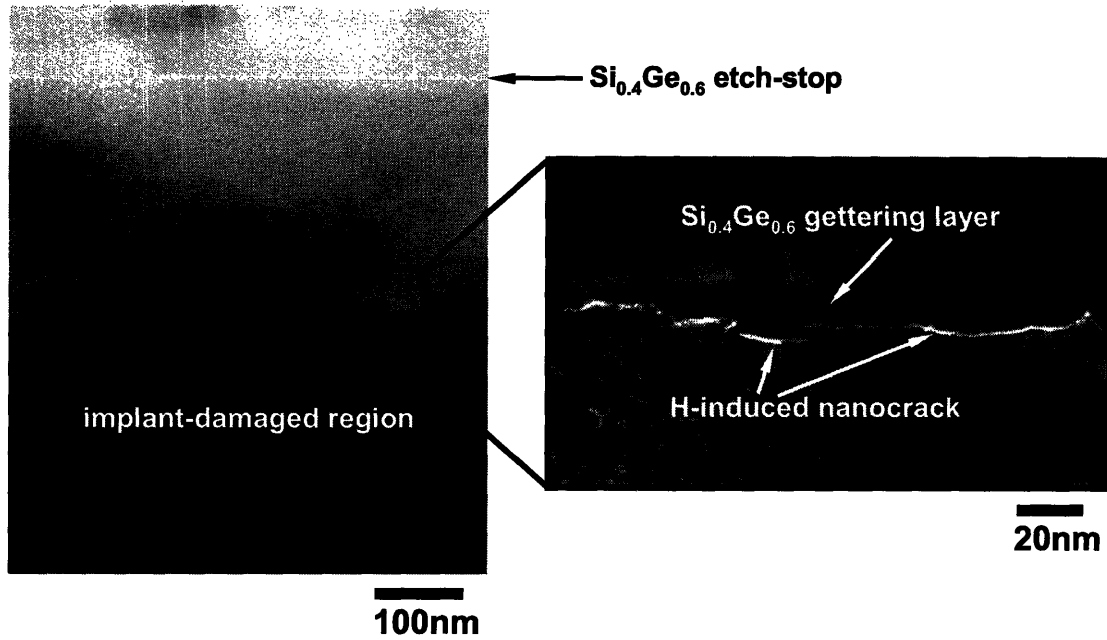


Figure 6.19: XTEM micrograph taken along the [110] pole showing a H⁺-implanted gettering layer after 300 °C annealing. Nanocracks propagate along the gettering layer leading to eventual layer separation.

Table 6.2 provides a summary of the evolution of platelet morphology in biaxially strained Si_{0.4}Ge_{0.6}/Ge after H⁺-implantation and annealing, as discussed in this section.

Anneal Treatment	Platelet morphology
As-implanted	Platelets have (100) and (010) habit planes, an aspect ratio of 1:1 and are confined to the strained host layer. Platelet diameter is limited to the thickness of the host layer.
200 °C, 1 hour	Platelets grow along the [100] and [010] directions while confined to the host layer. Host layer contains an equal distribution of platelets with (100) and (010) habit planes.
250 °C, 1 hour	Platelets discontinue growth in biaxial directions and extend into the adjacent Ge cladding layers. Most of the platelets have (100) and (010) habit planes, however {111} platelets also nucleate at the gettering layer and propagate into the adjacent Ge cladding.
300 °C, 1 hour	Nanocrack formation is observed at the gettering layer.

Table 6.2: Summary of platelet morphology evolution during annealing of H⁺-implanted gettering structures.

6.4.3: Hydrogen-induced relaxation of tensilely strained layers

The remarkably high density of platelets in tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ suggests the film undergoes significant relaxation upon hydrogen implantation and annealing. In order to characterize this phenomenon, Raman spectroscopy was used to measure the residual strain within the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layers after ion hydrogenation.

The degree of strain relaxation can also be used to determine the effective dilation vector of the platelet. Platelets are analogous to dislocations in that they are capable of relaxing tensile strain. For (100) and (010)-oriented platelets in a biaxially strained layer, the dilation parameter, b , is given by Equation 6.4 where δ is the elastic strain relief caused by a density of platelets, ρ_p having an average length $\bar{l}_p$ in the plane of the layer. The factor of 2 accounts for the two possible {100} platelet orientations observed in the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer.

$$b = \frac{2 \delta}{\rho_p \bar{l}_p} \quad \text{Equation 6.4}$$

The parameter δ is analogous to the plastic strain relief caused by misfit dislocations at a lattice-mismatched interface. Equation 6.5 expresses δ in terms of the residual elastic strain, ϵ , and the lattice mismatch, f , between the film and the substrate. For a $\text{Si}_{1-x}\text{Ge}_x$ layer deposited on a Ge substrate, $f = .0407(1 - x)$.

$$\delta = f - \epsilon \quad \text{Equation 6.5}$$

The residual strain analysis was performed on Sample 1 after implantation to a dose of $6 \times 10^{16} \text{ cm}^{-2} [\text{H}^+]$ at 100 keV through a 500 nm LTO layer. These conditions result in a hydrogen profile where the maximum hydrogen concentration coincides within

200 nm of the strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer as shown in Figure 6.8c. Anneals were then carried out at either 200 or 250 °C in N_2 ambient for 1 hour.

The buried $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer must be exposed at the surface for Raman analysis. Therefore all the samples were etched in hydrofluoric acid, then in 30 wt% KOH at 60 °C and finally in H_2O_2 to remove the LTO, Si passivation and Ge transfer layers, respectively. Chemical etching of the epitaxial layers was performed after the samples were annealed. Raman data was then collected using an Ar^+ ion laser (514.5 nm) with a spot power of 1 mW. The resulting spectra containing the Raman shifts for Ge-Ge vibrations in the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer and the Ge substrate are shown in Figure 6.20.

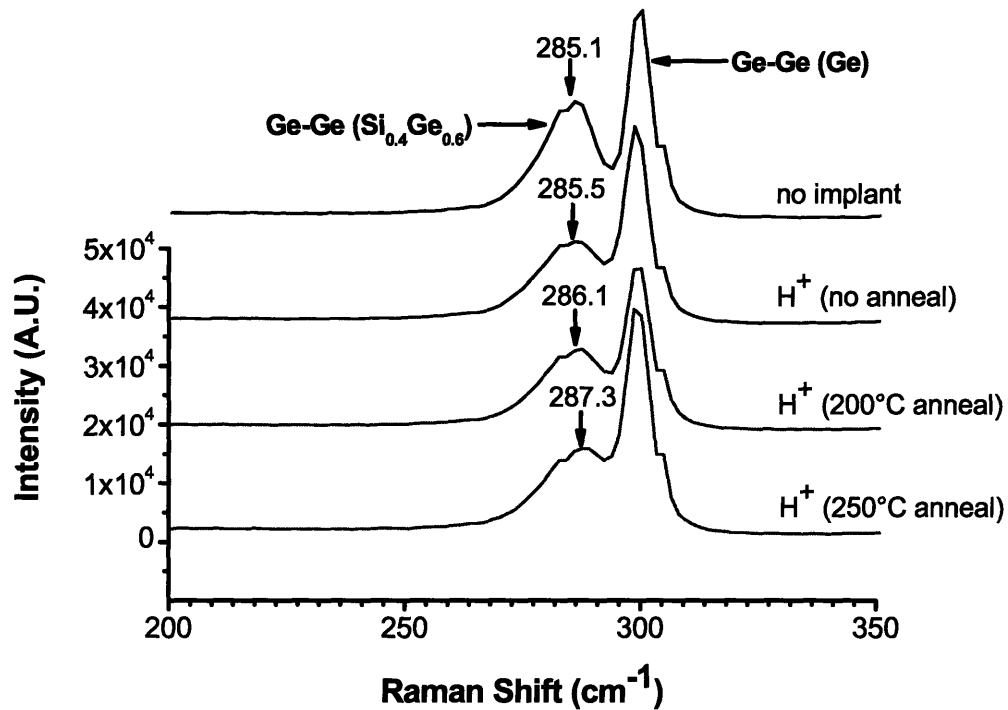


Figure 6.20: Raman spectra of tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ on relaxed Ge after various H^+ implantation and annealing treatments. The labeled Raman-shifts correspond to the Ge-Ge vibration in the strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer. As the annealing temperature is increased for the hydrogenated samples, the Ge-Ge ($\text{Si}_{0.4}\text{Ge}_{0.6}$) Raman-shift increases indicating the relaxation of tensile strain.

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The Raman shift of the Ge-Ge vibration in a strained $\text{Si}_{1-x}\text{Ge}_x$ layer is given empirically⁴⁸ by Equation 6.6, where x is the Ge fraction and ε is the residual elastic strain of the layer.

$$\omega_{\text{Ge-Ge}} = 280 + 20x - 409\varepsilon \quad \text{Equation 6.6}$$

The composition of the strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer is known from x-ray analysis of previously-grown calibration structures. Additionally, PVTEM measurement of the misfit dislocation density at the $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ interface indicates negligible relaxation of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer, yielding an elastic strain of $\varepsilon=1.61\%$ prior to H^+ implantation. Application of Equation 6.5 to the Ge-Ge ($\text{Si}_{0.4}\text{Ge}_{0.6}$) line in the no-implant spectrum of Figure 6.20 confirms this result, yielding an elastic strain of $\varepsilon=1.69\%$ assuming a Ge fraction of 0.6.

Using PVTEM measurements of the average size and density of the platelets in the strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer, Equations 6.4, 6.5 and 6.6 were used to determine the magnitude of their dilation vector, b . The results of this analysis are summarized in Table 6.3.

Sample description	$\delta/f(\%)$	$\bar{l}_p$ (nm)	ρ_p (cm^{-2})	b (Å)
No implant	0	N/A	N/A	N/A
H^+ implant, no anneal	2.5%	10.6 (± 2.2)	9.2×10^{10}	0.8
H^+ implant, 200 °C anneal	11.7%	39.8 (± 9.2)	* 3.4×10^{10}	2.8
H^+ implant, 250 °C anneal	29.4%	37.4 (± 6.6)	* 4.3×10^{10}	6.0

Table 6.3: Summary of hydrogen-induced $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ relaxation experiments.
***Small platelets ($\bar{l}_p \leq 11 \text{ nm}$) were not included in the density measurement.**

The as-implanted sample contains a large density of small (≤ 11 nm) platelets. After annealing the sample at 200 °C, a fraction of these undergo a ripening process and grow to an average length of 40 nm while their dilation vector increases to 2.8 Å. Concomitantly, 11.7% of the initial elastic strain in the layer is lost. The microstructure of the annealed samples contains a population of both small (10 nm) and large (40 nm) platelets. The undeveloped platelets were assumed to retain their original effective dilation vector of 0.8 Å. Since these only add a small contribution to the overall relaxation of the layer, they were not included in the strain relaxation analysis of the annealed samples. Increasing the annealing temperature to 250 °C induces further relaxation of the layer without an appreciable increase of platelet length or density. Instead, relaxation is accommodated by an increase of the dilation vector to 6 Å.

A comparison of the residual strain in the gettering layer with the platelet microstructure at different annealing temperatures suggests that the ripening process undergoes a mechanistic transition between 200 and 250 °C. During 200 °C annealing, the tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer relaxes via platelet elongation in the biaxial directions. The relaxation continues during 250 °C annealing. However, it is not accompanied by continued elongation of platelets along the strained layer. Instead, the dilation vector swells from 2.8 to 6 Å and the platelet begins to take on a circular shape, growing in the [001] direction (beyond the bounds of the strained layer), as shown in Figure 6.16. It will be shown analytically in Chapter 7 that these trends reduce the platelet's efficiency for releasing strain energy from the $\text{Si}_{0.4}\text{Ge}_{0.6}$ host layer, leading to a transition between strain-limited and diffusion-limited growth.

The monotonic increase of the dilation vector with annealing temperature is attributed to a build-up of gaseous H_2 within the platelet. As the annealing temperature is increased, hydrogen diffusing in the vicinity of the gettering layer becomes trapped and accumulates within the platelets, thus increasing their dilation vector. This is also correlated to the sudden decrease of detectable hydrogen within the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer upon annealing at 250 °C, as was observed in the corresponding SIMS data given in Figure 6.8b. This sequence of events illustrates the dual nature of hydrogen gettering in tensilely strained layers. First, the tensile strain of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer locally increases the solubility of interstitial hydrogen. Secondly, platelets preferentially nucleate within

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the layer and trap additional hydrogen causing the dilation vector of the platelets to increase. It is this second gettering mechanism that renders tensilely strained layers effective for layer transfer applications since platelets eventually grow into nanocracks causing separation of the surface layer from the substrate.

Residual strain analysis also reveals that platelets are highly effective for relaxing tensile strain. Analogous to the relation of a dislocation to its Burgers vector, the strain energy of a platelet follows a parabolic dependence on its dilation vector. Meanwhile, the strain energy released from the layer follows a weaker, linear relation with b . Therefore from a thermodynamic argument, platelets relax strain most efficiently when their dilation parameter is small. The strain energy of misfit dislocations follows the same trend. However, the smallest Burgers vector available to a misfit dislocation in diamond cubic crystals is $a/2\langle 110 \rangle$, corresponding to a value of 3.9 Å for $\text{Si}_{0.4}\text{Ge}_{0.6}$. Conversely, the dilation vector of a platelet is variable and starts off at a value of only 0.8 Å at the embryonic stage of formation. Such a small dilation vector greatly reduces the critical thickness of the layer and also contributes to a small energy barrier for nucleation. The ultimate result is an extremely large density ($\sim 10^{11} \text{ cm}^{-2}$) of platelets in strained layers that would otherwise be stable against relaxation by the dislocation mechanism. During ripening, the platelets elongate in the biaxial directions until their dilation vector increases to a point where strain-driven growth is no longer favorable. At this point (during 250 °C annealing), interstitial H_2 diffusion is activated and platelet growth becomes driven by formation of gaseous H_2 within the defects. This leads to a decrease of detectable hydrogen in SIMS and marks the transition between strain-driven and diffusion-driven growth as indicated in Figure 6.21.

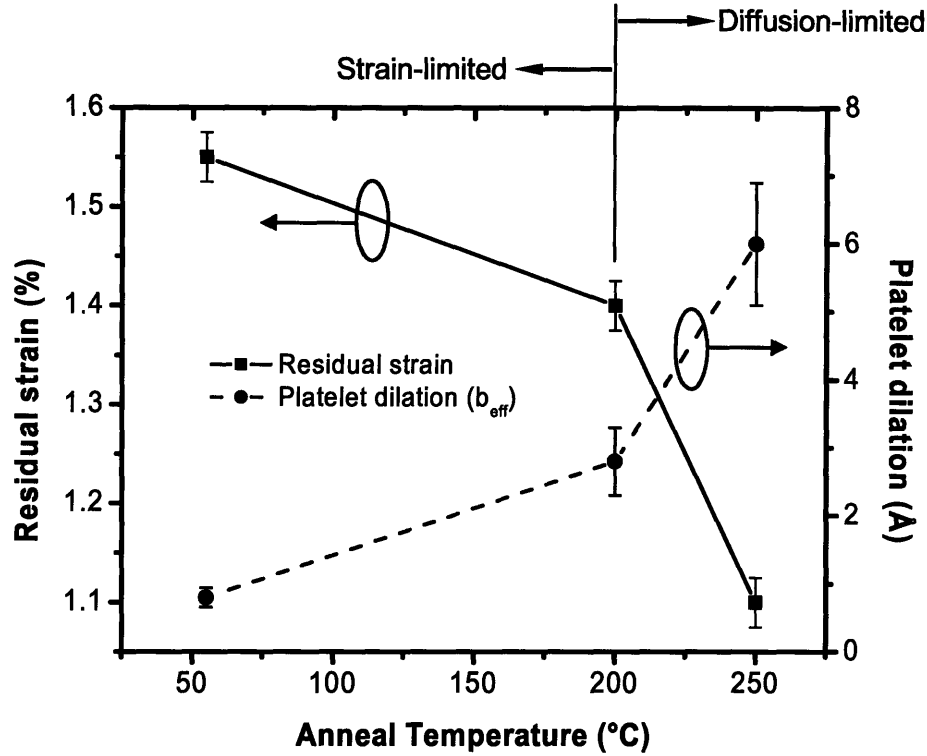


Figure 6.21: Plot of $\text{Si}_{0.4}\text{Ge}_{0.6}$ relaxation and platelet dilation vector as a function of post-implant annealing temperature. The platelet strain energy increases with dilation vector and decreases with layer strain, leading to a transition from strain-limited to diffusion-limited growth.

Relaxation data for $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ layers indicate that a strain of at least 0.5% can be relaxed over a distance of 11 nm through introduction of hydrogen, corresponding to a strain relaxation rate of 45%/μm. In glaring contrast, a $\text{Si}_{1-x}\text{Ge}_x$ compositionally graded buffer is relaxed at a rate of only 0.4%/μm by the dislocation mechanism. The source of this efficacy for strain relaxation by platelets is the small (0.8 Å) dilation vector and a negative surface energy that are characteristic of a platelet undergoing nucleation. These features will be taken into account in the derivation of the expression for platelet free energy in Chapter 7.

Platelets therefore offer a novel alternative to dislocations for efficient relaxation of tensile strain. This introduces an aspect of hydrogen-induced strain relaxation that is unrelated to layer exfoliation applications. Relaxation of tensilely mismatched layers by

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compositional grading has historically been impeded by the formation of stacking faults via separation of $a/2\langle 110 \rangle$ threading dislocations into two Shockley partials.¹¹¹ These stacking faults inhibit dislocation glide and give rise to highly defective films. Although stacking fault formation can be inhibited by maintaining low strain levels throughout the graded buffer, this approach requires thick epitaxial layers which are impractical for integration with CMOS. On the other hand, introduction of hydrogen by ion implantation or plasma immersion may provide a highly effective alternative to dislocations for relaxing tensile strain in lattice-mismatched layers.

A possible limitation to the relaxing of tensilely strained layers with platelets stems from the fact that these defects propagate through the thickness of the strained layer. This is in contrast to misfit dislocations which are confined to the mismatched interface. It is currently unknown whether these defects can be eliminated from the surface device layer in virtual substrates fabricated in this manner.

6.5: Application of H-gettering structures to layer transfer technology

The potential benefits of H-gettering structures for layer transfer applications are twofold. First, a reduction in H^+ dose for layer exfoliation can significantly reduce the costs of layer transfer processes. Secondly, a reduction in thermal budget enables bonding of thermally mismatched wafers. This section will characterize the blistering kinetics of H^+ -implanted Ge samples containing gettering layers and compare the results to conventional samples without a gettering layer. The results will reveal how the blistering kinetics of gettering structures can be optimized by tailoring their thermal history after ion implantation.

6.5.1: Blistering kinetics of H-gettering structures

In order to characterize the benefits of gettering structures for layer exfoliation, a surface blistering experiment was devised. Samples 1 and 2 were implanted with H^+ to a

dose of $6 \times 10^{16} \text{ cm}^{-2}$ at 100 keV, through a 300 nm LTO screen yielding the corresponding hydrogen profiles shown in Figure 6.8(c.). After implantation, the LTO layer was stripped in HF and the samples were subjected to various annealing treatments varying between 200 and 300 °C in a tube furnace under N_2 atmosphere. These anneals, denoted as “pre-anneals” were not sufficient to induce observable surface blisters in any of the implanted samples. The samples were then annealed on a hotplate at temperatures exceeding the pre-anneal temperature while observing the surface with a stereomicroscope. The onset of blistering was defined as the hotplate annealing time required to observe a surface blister at a magnification of 70×. Figure 6.22 is an optical micrograph using differential interference contrast (DIC) showing a Ge surface shortly after the onset of blistering.

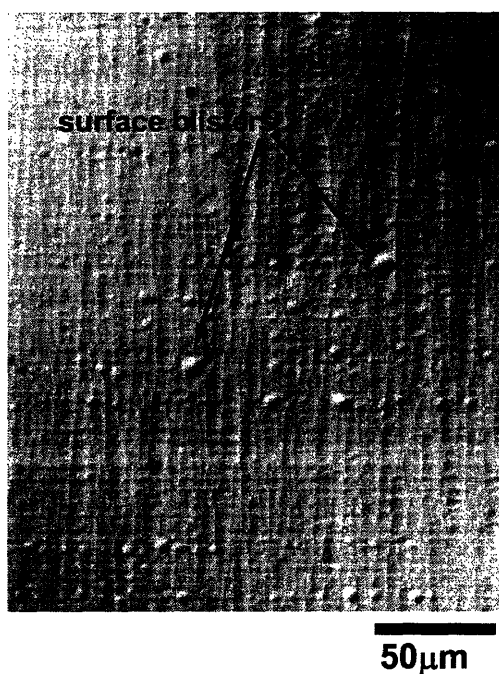


Figure 6.22: Optical DIC micrograph showing the onset of surface blistering of a H^+ -implanted virtual substrate containing a H-gettering layer.

Figure 6.23 shows the effect of a tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer on the blistering kinetics of a Ge virtual substrate implanted to a dose of $6 \times 10^{16} \text{ cm}^{-2} [\text{H}^+]$. The Figure also highlights the difference in blistering time as a function of pre-anneal

temperature. The effect of the gettering layer is dramatic for the plotted conditions. A pre-anneal of 200 °C, denoted by the circles in Figure 6.23, yield similar blistering characteristics for samples with and without a gettering layer. The blistering time decreases with temperature in Arrhenius fashion for $T < 440$ °C. This general trend is consistent with data published in the literature for various semiconductor materials.^{82, 83} However, as the hotplate temperature is increased beyond 440 °C, the blistering time is observed to *increase* with temperature.

The effect of increasing the pre-anneal temperature is strikingly different between the two samples. A pre-annealing temperature of 250 °C, denoted by squares, significantly accelerates the blistering kinetics of the gettering structure while significantly degrading that of the structure without a gettering layer. In fact, the only conditions that yielded modest blistering in the sample without a gettering layer required ~20 hours of annealing at 350 °C as shown in Figure 6.23. On the other hand, the 250 °C pre-annealed gettering structure displayed a monotonic decrease in blistering time with temperature beyond 440 °C. Though not shown in the Figure, pre-annealing the gettering structure at 300 °C resulted in similar blistering behavior as for the 250 °C pre-annealed sample. These data are provided in Appendix A. Without a pre-anneal treatment, none of the implanted samples readily blistered at the annealing times and temperatures investigated in this experiment (irrespective of the presence of a gettering layer).

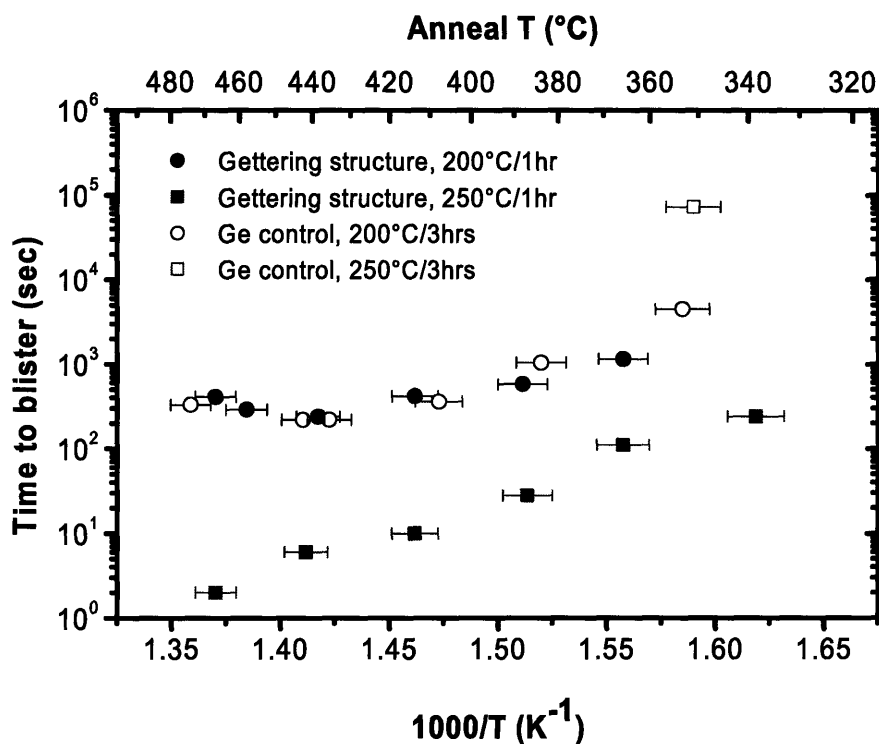


Figure 6.23: Arrhenius plots of blistering time vs. temperature for Samples 1 (without a gettering layer) and Sample 2 (with a gettering layer) subjected to two different pre-anneal temperatures. Both samples were implanted with H^+ to a dose of $6 \times 10^{16} \text{ cm}^{-2}$. The blistering kinetics are not affected by the gettering layer after pre-annealing at 200 °C. However, pre-annealing the samples at 250 °C degrades the blistering kinetics of Sample 1 while enhancing those of Sample 2. The ability of the gettering layer to trap mobile H_2 molecules explains the observed disparity between the two samples.

The blistering behavior illustrated in Figure 6.23 can be explained in terms of hydrogen gettering phenomena. As mentioned earlier, the underlying principle to layer exfoliation is not only the limited solubility of hydrogen in Si or Ge, but also the ability of the target material to trap the implanted hydrogen so as to prevent its effusion during annealing. Conventional layer exfoliation relies on the defects generated during ion implantation to getter the hydrogen during post implantation annealing. Platelets therefore play a critical role in the blistering mechanism for H^+ -implanted samples. The samples containing a gettering layer provide a favorable site for platelet nucleation which

in turn serves as a sink for hydrogen during sample annealing. By pre-annealing the implanted structure, the platelets within the gettering layer can be made to grow without significant loss of hydrogen to surface effusion. This is manifested by the $\sim 100\times$ reduction in blistering time by increasing the pre-anneal temperature from 200 to 250 °C. Effusion of hydrogen also accounts for the two kinetic regimes observed in the 200 °C pre-anneal samples. Blistering kinetics at hotplate annealing temperatures < 440 °C are likely limited by the rupture of lattice bonds during growth of pressurized defects as previously reported.^{79, 82, 83} However, blistering at $T > 440$ °C is limited by hydrogen effusion, particularly for the sample without a gettering layer. The pre-anneal temperature has a profound effect on the temperature at which the transition between effusion-limited and platelet growth-limited blistering occurs.

6.5.2: Anomalous blistering kinetics of hydrogen-implanted germanium

The small annealing window that resulted in blistering of the Ge sample without a gettering layer is somewhat anomalous given that blistering of H^+ -implanted Ge samples has been reported⁸³ at a dose well below $6 \times 10^{16} \text{ cm}^{-2} [H^+]$. However, the published results for hydrogen-implanted Ge contain a significant amount of variation. For example, Tong, *et al.*⁸² quote a blistering activation energy of 1.0 eV while Bedell *et al.*⁸³ provide a value of 1.7 eV for an implantation dose of $1 \times 10^{17} \text{ cm}^{-2} [H^+]$. In contrast, the blistering results for Si have very little discrepancy between the two authors.

Recent investigations of Ge exfoliation also have reported a shelf life dependence on Ge blistering kinetics.¹¹² This investigation revealed that the blistering kinetics of hydrogen-implanted Ge is greatly reduced after a shelf life of 6 weeks. The investigators have attributed this phenomenon to H_2 effusion during the storage period; however they provide no data comparing the H-content of the implanted sample during that time to support their claim. The blistering experiments in the present study were performed within 2-4 weeks of ion implantation. However, hydrogen forward scattering (HFS) analysis performed on the as-implanted Ge samples 12 weeks after H^+ implantation

revealed a total dose of $6.2 \times 10^{16} \text{ cm}^{-2}$, indicating that hydrogen was not lost during the 12 week storage period.

Although the anomalous blistering behavior of Ge is not well understood, it is reasonable to believe that exfoliation kinetics depend on implantation conditions by influencing the degree of damage imparted to the target substrate. A large degree of defects created during the implantation process can influence the intrinsic gettering behavior of the donor substrate, thus affecting its blistering behavior during post-implant annealing. A strong dependence of blistering kinetics on H^+ or He^+ fluence has already been well documented.^{82, 83, 87} This trend was related to the weak dependence of implantation damage at low ion dose compared to a super-linear increase of implant damage at high implantation dose. The transition between these regimes was found to be a function of target temperature.¹¹³ The use of gettering layers could lessen the effects of implantation conditions and stabilize the layer transfer process by providing a preferential nucleation site for blister formation.

6.6: Conclusion

Tensilely strained semiconductor layers were shown to getter implanted hydrogen, making them useful for applications utilizing the layer exfoliation process. The gettering behavior of such layers is comprised of two independent mechanisms. First, tensile strain enhances the solubility of interstitial impurities, providing a locally reduced chemical potential for dissolved hydrogen. Second, tensile strain drives preferential nucleation of platelets within the gettering layer. These platelets are efficient gettering centers for hydrogen and are a necessary component of the exfoliation process. These phenomena can be combined in an engineered donor substrate for highly efficient transferal of Ge and other high performance materials onto a silicon substrate.

Chapter 7: Modeling H-induced platelet nucleation

Layer exfoliation technology is largely enabled by the entrapment of implanted hydrogen molecules by defects below the wafer surface. By incorporating a tensilely strained layer into the donor wafer, the implanted hydrogen concentration is locally enhanced by a self-reinforcing process of gettering and platelet nucleation, leading to enhanced blistering kinetics.

This chapter develops an analytical model for the free energy of platelet formation revealing the two kinetic regimes observed during platelet growth. It is shown that platelet nucleation is a strain-driven process while growth and eventual coalescence into voids is kinetically-limited by hydrogen dissolution and formation of SiGe-H surfaces. By tailoring the annealing temperature of H-implanted gettering structures, the platelets can be made to grow with minimal loss of H_2 due to surface effusion, thus maximizing their benefits in layer transfer applications.

7.1: Introduction

Platelet growth in the presence of hydrogen is a strain-driven phase transformation analogous to stress-induced hydride formation at a crack tip. In this respect, layer exfoliation is a manifestation of the hydrogen embrittlement phenomena that have been observed in metals for decades. An illustration of platelet growth is provided in Figure 7.1. Upon annealing, supersaturated hydrogen molecules condense at platelets causing them to become pressurized with gaseous H_2 . This produces a tensile strain field along the periphery of the defect, thereby enhancing the local concentration of interstitially dissolved H_2 . The platelet then propagates into the tensilely strained region ahead of the 'crack tip', releasing strain energy as the defect grows. This process continues until the microscopic platelets coalesce into macroscopic cracks, thereby causing separation of a surface layer from the substrate.

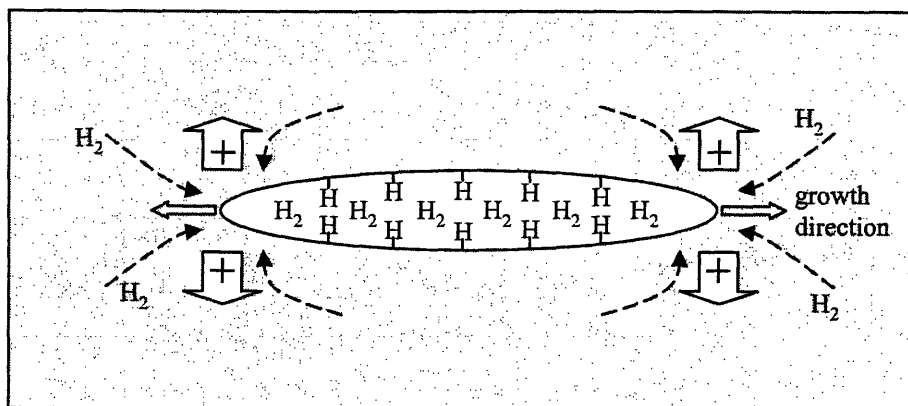


Figure 7.1: Schematic illustration of platelet expansion into a cavity. The growth mechanism is analogous to H-induced fracture in metals. Tensile strain energy is released as the 'crack' propagates into the H-rich region at the periphery of the platelet. This process continues as the platelet evolves into an internal cavity pressurized with gaseous H_2 .

In Chapter 6, the transition from strain-limited to diffusion limited growth was characterized by the increase of the platelet dilation vector during annealing. Previous experimental studies^{91,93,97,114} concerning microstructure and growth are limited to characterization of platelets in bulk wafers. In these instances, measurement of the

magnitude of the dilation vector is limited to HRTEM imaging.⁹³ In this study, the platelets are characterized while confined to a tensilely strained gettering layer, thereby providing a reference point for observing subtle changes in platelet microstructure during annealing. Furthermore, the dilation vector can be measured as a function of annealing temperature with strain relaxation and platelet density measurements. These experimental details are invaluable for formulating a mathematical model for platelet nucleation and growth.

7.2: Analytical nucleation model

Platelet nucleation and growth can be described using a simple Volmer-type⁹¹ theory for a circular precipitate:

$$\Delta G(r) = 2\pi r^2 \Gamma_p - \beta \pi r^2 b \frac{P}{k_B T} \Delta H_s + 2\pi r U_1 + \frac{\mu b^2 r (2 - \nu)}{4 (1 - \nu)} \ln \left(\frac{8r}{e^2 r_0} \right) \quad \text{Equation 7.1}$$

Equation 7.1 gives an expression for the free energy change of a circular platelet with a radius, r . The expression contains terms that account for surface, volume, perimeter and strain energy contributions, respectively. The surface energy, Γ_p , is defined as the formation energy of a hydrogen-terminated semiconductor surface. This term is always negative since both Si-H and Ge-H bonds are stable relative to the energy between two corresponding semiconductor atoms.⁸⁷ This term contains a factor of 2 which accounts for the fact that a platelet is composed of two H-terminated surfaces.

The second term accounts for the energy released during the dissolution of molecular hydrogen within the platelet volume, given by $\beta \pi r^2 b$, where β is a geometric factor and b is the magnitude of the dilation vector of the platelet. The number of H_2 molecules inside the platelet is given by $P/(k_B T)$, where P is the pressure inside the cavity. ΔH_s is the enthalpy of solution for interstitial H_2 in the semiconductor.

The third term accounts for the perimeter energy of the platelet where U_l is an energy per unit length.

A direct correlation can be made between the strain field of a platelet and that of a dislocation loop by considering the sources of lattice distortion in each case, as illustrated in Figure 7.2. For a dislocation loop in Si (Figure 7.2a), the lattice deformation is caused by an extra plane of silicon atoms inserted into the crystal. Alternately, for the case of a platelet (Figure 7.2b), silicon atoms are replaced with hydrogen atoms that serve to terminate the dangling bonds at the core of the defect. Hydrogen can also be present in the form of gaseous H_2 molecules- the amount of which will affect the dilation vector of the platelet. The magnitude of the platelet dilation vector is not well defined, nor is it confined to specific crystallographically-allowed values, as is the case for dislocation loops. However, both platelets and dislocation loops induce a strain field that can be described using the Volterra deformation model leading to identical mathematical expressions for their strain energy. Consequently, the final term is the intrinsic strain energy of the platelet due the distortion of the surrounding lattice. This contribution is modeled as the energy of a dislocation loop¹¹⁵ with a Burgers vector, b , normal to the platelet habit plane. For the purpose of this work, the dilation vector and the Burgers vector are functionally equivalent. The strain energy term contains the elastic constants, μ and ν which correspond to the shear modulus and Poisson's ratio, respectively, e is the Naperian logarithmic base and r_0 is the core parameter of the dislocation after which the platelet is modeled.

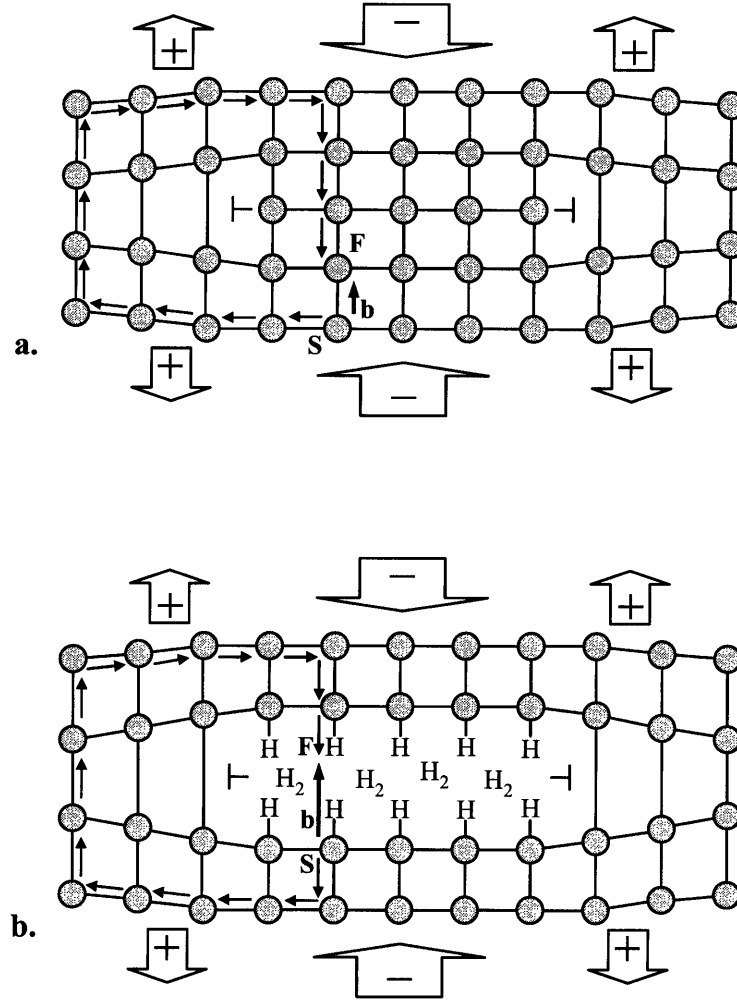


Figure 7.2: (a.) Lattice distortion resulting from an extra plane of silicon atoms of a dislocation loop and (b.) H-terminated silicon surfaces of a platelet. The extra plane of atoms extends into the page for both cases. The figure also depicts a Burgers circuit created with the RHSF convention in the deformed crystal. Note that the dilation vector is not well defined for the case of a platelet due to the lattice discontinuity at the platelet core.

The perimeter energy of the platelet can be directly correlated to the core energy of a dislocation loop. Since this term scales linearly with the radius of the platelet, r , it can be incorporated with the platelet strain energy within a cutoff radius term, r_0 . A cutoff of $b/3$ is typically used for diamond cubic materials¹¹¹, reducing Equation 7.1 to:

$$\Delta G(r) = 2\pi r^2 \Gamma_p - \beta \pi r^2 b \frac{P}{k_B T} \Delta H_s + \frac{\mu b^2 r}{4} \frac{(2-\nu)}{(1-\nu)} \ln \left(\frac{24r}{e^2 b} \right) \quad \text{Equation 7.2}$$

An alternative description of platelet growth can be obtained from a fracture mechanics approach as demonstrated by Freund¹¹⁶ and Yang¹¹⁷. Both authors modeled platelet growth using the Griffith criterion approach by comparing the energy released by the propagation of a H₂-pressurized microcrack with the energy necessary to create two semiconductor surfaces. This approach works well for large platelets that contain significant amounts of gaseous hydrogen in which case they can be modeled as microcracks. The model presented in Yang's paper is applied to internal cracks which are 0.1-10 μm in diameter and have a microscopic crack opening profile. In contrast, the platelets that are being modeled in this chapter have a radius of ~10 nm and have a maximum opening profile, i.e. dilation vector of 6 Å. Consequently, the present model will be applied to platelets where the energy released by their formation is more accurately represented by a dislocation loop.

7.2.1: Surface energy of a platelet

Published FTIR data from hydrogen-implanted Si has shows a direct correspondence between the absorption lines from {001}¹¹⁸ and {111}⁷⁶ platelets with HF treated (001) and (111) silicon surfaces, respectively.⁷⁹ These results validate modeling the surface energy of the platelet as the formation of two internal, H-terminated semiconductor surfaces. Such a reaction proceeds as the following for Si and Ge crystals:



The superscript in H₂^{i_{Si,Ge}} indicates that the H₂ molecule is dissolved interstitially. At room temperature, hydrogen dissolved in Si or Ge is most stable in its molecular form and occupies a tetrahedral interstitial site. The bond length of H₂ⁱ in this configuration is increased relative to H₂ in vacuum, thereby reducing its dissociation energy from 4.5 to

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3.8 eV.⁸⁷ Despite the energy required to break a Si-Si or Ge-Ge bond and to dissociate a H_2^i molecule, the silanic and germanic bonds are relatively stable, and therefore the reactions given in Equation 7.3 and 7.4 proceed. Furthermore, the dissolution of $H_2^{i_{Si,Ge}}$ is an exothermic process, thus providing additional thermodynamic driving force for platelet nucleation.

Ab-initio calculations for the formation of extended hydrogen complexes have estimated the net energy release of Equation 7.3 to about 0.15 eV.¹¹⁹ A similar value is expected for Ge platelet formation due to comparable formation energies for the Si-H (−3.4 eV)¹²⁰ and Ge-H (−3.0 eV)¹²¹ as well as the Ge-Ge (−1.9 eV)¹²² and Si-Si (−2.3 eV)¹²³ bonds.

Both {100} and {111} silicon surfaces terminated with hydrogen are bulk-like in nature where all the dangling bonds are saturated with hydrogen atoms.¹²⁴ Therefore, the surface energy released during nucleation of an internal surface of hydrogenated Si or Ge is given by Equation 7.5, where $\rho_b^{\{hkl\}}$ is the surface bond density of an unreconstructed {hkl} surface.

$$\Gamma_p^{\{hkl\}} = -0.08 \cdot \rho_b^{\{hkl\}} \quad \text{eV/nm}^2 \quad \text{Equation 7.5}$$

Table 7.1 contains calculated the values for $\rho_b^{\{hkl\}}$ and $\Gamma_p^{\{hkl\}}$ for {001} and {111} surfaces.

{hkl}	Si		Ge	
	$\rho_b \text{ (cm}^{-2}\text{)}$	$\Gamma_p \text{ (eV/nm}^2\text{)}$	$\rho_b \text{ (cm}^{-2}\text{)}$	$\Gamma_p \text{ (eV/nm}^2\text{)}$
{001}	1.35×10^{15}	−1.08	1.25×10^{15}	−1.00
{111}	1.56×10^{15}	−1.25	1.44×10^{15}	−1.15

Table 7.1: Surface energy parameters used in calculating the free energy of platelet formation.

7.2.2: Volume energy of a platelet

It was observed in Section 6.4.3 that the dilation vector of platelets in tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ increases with annealing temperature. Other studies of platelets in relaxed semiconductors suggest a similar result, where published values vary between 0.1 and 1 nm.^{93, 114} In addition to the characteristic Si-H bonds, spectroscopic measurements have shown that platelets contain molecular H_2 after annealing at temperatures $>100^\circ\text{C}$.¹²⁵ H_2 molecules can accumulate within a platelet causing an increase of the dilation vector and an energy release corresponding to the enthalpy of H_2 solution. Consequently, as the magnitude of b and the condensation of gaseous H_2 become significant, the free energy change during platelet ripening eventually becomes dominated by the volume, i.e. second term in Equation 7.2.

7.2.3: Strain energy of a platelet in a relaxed layer

The strain energy, i.e. third term in Equation 7.2 assumes that the platelet medium is elastically isotropic. However, monocrystalline semiconductors are perceptibly anisotropic with elastic constants that vary with orientation. Table 2.1 contains the stiffness constants for Si and Ge that are needed to calculate the strain energy of a platelet. An isotropic shear modulus can be calculated in terms of these constants depending on platelet orientation. Since the dilation vector of a platelet is purely edge in nature, the modulus used for calculating its intrinsic strain energy should be the one for shear applied to the plane perpendicular to the platelet habit plane and in the direction of the dilation vector. For the case of a $\{100\}$ platelet:

$$\mu^{\{100\}} = \frac{1}{C_{44}} \quad \text{Equation 7.6}$$

7.2.4: Strain energy of a platelet in a strained layer

The free energy expression given by Equation 7.2 can be modified to describe platelet formation in the presence of strain. The coherency energy released during the formation of an arbitrarily oriented platelet in a strained layer is given by:

$$W_{\varepsilon}(r) = -\pi r^2 \left[\mu b \frac{(1+\nu)}{(1-\nu)} \right] \varepsilon \cos \lambda \cos \phi \quad \text{Equation 7.7}$$

The $\cos \lambda \cos \phi$ term in Equation 7.7 accounts for the orientation of the platelet relative to the direction of strain. This term is historically known as the Schmid factor which is used to resolve the shear component of the stress acting on a dislocation in the direction of slip. For the case of a platelet within a layer under biaxial strain¹⁸, λ is the angle between the dilation vector and a line in the film plane which is perpendicular to the intersection of the platelet habit plane and the surface. ϕ is the angle between the surface and the normal to the platelet habit plane. Adding the strain energy released by a platelet to the free energy expression for the relaxed case, Equation 7.8 is obtained. This gives the free energy of platelet formation within a strained layer.

$$\Delta G(r) = 2\pi r^2 \Gamma_p - \beta \pi r^2 b \frac{P}{k_B T} \Delta H_s + \frac{\mu b^2 r}{4} \frac{(2-\nu)}{(1-\nu)} \ln \left(\frac{24r}{e^2 b} \right) - \pi r^2 \left[\mu b \frac{(1+\nu)}{(1-\nu)} \right] \varepsilon \cos \lambda \cos \phi$$

Equation 7.8

Inclusion of the released strain energy term reduces the free energy of platelet formation relative to the relaxed case, resulting in preferential nucleation of platelets having a dilation vector component in the plane of the strained layer. For a (100) or (010) platelet, where the dilation vector is parallel to the film plane, $\lambda = \phi = 0$, resulting in a Schmid factor of 1. Platelets with a {111}-orientation have $\cos \lambda = \cos \phi = \sqrt{6}/3$,

resulting in a Schmid factor of $2/3$. Nucleation of $\{100\}$ platelets with a dilation vector parallel to the biaxial plane are energetically favorable to $\{111\}$ platelets from the perspective of efficient strain relief. Therefore, the “slip system” for strain relaxation of tensile layers via the platelet mechanism is $\langle 100 \rangle \{100\}$.

The intrinsic strain energy of a platelet scales with b^2 while the energy released by their formation has a weaker, linear dependence on b . Therefore platelets undergoing nucleation will tend to minimize their dilation vector. Platelets with a small dilation vector have limited capacity for condensed H_2 molecules; therefore the volume term in Equation 7.8 is negligible for platelet nucleation.

The energy barrier for platelet nucleation is found by maximizing Equation 7.8 relative to the platelet radius. Neglecting the volume term, the ensuing expression for the critical radius (r_c) at which further growth becomes energetically favorable is given by Equation 7.9. This expression is analogous to the Matthews-Blakeslee criterion for dislocation formation since the thickness of the strained layer must be at least $2r_c$ to accommodate a critically-sized platelet, i.e. platelets will not form in layers that are less than $2r_c$ thick. The corresponding activation energy is found by setting $\Delta G_A = \Delta G(r_c)$.

$$r_c = \frac{b^2 \mu (2 - \nu) \left[1 + \ln \left(\frac{24 r_c}{b e^2} \right) \right]}{8\pi [2\Gamma_p (1 - \nu) - b\epsilon\mu(1 + \nu) \cos \lambda \cos \phi]} \quad \text{Equation 7.9}$$

7.3: Numerical nucleation model

The nucleation model presented in the previous section is most conveniently applied to circular platelets in infinitely thick strained layers. However, conventional strained layers have finite thickness and platelets within such layers are generally not circular in shape. Hence, the evolution of platelet microstructure in real systems is generally more complex than what can be easily handled with a closed-form model.

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The particular problem with modeling platelet free energy lies within the strain terms of Equation 7.8. The general approach for determining the strain energy of a platelet involves integration of the strain energy density function over the volume of the defect:

$$W = \frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \sigma_{ij}(x, y, z) \epsilon_{ij}(x, y, z) dx dy dz \quad \text{Equation 7.10}$$

For regularly-shaped platelets in an infinite medium, a closed-form solution to Equation 7.10 can be generally obtained. However, as the diameter of the platelet approaches the boundary of a layer with finite thickness, its strain field interacts with the adjacent cladding layers, requiring the volume integral to be performed piecewise. Furthermore, strain relief only occurs in the central region of a radially expanding platelet. Therefore, a platelet is not expected to grow beyond the boundaries of the strained layer in order to maximize strain relaxation efficiency. It was shown in Chapter 6 that platelets are indeed confined to tensilely strained layers at annealing temperatures of ≤ 200 °C, growing only in the biaxial directions parallel to the film plane. Consequently, a numerical approach is more convenient for evaluating the energy of platelets in gettering layers where strain profiles are discontinuous and platelet geometry is complex. Such an approach will be applied in this section.

Numerical evaluation of the platelet strain energy will be applied to the three cases illustrated in Figure 7.3. First, the strain energy will be calculated for a platelet in relaxed Ge as shown in Figure 7.3a. Then the case for tensilely strained layers will be considered for platelet growth under two geometric constraints. In both cases, the platelet is centered within the tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer. Figure 7.3b illustrates a platelet growing unconfined to the strained layer. An alternate scenario is depicted in Figure 7.3c where the growing platelet is no longer confined to the strained layer upon reaching a Ge/ $\text{Si}_{0.4}\text{Ge}_{0.6}$ interface.

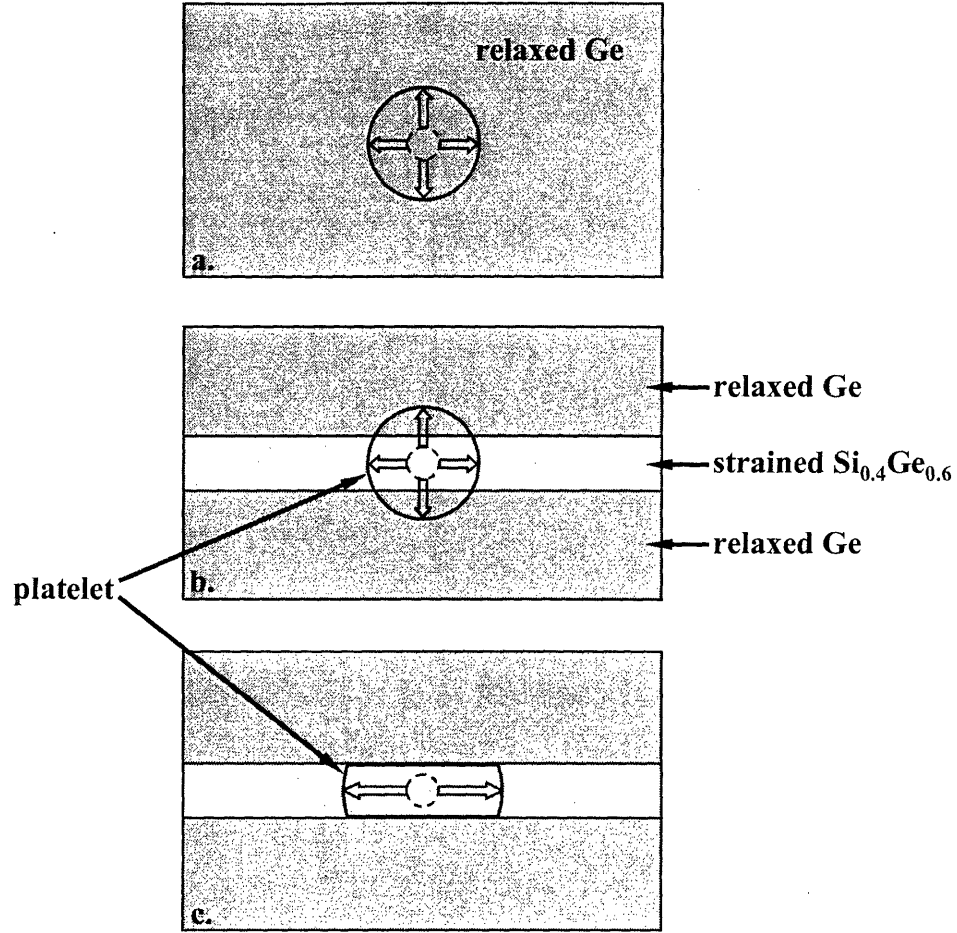


Figure 7.3: Schematics of the three cases considered in the platelet nucleation model. 7.3a illustrates a platelet within a relaxed Ge layer. 7.3b and 7.3c illustrate a platelet within a strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer growing in unconfined and confined configurations, respectively. The dilation vector of the platelet in each of these illustrations points into the page.

The remainder of this section will outline the formulation of the numerical strain energy model, which was constructed using MathematicaTM computational software. The Mathematica notebook containing the simulation source code is provided in Appendix B.

7.3.1: Numerical evaluation of platelet strain energy in a relaxed layer

Closed-form solutions for the strain fields of a circular dislocation loop have been derived by Khraishi, *et al.*¹²⁶. The platelet is modeled as a prismatic dislocation loop

lying in the x-y plane with a dilation vector having a pure edge component oriented in the z direction. The geometrical aspects of the present model are given in Figure 7.4.

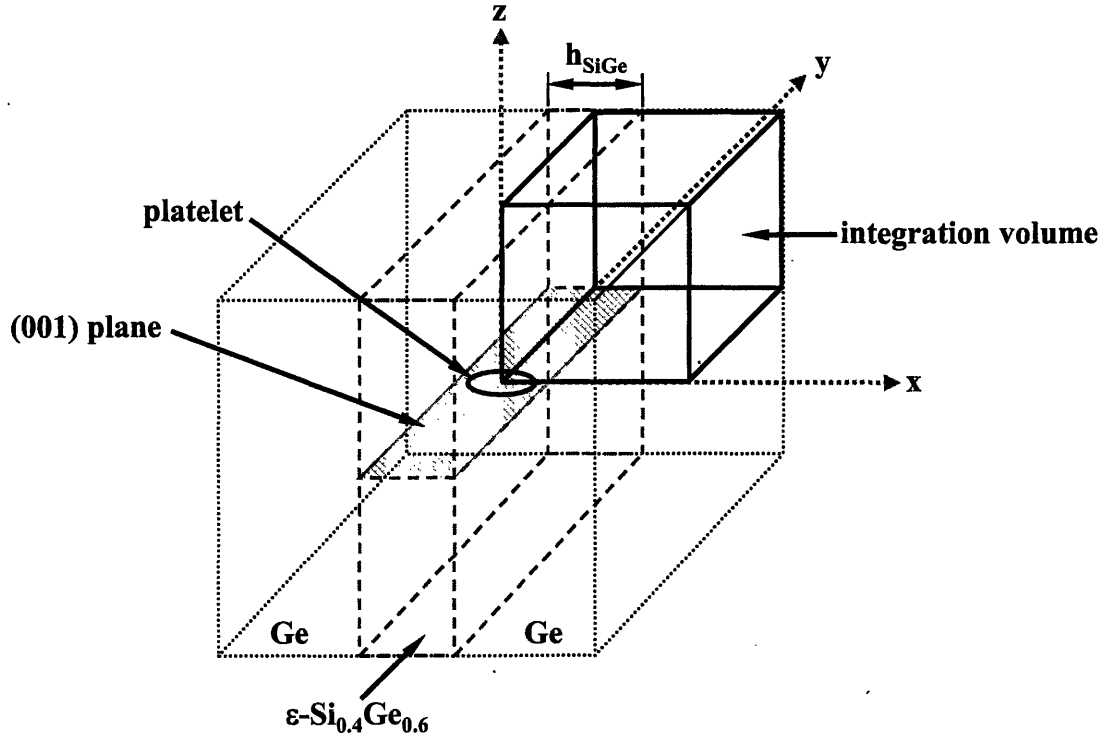


Figure 7.4: Geometric aspects of the platelet simulation model. The platelet is centered at $x=y=z=0$ and has an x-y habit plane. The volume over which the strain energy density is integrated reduced to 1/8 of the total volume owing to the symmetry of the model.

The stress components of the platelet were found using Khraishi's isotropic solution for strain with the anisotropic form of Hooke's Law, where C_{ijkl} is the stiffness tensor:

$$\sigma_{ij} = C_{ijkl} \epsilon_{kl} \quad \text{Equation 7.11}$$

The strain energy density is calculated at a point (x, y, z) by summing the individual strain energy components of the platelet. For a platelet in a relaxed semiconductor, the strain energy density function is given by Equation 7.12 where $(\sigma_p)_{ij}$ and $(\epsilon_p)_{ij}$ are the stress and strain components, respectively, of a platelet having a radius r .

$$w_0(x, y, z, r) = \frac{1}{2} \sum_i \sum_j (\sigma_p)_{ij}(x, y, z, r) (\epsilon_p)_{ij}(x, y, z, r) \quad \text{Equation 7.12}$$

The strain energy of a platelet in a relaxed layer is found by integrating Equation 7.12 over the volume of the platelet, yielding Equation 7.13 where z_0 is the integration cutoff parameter of the platelet core. For diamond cubic semiconductors, a cutoff parameter of $b/3$ was used. Owing to the symmetry of the model, the integral was evaluated only in one octant of the volume as indicated in Figure 7.4. Therefore, the factor of 8 accounts for the volume not included in the integral, yielding the strain energy for the entire platelet.

$$W_0(r) = 8 \int_{z=z_0}^{\infty} \int_{y=0}^{\infty} \int_{x=0}^{\infty} w_0(x, y, z, r) dx dy dz \quad \text{Equation 7.13}$$

Evaluation of Equation 7.13 was carried out numerically by discretization of the spatial coordinates of the volume integral. Since the platelet energy density changes rapidly near the core of the platelet, a small element size was required to maintain accuracy of the result. However, to minimize computation time, the volume element was reduced only within $3b$ of the platelet core. The platelet strain energy is now given by Equation 7.14, where the superscript N indicates that the integral is evaluated numerically. The first summation accounts for the energy near the platelet core using a

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small volume element, ΔV_c , while the second summation incorporates the remainder of the volume using a larger element size, ΔV .

$$W_0^N(r) = 8 \left[\sum_k \sum_j \sum_i w_0(x_c(i), y_c(j), z_c(k), r) \Delta V_c + \sum_k \sum_j \sum_i w_0(x_c(i), y_c(j), z_c(k), r) \Delta V \right] \quad \text{Equation 7.14}$$

After numerical evaluation of the strain energy, the free energy of a platelet in a relaxed layer is given by:

$$\Delta G_0^N(r) = 2\pi r^2 \Gamma_p - \beta \pi r^2 b \frac{P}{k_B T} \Delta H_s + W_0^N(r) \quad \text{Equation 7.15}$$

This expression is the numerical equivalent to the closed form model given by Equation 7.2. The closed-form solution is suitable for platelets in bulk semiconductors; however the numerical version was derived here for completeness. The same procedure will now be used to obtain the numerical free energy model for a platelet within a strained layer.

7.3.2: Numerical evaluation of platelet strain energy in a strained layer

The energy of a platelet within a strained layer was evaluated by superimposing the strain fields of the platelet with the strain profile of a $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer buried in relaxed Ge. The combined platelet/film strain field is given by Equation 7.16, where $(\epsilon_p)_{ij}$ and $(\epsilon_F)_{ij}$ are the strain components of the platelet and $\text{Ge}/\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ structure, respectively.

$$(\epsilon_{PF})_{ij}(x, y, z, r) = (\epsilon_p)_{ij}(x, y, z, r) + (\epsilon_F)_{ij}(x, y, z) \quad \text{Equation 7.16}$$

The stress components of the platelet within the strained layer are found using Hooke's law, leading to the combined strain energy density:

$$w_{PF}(x, y, z, r) = \frac{1}{2} \sum_i \sum_j (\sigma_{PF})_{ij}(x, y, z, r) (\epsilon_{PF})_{ij}(x, y, z, r) \quad \text{Equation 7.17}$$

The free energy of platelet formation only accounts for changes in the energy of the system. Therefore, the net strain energy of a platelet within a strained layer is defined as the change in energy of the platelet and film relative to the system before introduction of the platelet. This is given by Equation 7.18, where $W_{PF}(r = 0)$ is the volume integral of Equation 7.17 prior to insertion of the platelet.

$$W_\epsilon(r) = 8 \int_{z=z_0}^{\infty} \int_{y=0}^{\infty} \int_{x=0}^{\infty} w_{PF}(x, y, z, r) dx dy dz - W_{PF}(r = 0) \quad \text{Equation 7.18}$$

In a manner similar to the relaxed case, Equation 7.18 was evaluated numerically yielding the strain energy of a platelet within a strained layer. The strain energy of the confined platelet depicted in Figure 7.3c was found by defining each of the strain components of the platelet in piecewise fashion as detailed in the Mathematica code given in Appendix B. Replacing the strain terms in Equation 7.8 with the numerically evaluated result, the free energy of platelet formation within a strained layer is given by:

$$\Delta G_\epsilon^N(r) = 2\pi r^2 \Gamma_p - \beta \pi r^2 b \frac{P}{k_B T} \Delta H_s + W_\epsilon^N(r) \quad \text{Equation 7.19}$$

The difference between the relaxed and strained case lies solely in the final term of Equation 7.19. The numerical version of the platelet free energy enables its evaluation

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for platelets geometrically confined to strained gettering layers. The simulation results for platelets in relaxed and strained layers will now be presented.

7.4: Simulation results

This section will apply the thermodynamic models that were developed for platelet formation in tensilely strained layers. The simulation results for nucleation and growth will be considered independently.

In Section 7.4.1, the closed-form expression for the free energy will be used to study platelet nucleation. Assuming that the platelet nuclei are much smaller than the thickness of a tensilely strained gettering layer, the closed form solution (Equation 7.8) will be used. An approximate value for Γ_p was derived in Section 7.2.1 based on the formation energy of a hydrogen-terminated semiconductor surface. For platelets growing in a $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer, linear interpolation yields a value of $\Gamma_p^{\{100\}} = -1.03 \text{ eV/nm}^2$.

Furthermore, platelets have a small dilation vector during nucleation and contain negligible amounts of condensed hydrogen. Therefore, the volume term offers only a small contribution to the free energy and will be neglected in the nucleation model.

In Section 7.4.2, the numerical model will be applied to evaluate the strain energy of a platelet during ripening. During ripening, the dilation vector increases by nearly an order of magnitude as molecular hydrogen condenses within the platelet. In this case, it is necessary to include the volume term to account for the energy released by the phase separation of gaseous H_2 from the semiconductor. However, a comparison of the strain energy under various geometric constraints is sufficient to reveal a transition between strain-limited and diffusion-limited growth. Therefore, only the strain energy will be considered for platelet growth simulations.

7.4.1: Platelet nucleation

Figure 7.5 is a plot of the free energy of platelet formation as a function of radius for dilation vectors of 0.8 and 1.0 Å. The tensile strain in both cases is 1.6%, corresponding to the lattice mismatch between $\text{Si}_{0.4}\text{Ge}_{0.6}$ and Ge. The Figure illustrates the reduction of the activation energy for platelet nucleation within a tensilely strained layer, thus resulting in their preferential nucleation.

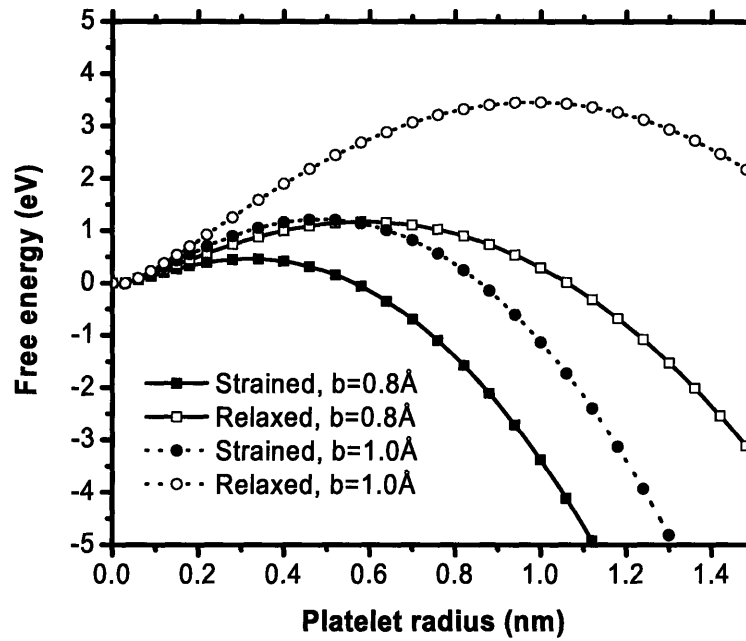


Figure 7.5: Simulation of free energy vs. radius for {100} platelets showing the dependence of the dilation vector on platelet nucleation. $\epsilon=1.6\%$ in both strained layer cases.

The activation energy for the nucleation of a platelet increases with its dilation vector, as shown in Figure 7.6. The plot also illustrates the superior effectiveness of a platelet over a dislocation for the relaxation of tensile strain. The Burgers vector for a dislocation loop is fixed by the lattice constant and crystal structure yielding a value of 3.9 Å for SiGe. This results in an activation energy of >200 eV for the nucleation of a dislocation loop in a $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ layer. While homogeneous nucleation of dislocation

loops is energetically prohibitive, misfits readily form via heterogeneous sources in strained layers grown beyond their critical thickness. The activation energy for glide of an existing dislocation in Ge is 1.65 eV.¹²⁷ In comparison, the activation energy for a platelet with $b=0.8$ Å is only 0.46 eV, thus providing a low energy path for strain relaxation. As a result, massive nucleation of $>10^{10}$ cm⁻² platelets is observed in the gettering layer after annealing at $T < 200$ °C. Furthermore, the critical radius of a platelet with $b=0.8$ Å is only 0.34 nm allowing them to form in very thin layers that would otherwise be stable against relaxation by the aforementioned dislocation mechanism.

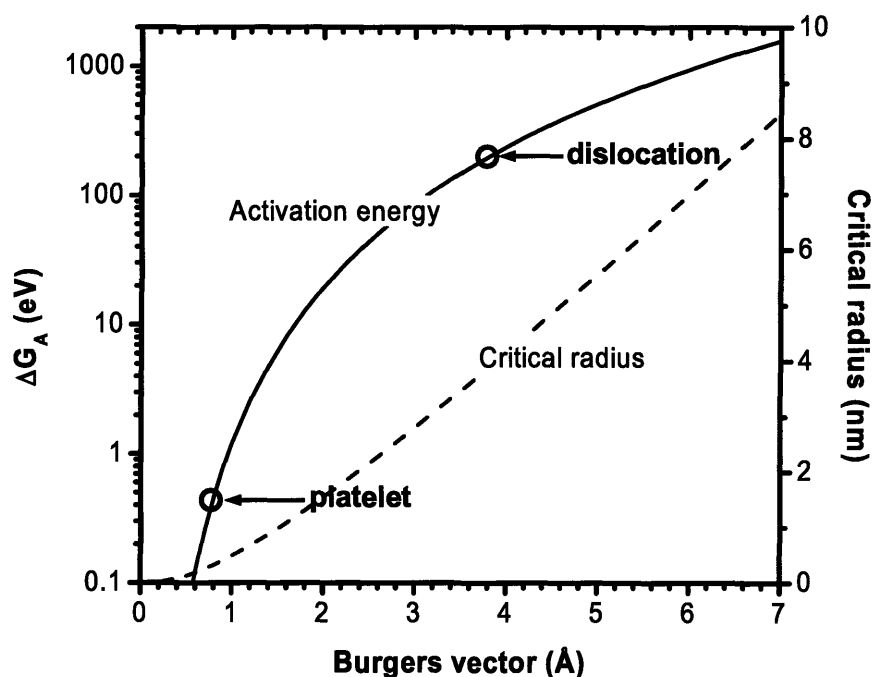


Figure 7.6: Plot of activation energy and critical radius vs. dilation vector for nucleation of a platelet within a tensilely strained Si_{0.4}Ge_{0.6}/Ge layer. Platelets have a dilation vector of 0.8 Å during nucleation and therefore result in a low energy path for relaxation of tensilely strained layers.

The smallest dilation vector that was observed for platelets in tensilely strained Si_{0.4}Ge_{0.6} was 0.8 Å. This also appears to be the ideal value for preferential nucleation relative to a relaxed semiconductor. Figure 7.7 is a plot of activation energy as a function of tensile strain for platelets with dilation vectors varying between 0.6 and 1.0 Å. The

activation energy for a platelet with $b=0.6 \text{ \AA}$ is only $\sim 0.25 \text{ eV}$ and has a relatively weak dependence on strain. The free energy of platelets with a very small dilation vector is dominated by the surface energy term; therefore strain has little influence on their nucleation. Therefore, platelets with $b=0.6 \text{ \AA}$ would nucleate everywhere within the gettering structure, regardless of strain. On the other hand, platelets with $b=1.0 \text{ \AA}$ have a large activation energy and are unlikely to nucleate spontaneously at low temperature. This result provides validation to the model presented herein as platelets with a dilation vector of $0.8 \text{ \AA}$ were experimentally observed to preferentially nucleation in tensilely strained layers.

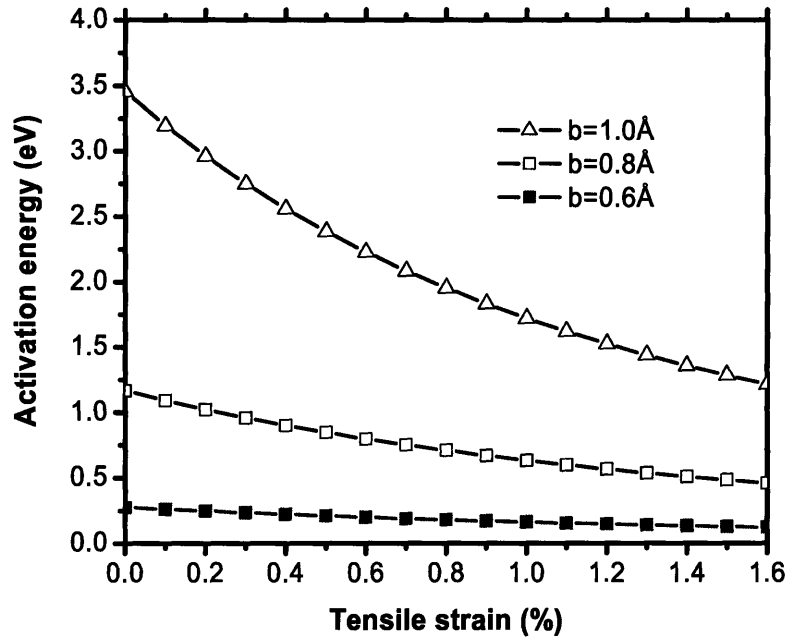


Figure 7.7: Plot of activation energy vs. strain for nucleation of a platelet within a tensilely strained layer. Nucleation of platelets with $b=0.6 \text{ \AA}$ has a weak dependence on strain while platelets with $b=1.0 \text{ \AA}$ have a large activation energy. Experimentally, a dilation vector of $0.8 \text{ \AA}$ is observed for preferential nucleation of platelets in tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layers.

7.4.2: Platelet growth

Upon annealing, the platelets in strained $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ layers ripen to an average length of ~ 40 nm while also undergoing other structural changes. Simulations of platelet strain energy in tensilely strained layers reveal how platelet morphology is affected by the residual strain in the layer and the dilation vector of the platelet. Figures 7.8 and 7.9 are strain contour plots of the superimposed strain fields of a platelet and a tensilely-strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer clad by two relaxed Ge layers. The contours represent lines of constant strain where the incremental strain between each line is constant. The dashed and solid lines indicate tensile (positive) and compressive (negative) values of strain, respectively. The strain fields were plotted using the same conventions given in Figure 7.4. All of the plots are for a platelet with $b=3$ Å and radius, $r=10$ nm where the major radius is used for a platelet confined to the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer. The strain of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer is 1.6%, corresponding to its lattice constant mismatch relative to the relaxed Ge clad.

Figure 7.8 represents the ϵ_{zz} component of strain along the x and y directions at $z=b$ while Figure 7.9 contains plots of the same strain component along the x and z directions at $y=0$. At $r \geq h_{\text{film}}/2$, the confined platelet ceases to expand in the x direction as seen in Figures 7.8b and 7.9b.

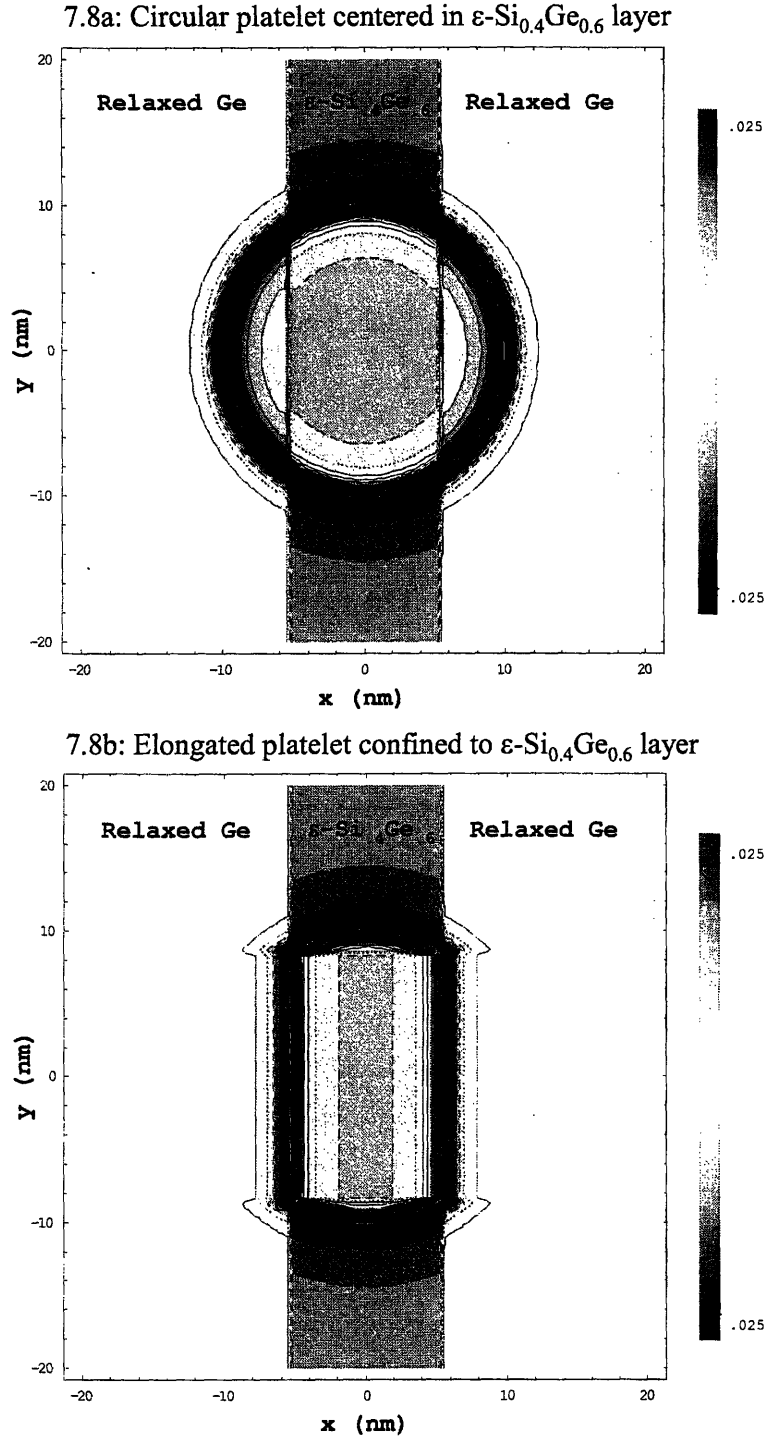


Figure 7.8: Strain contour plots of the ϵ_{zz} component of a platelet unconfined (a.) and confined (b.) to a tensilely strained Si_{0.4}Ge_{0.6} layer. The strain distribution for both cases is plotted as a function of x and y at $z=b$. In these plots, $b=3$ Å (pointing into the page), $\epsilon_{\text{film}}=1.6\%$, $h_{\text{film}}=11$ nm and $r=10$ nm. The areas shaded in red and with dashed contour lines represent regions of tensile strain while those shaded in blue with solid contour lines represent regions of compressive strain.

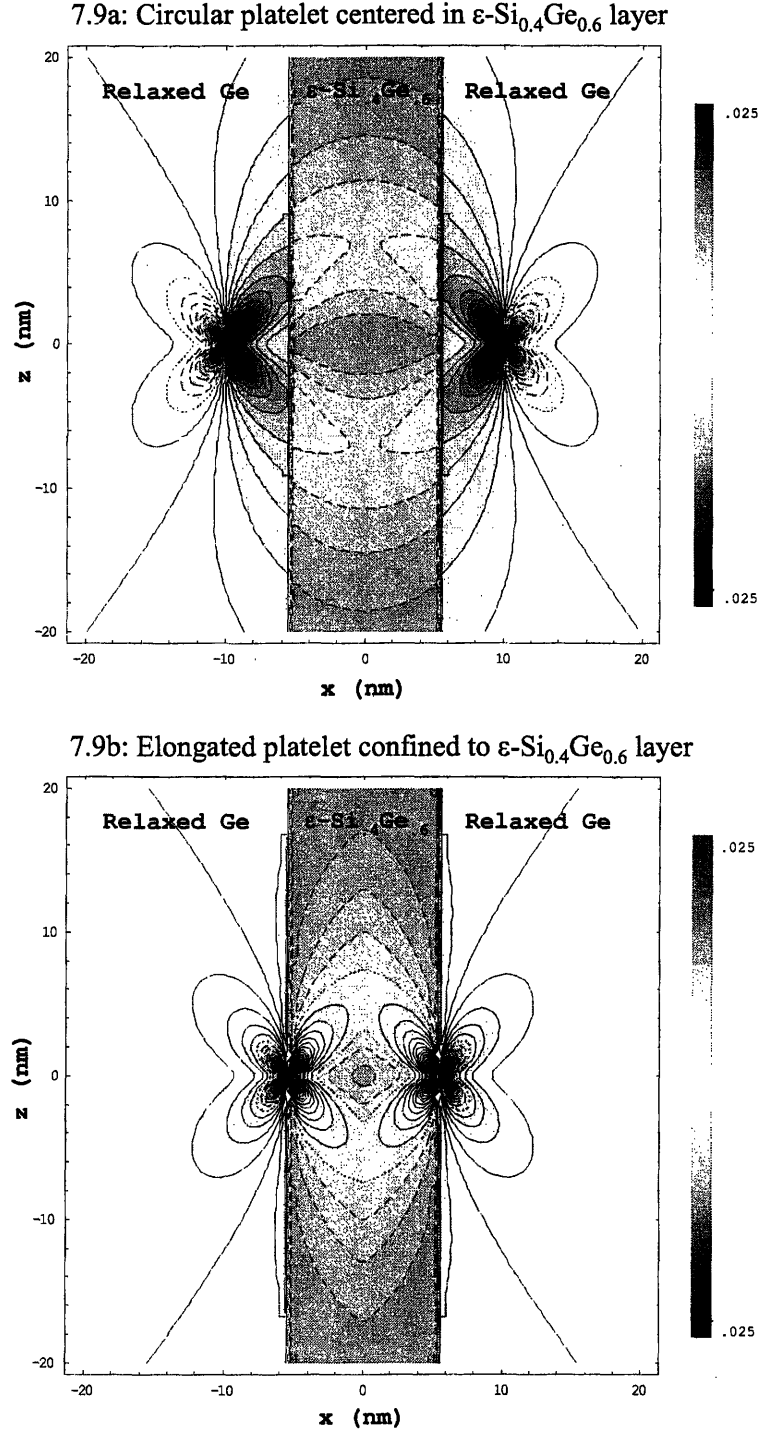


Figure 7.9: Strain contour plots of the ϵ_{zz} component of a platelet unconfined (a.) and confined (b.) to a tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer. The strain distribution for both cases is plotted as a function of x and z at $y=0$. In these plots, $b=3 \text{ \AA}$ (pointing along the z -axis), $\epsilon_{\text{film}}=1.6\%$, $h_{\text{film}}=11 \text{ nm}$ and $r=10 \text{ nm}$. The areas shaded in red and with dashed contour lines represent regions of tensile strain while those shaded in blue with solid contour lines represent regions of compressive strain.

The strain interactions of a platelet with $b=3 \text{ \AA}$ decay significantly over a distance of $\sim 5 \text{ nm}$ from its core. This can be implied from the strain contour plots or by an image force calculation. The interaction of a platelet (or a dislocation) with the sample surface depends on its dilation vector and the distance to the free surface. The image force per unit length on an edge dislocation at a distance h from the surface is given by Equation 7.20.

$$\frac{F_x}{L} = \frac{\mu b^2}{4\pi(1-\nu)h} \quad \text{Equation 7.20}$$

The force per length necessary to propagate a dislocation during heteroepitaxial growth is typically 1000 dyne/cm .¹⁸ Applying this criterion to the largest experimentally observed dilation vector ($b=6 \text{ \AA}$), only platelets within $\sim 4 \text{ nm}$ of the surface are expected to have a strong image force. Since the gettering layer is buried by 500 nm of relaxed Ge and the largest experimentally observed platelet has a radius of only $\sim 25 \text{ nm}$, an image platelet was not included in the strain energy calculation.

The strain energy released by a platelet scales with its area. As a result, any energy comparison of platelets under various geometric constraints requires them to have the same surface area. Therefore, an effective radius (r_{eff}) was defined for the confined platelet in such way that if the platelet was circular, it would have the same surface area as the confined platelet. The effective radius for a confined platelet is given by Equation 7.21, where A is the surface area of the platelet. For a platelet having a circular shape, r_{eff} is simply equal to the radius, r . This convention allows for a direct strain energy comparison of two platelets, regardless of whether they are growing confined to the $\text{Si}_{0.4}\text{Ge}_{0.6}$ host layer.

$$r_{\text{eff}} = \sqrt{\frac{A}{\pi}} \quad \text{Equation 7.21}$$

Chapter 7: Modeling H-induced platelet nucleation

The strain energy of the platelet was simulated for cases with varying dilation vector and biaxial strain. Figure 7.10 is a plot of the platelet strain energy as a function of the effective radius with the strain of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer fixed at 1.6%. Similar to a dislocation loop, a platelet in a relaxed layer will minimize its energy by forming a circular shape. Since the perimeter energy scales as b^2 , the tendency to form circular platelets is particularly strong for large values of b . However, during strain-driven growth within a gettering layer, the lowest energy shape of a platelet is that which relaxes strain most efficiently. This yields a platelet with an elongated shape, as it grows confined to the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer. As the dilation vector increases, the intrinsic term dominates the strain energy of the platelet. At this point, growth is driven primarily by the surface and volume contributions to the free energy and the unconfined configuration becomes favored as the platelet strives to minimize its total energy.

According to the results plotted in Figure 7.10, platelets with $b < 2 \text{ \AA}$ growing confined to the strained layer cause a negative change in strain energy. Therefore, strain relaxation is sufficient for the propagation of platelets with a small dilation vector thus defining the strain-limited growth regime. However, platelets with $b \geq 3 \text{ \AA}$ have positive strain energy and require a contribution from the surface and volume terms to propagate. Since these terms require H_2 diffusion, the growth regime of platelets with $b \geq 3 \text{ \AA}$ is characterized as being diffusion-limited.

Figure 7.10 also illustrates a cross-over point where the strain energy of a platelet is equal in both confined and unconfined configurations. A platelet with an effective radius below the cross-over point is therefore expected to grow confined to the strained layer. The cross-over radius increases with decreasing dilation vector and does not occur for $b = 2 \text{ \AA}$. These results are in good agreement with experiment which revealed that platelet growth along an 11 nm thick strained $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ layer terminates upon reaching a dilation vector of $2.8 \text{ \AA}$. The dilation vector was observed to increase during 250°C annealing, resulting in platelet growth beyond the boundaries of the strained layer.

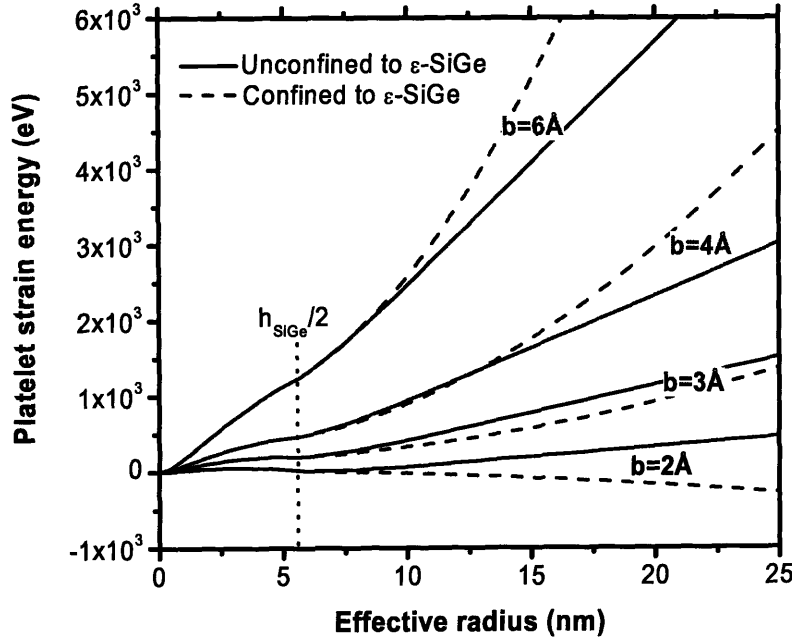


Figure 7.10: Net strain energy of a platelet growing in confined and unconfined configurations relative to a strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer with thickness $h_{\text{SiGe}}=11$ nm. Confined growth is preferred for platelets with small radius and dilation vector. Confined platelets with $b=2$ Å have negative strain energy and are always preferred over the unconfined shape.

The transition between strain-limited and diffusion-limited growth also depends upon the tensile strain of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer, which was observed to relax during platelet growth. Figure 7.11 is a series of strain energy plots for platelets with $b=3$ Å within layers containing various degrees of tensile strain. The amount of strain energy released by a platelet is limited by the initial strain of the layer. As expected, the confined platelet is stable in highly strained layers. As the strain in the gettering layer is relaxed, the strain energy of an unconfined platelet approaches that of a relaxed layer and the effective radius at which cross-over between confined and unconfined growth occurs is reduced.

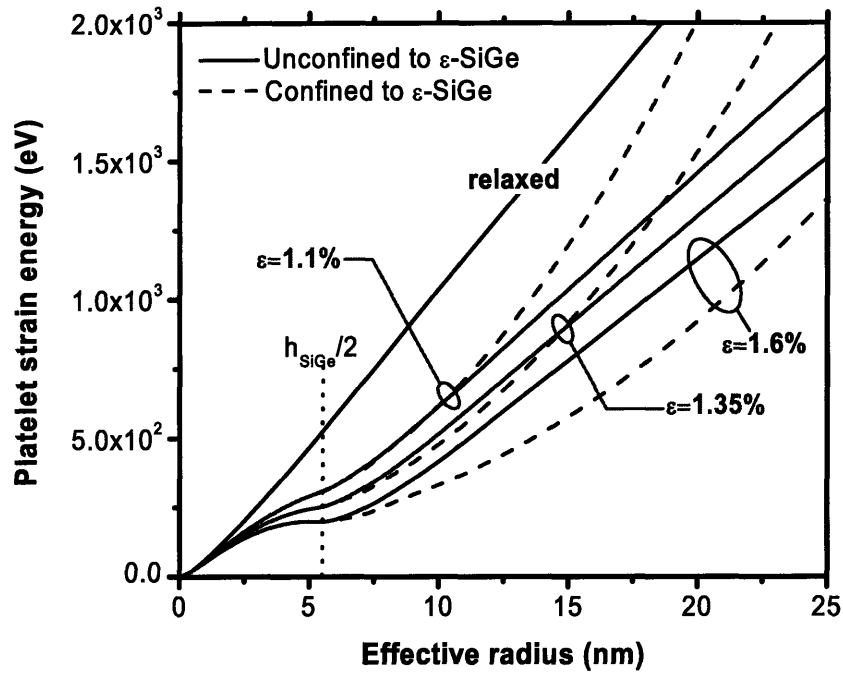


Figure 7.11: Net strain energy of a platelet growing in confined and unconfined configurations relative to a strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer with thickness $h_{\text{SiGe}}=11$ nm. The confined shape is preferred for a platelet growing within highly strained layers.

7.5: Conclusion

Platelet formation within tensilely strained layers can be categorized into three kinetic regimes: nucleation, strain-limited and diffusion-limited growth. The first two are driven by the relaxation of tensile strain, analogous to the nucleation of misfit dislocations in strained layers of supercritical thickness. Diffusion-limited growth and eventual surface blistering are the result of H_2 segregation at the platelet, which requires diffusion of interstitial hydrogen. Each of these regimes is characterized by changes in platelet morphology, as summarized in Figure 7.12.

After nucleation at low temperature ($\sim 50^\circ\text{C}$), the platelet expands to a diameter that is limited by the thickness of the gettering layer, as denoted by the circular platelet in Figure 7.12a. Upon 200°C annealing, strain-limited growth occurs which is characterized by platelet extension along the biaxial directions of the gettering layer.

Strain-limited growth continues until the platelet reaches its critical dilation vector (2.8 Å for an 11 nm thick $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer). During 250 °C annealing, the dilation vector continues to increase and platelet growth becomes driven by hydrogen condensation. Consequently, the platelet grows beyond the boundaries of the strained layer, as shown in Figure 7.12b, to minimize its perimeter energy.

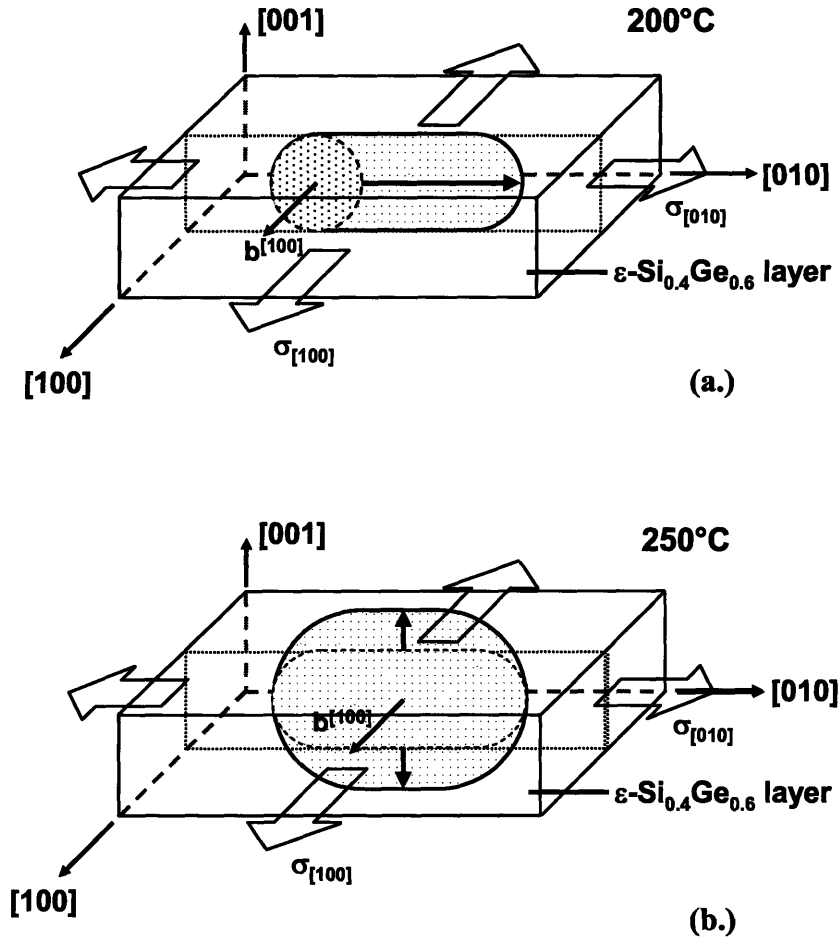


Figure 7.12: Evolution of platelet microstructure during annealing of hydrogen-implanted strained $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$. (a.) Platelet morphology after nucleation (~ 50 °C) by *in-situ* annealing during the H^+ implantation step. After subsequent annealing at 200 °C, the platelets extend along the strained layer. (b.) After annealing at 250 °C, the dilation vector increases beyond the critical value for strain-limited growth and the platelet expands beyond the bounds of the strained layer. Meanwhile, the length of the platelet in the plane of the $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer does not change appreciably.

Chapter 7: Modeling H-induced platelet nucleation

The simulation results presented in this chapter provide valuable insight to the mechanism of platelet-induced strain relaxation. The effectiveness of hydrogen-gettering structures lies in the fact that the dilation vector of a platelet during nucleation is only 0.8 Å. The considerably larger Burgers vector of a dislocation (3.9 Å) allows growth of a gettering layer that is approximately 20× thicker than the critical thickness for platelet nucleation. Upon hydrogen implantation, massive preferential nucleation of platelets is observed in the gettering layer leading to subsequent gettering and improved exfoliation kinetics. In the conventional dislocation mechanism, the only parameters that control strain relaxation are thickness and strain. For the case of platelet growth, the dilation vector is also variable where strain-driven growth occurs for dilation vectors below a critical value. By maximizing the strain and thickness of the gettering layer, the strain-limited regime can be extended to platelets with larger dilation vectors, thus increasing the platelet density and effectiveness of the structure for layer transfer applications.

Chapter 8: Summary and conclusions

The continued increase of the speed and functionality of microelectronic circuits can only be sustained with a concomitant advancement of substrate technologies. As we approach the end of traditional CMOS scaling, further performance improvements will become limited by the intrinsically poor carrier mobility and indirect bandgap of silicon. Engineered substrates enable the flexibility to combine the functionality of several materials onto a single substrate, thereby significantly extending the technology roadmap for silicon.

While this thesis focuses specifically on the integration of relaxed Ge layers with silicon, the techniques which were used are applicable to virtually any material system. Other potential applications include transferal of strained Ge layers or compound semiconductors to large diameter silicon wafers. Furthermore, Chapters 4 and 5 addressed the challenge pertaining to planarization of thin Ge layers which ultimately led to new research directions geared toward improving our understanding of the layer exfoliation process. This chapter summarizes these discoveries and provides a guide for future work.

8.1: Review of results

This work represents the first demonstration of epitaxial Ge transfer onto an arbitrarily-sized silicon wafer using a combination of graded buffer growth and direct wafer bonding. By bonding virtual substrates rather than bulk wafers, the problems of thermal expansion and wafer size mismatch were effectively eliminated. However, prior to applying this method, the challenge of virtual substrate planarization was addressed. This motivated the use of CMP layers for virtual substrate planarization and etch-stop structures for exfoliation damage removal from transferred Ge layers. The incorporation of the etch-stop ultimately led to the discovery of hydrogen gettering by tensilely-strained layers. These gettering structures were demonstrated to improve H-exfoliation kinetics and provided a fundamental understanding of platelet formation in hydrogen-implanted semiconductors. These constitute the three major aspects of the thesis, which are now summarized.

8.1.1: CMP layers for planarization of virtual substrates

The difficulty in planarizing Ge virtual substrates stems from the solubility of GeO_2 in aqueous CMP slurries. This necessitates the use of a deposited CMP layer which enables planarization of virtual substrates prior to bonding. Both oxide and epitaxial Si layers were used to produce surface planarity sufficient for direct bonding of wafers. However, deposited oxide CMP layers require thermal densification at $\geq 600^\circ\text{C}$ prior to bonding which may be unsuitable for some thermally-limited applications, particularly those incorporating SiGe etch-stop layers for selective chemical removal of H-induced exfoliation damage.

8.1.2: Ge donor structures

Layer transfer from Ge virtual substrates requires the H-exfoliation technique since chemical etch-back cannot be readily applied to structures graded to pure Ge. However, this method is accompanied by mechanical damage at the surface of the

transferred layer which must be removed prior to device integration. CMP techniques cannot be readily applied to thin Ge layers, precluding their application to exfoliation damage removal. Consequently, the motivation of Ge donor structures that utilize an etch-stop layer is also rooted in the fact that Ge is difficult to CMP. Incorporation of an etch-stop layer within the uniform composition cap provides a solution to this quandary, allowing the removal of ion damage with a selective etch instead of CMP. In addition, donor structures provide the added benefit of enabling transfer of epitaxially-defined layers, allowing the fabrication of arbitrarily thin device layers on a silicon substrate.

Design of the composition and thickness of the etch-stop layer involves consideration of etch selectivity, critical thickness and thermal stability of the structure as discussed in Chapter 5. Ge etch selectivity relative to $\text{Si}_{1-x}\text{Ge}_x$ layers increases drastically as the Ge fraction of the layer is reduced below $x \sim 0.8$. However, this is accompanied by a concomitant reduction of the critical thickness of the layer, thus limiting its thickness prior to dislocation nucleation. In practice, a $\text{Si}_{0.4}\text{Ge}_{0.6}$ layer provides a Ge etch selectivity of $>150:1$ and can be grown 2-3 times beyond the critical thickness without a degradation of material quality. Some interdiffusion of the etch-stop layer occurs during thermal treatment of the donor structure, however acceptable etch selectivity is maintained until the Ge fraction of the layer exceeds ~ 0.8 . Consequently, a 50% reduction of the Si content is tolerable for maintenance of etch selectivity. Interdiffusion of this extent is observed only after annealing at 650°C for ~ 12 hours, yielding more than sufficient thermal stability of a $\text{Si}_{0.4}\text{Ge}_{0.6}$ etch-stop structure to allow densification of deposited oxide CMP layers.

Growth of donor structures on bulk Ge wafers requires homoepitaxial initiation on a Ge surface. UHVCVD enables the low growth temperatures necessary for initiation of two-dimensional film morphology, leading to planar transfer structures on bulk Ge wafers. Bulk wafers do not suffer from the crosshatched surface roughness inherent to compositionally graded buffers, thereby allowing layer transfer without prior CMP. Homoepitaxial initiation will also be necessary for growth on planarized virtual substrates when the maturation of CMP technology enables the polishing of thin Ge films.

8.1.3: Hydrogen gettering structures

Tensile gettering layers accelerate layer exfoliation kinetics by providing a gathering site for the implanted hydrogen molecules. The gettering behavior of these layers occurs by two independent mechanisms. First, tensile strain increases the solubility of interstitial hydrogen. Secondly, tensile strain also drives the nucleation of platelets which serve as secondary gettering sites for hydrogen. By annealing the implanted gettering structure at low temperature (~ 250 °C), the platelets within the strained layer are compelled to grow with minimal loss of hydrogen to surface effusion. The density of these preferentially nucleated platelets was observed to exceed 10^{10} cm^{-2} in structures implanted to a dose of $6 \times 10^{16} \text{ cm}^{-2} [\text{H}^+]$. The tendency of H_2 entrapment within platelets causes gettering structures to significantly improve the exfoliation efficiency of the implanted hydrogen molecules.

Supplemented with experimental data from Chapter 6, strain energy simulations revealed that the functionality of tensilely strained H-gettering layers lies in the small dilation vector of platelet nuclei. Platelets are analogous to dislocations in that they have the ability to relax strain; however unlike dislocations, platelets are limited to relaxation of tensile strain. Also unlike a dislocation, the dilation vector of a platelet is only $0.8 \text{ \AA}$ during nucleation. This is in stark contrast to the $3.9 \text{ \AA}$ Burgers vector of a dislocation in $\text{Si}_{0.4}\text{Ge}_{0.6}$, leading to a significantly lower activation energy value for platelet nucleation. As a result, tensilely strained layers at the critical thickness for dislocation formation are $\sim 20\times$ thicker than the critical value for platelets. Upon introduction of hydrogen, the latter mechanism becomes active resulting in rampant nucleation of platelets in the tensilely strained gettering layer.

The evolution of platelet microstructure confirms that the layer exfoliation process is driven by the energy released during condensation of gaseous hydrogen within the platelet. During annealing, the platelet fills with gaseous hydrogen and its dilation parameter increases to a point where platelet growth can no longer be justified with a strain relaxation argument. Instead, the free energy of platelet formation becomes driven by the enthalpy of hydrogen dissolution and formation of hydrogenated semiconductor surfaces.

8.2: Suggestions for future work

8.2.1: Advanced donor structures

The Ge layers transferred in this work were strain relaxed, however strained Ge on insulator (ϵ GOI) structures have the greatest potential for high carrier mobility device applications. This is particularly true for the case of tensilely strained layers where both electron and hole mobility enhancements are expected.¹²⁸ Compressive ϵ GOI has been demonstrated by Nakahara, *et al.* using a Ge condensation technique¹²⁹, however only transferal of epitaxially strained Ge layers guarantees high quality, low defect density films. Epitaxially strained Ge layers can only be transferred from virtual substrates, which will inevitably require planarization prior to bonding. Consequently, strained Ge donor structures may also require use of CMP layers necessitating thermal densification treatments. The thermal budget of strained Ge structures was shown to be lower compared to the relaxed case; therefore a triple channel structure⁷⁴ (consisting of strained Ge clad by two strained Si layers) appears to be the most optimistic of all strained GOI solutions.

In addition to high electron mobility, tensilely strained Ge is also predicted to become a direct bandgap semiconductor at a strain of $\sim 1.6\%$.¹²⁸ Tensilely strained Ge requires growth on virtual substrates having a larger lattice constant than Ge, namely InGaAs. Relaxed InGaAs or tensilely-strained Ge layers can be transferred to Si substrates from compositionally graded $\text{In}_{1-x}\text{Ga}_x\text{As}$ layers grown on Ge virtual substrates or from previously transferred Ge films. All of these constitute advanced substrate technologies that have potentially important applications for both electronic and photonic device integration with silicon.

8.2.2: Novel hydrogen gettering structures

While platelets are a necessary component to hydrogen-induced layer exfoliation, not every platelet contributes to surface blistering. A portion of the implanted hydrogen is bound at the internal surfaces of these platelets and consequently does not contribute to

Chapter 8: Summary and conclusions

microcrack pressurization. Tensilely strained gettering layers improve the exfoliation efficiency by providing a large density of platelets near the vicinity of the maximum implanted hydrogen concentration. Further improvements in efficiency could be achieved by preventing platelet nucleation everywhere else. In theory, this could be achieved with compressively strained layers placed on either side of the gettering layer. Compressive strain is predicted to increase the free energy of platelet formation according to the model presented in Chapter 7. Furthermore, the solubility of interstitial hydrogen is reduced in the compressively strained regions compelling implanted hydrogen to migrate to the platelets in the adjacent tensilely-strained gettering layer.

Experiments of this nature may also shed insight to the role of point defects on the hydrogen gettering behavior of tensilely strained layers. It is unclear whether vacancies are involved in the transport of hydrogen to the platelets in the gettering layer. As discussed in Chapter 6, vacancies are expected to migrate away from tensile strain; nonetheless the denuded zone that appears at either side of the gettering layer suggests point defect depletion during annealing of hydrogen implanted gettering structures. The role of point defects can be studied with positron annihilation experiments¹³⁰ to provide vacancy profiles as a function of annealing temperature or by monitoring hydrogen redistribution in the presence of both tensile and compressive strain fields.

Appendix A: Data supplement to Chapter 6

This appendix provides additional experimental data which were omitted from Chapter 6 for clarity. Most of the results presented here are for Sample 3 (a H-gettering structure containing a 24 nm thick strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer). However, additional blistering kinetics data for Sample 2 (a H-gettering structure containing an 11 nm thick strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer) are also provided.

The appendix is separated into three sections. Section A1.1 provides a physical description of Samples 2 and 3. Section A1.2 provides SIMS data for Sample 3 showing the redistribution of ion-implanted hydrogen in the structure as a function of annealing temperature. Finally, Section A1.3 provides additional surface blistering data for Samples 2 and 3 after H^+ implantation.

The purpose of Appendix A is to provide supplemental data which may be useful for planning experiments in future work. It is not intended to present new concepts; however introduction of a new sample warrants comparison to the results presented in Chapter 6. These comparisons will be made in Sections A1.2 and A1.3.

A1.1: Description of Samples 2 and 3

Figure A1.1 is a schematic of the samples discussed in this appendix. Sample 2 was discussed extensively in Chapter 6. It is reintroduced here as a reference for the experimental data presented in subsequent sections of the appendix.

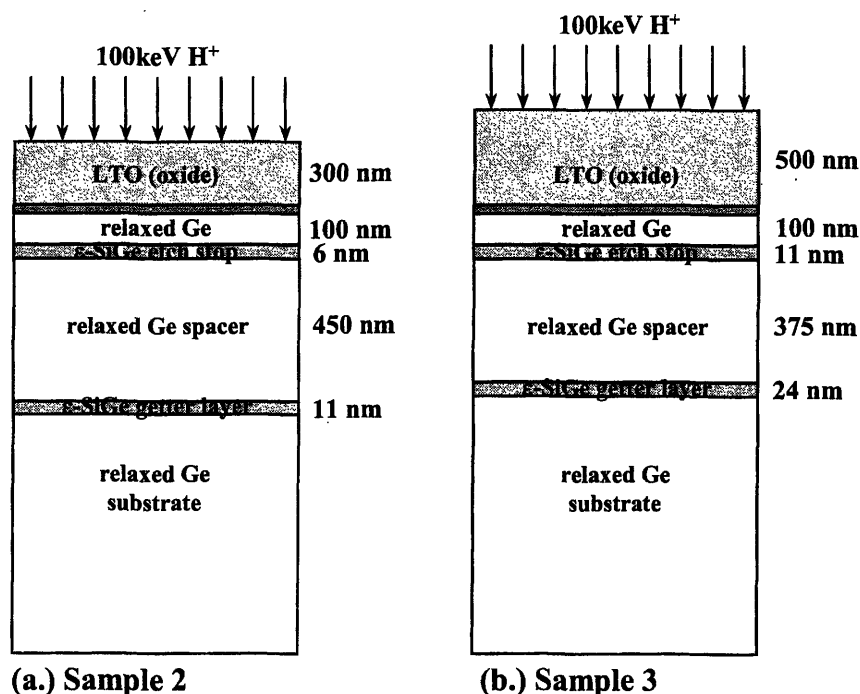


Figure A1.1: Schematics of the H gettering structures discussed in Appendix.

Samples 2 and 3 were both implanted with H⁺ ions to a dose of $6 \times 10^{16} \text{ cm}^{-2}$ at an energy of 100 keV. However, the thickness of the LTO screen for Sample 2 was 300 nm while that of Sample 3 was 500 nm. This resulted in an implanted H distribution in Sample 3 where the maximum H concentration is located precisely upon the Si_{0.4}Ge_{0.6} gettering layer as shown in Figure A1.3.

A1.2: Hydrogen redistribution during annealing

A1.2.1: Sample 2 (11 nm $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer)

Sample 2 comprises an 11 nm thick $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer buried within the uniform composition cap a of relaxed Ge virtual substrate. The behavior of this sample after H^+ ion implantation was discussed in Chapter 6.

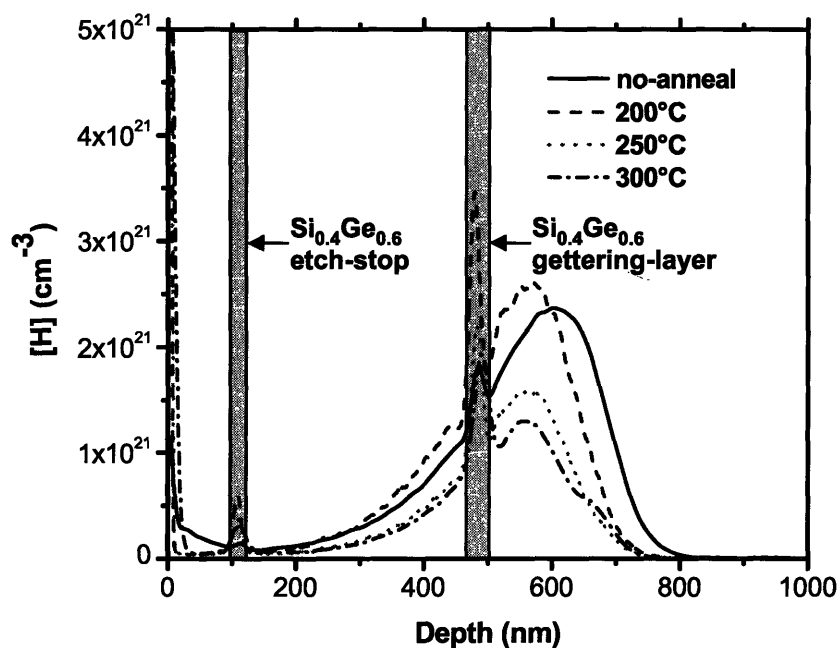


Figure A1.2: SIMS data showing evolution of the $[\text{H}]$ distribution in Sample 2 during low temperature annealing for 1 hour. These profiles are identical to those presented in Figure 6.9b.

A1.2.2: Sample 3 (24 nm $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer)

Sample 3 is similar to Sample 2 except that it comprises a thicker (24 nm) tensilely strained $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer buried below within the relaxed Ge layers. In addition, this sample was implanted through a 500 nm thick LTO screen layer, placing the maximum concentration of the [H] distribution in very close proximity to the strained layer as shown in Figure A1.3.

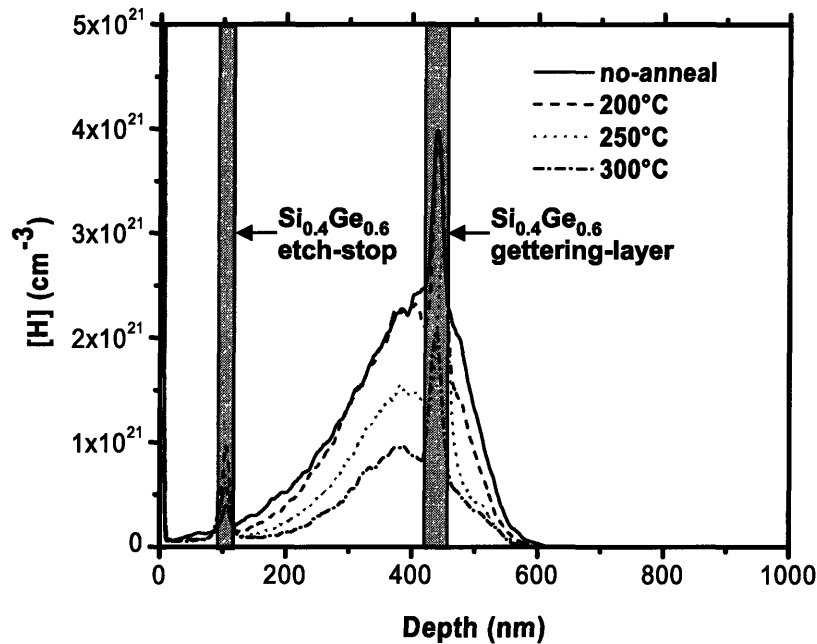


Figure A1.3: SIMS data showing evolution of the [H] distribution in Sample 3 during low temperature annealing for 1 hour.

The maximum hydrogen concentration occurs at the gettering layer upon annealing at a temperature ≤ 200 °C. The apparent decrease in H content after annealing at ≥ 250 °C is due to diffusion of interstitial H_2 which segregates to nearby platelets, as discussed in Chapter 6. Unlike Sample 2, placement of the gettering layer at the depth of highest hydrogen concentration leads to a higher accumulation of hydrogen in the as-implanted state. This is evident upon comparison of the unannealed [H] distributions in

Figures A1.2 and A1.3. However, the relation of the [H] profile in Sample 3 in comparison to that of Sample 2 degrades the gettering efficiency of the strained layer. This behavior is evidenced in the blistering data presented in the next section.

A1.3: Blistering kinetics of H-gettering structures

A1.3.1: Sample 2 (11 nm Si_{0.4}Ge_{0.6} gettering layer)

Figure A1.4 plots the blistering time as a function of annealing temperature of Sample 2 for pre-anneal treatments ranging from 200 to 300 °C for 1 hour. The experimental procedure that was used to obtain these results is presented in Section 6.5.1. This plot is identical to Figure 6.22 except that Sample 1 is omitted here and data for additional pre-anneal temperatures are provided.

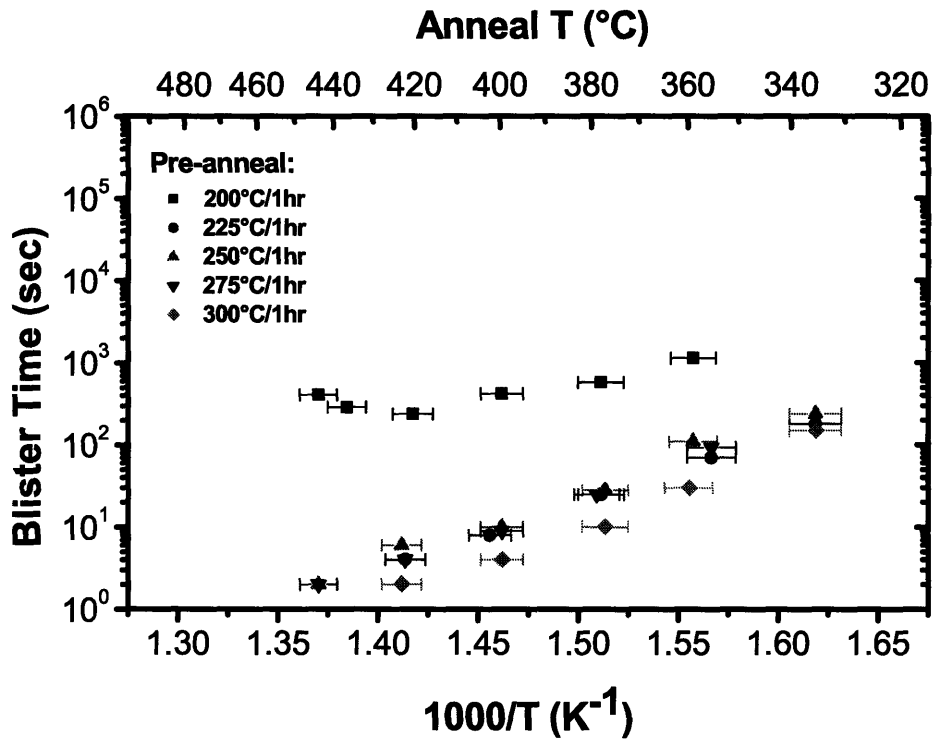


Figure A1.4: Arrhenius plots showing blistering time vs. hotplate annealing temperature for Sample 2 subjected to various pre-anneal temperatures.

Figure A1.5 provides blistering data for Sample 2 plotted as a function of pre-annealing temperature where each curve represents a constant hotplate anneal temperature. A plot of the data in this format will reveal a subtle difference pertaining to the gettering efficiency of Samples 2 and 3. Each curve represents a constant hotplate annealing temperature.

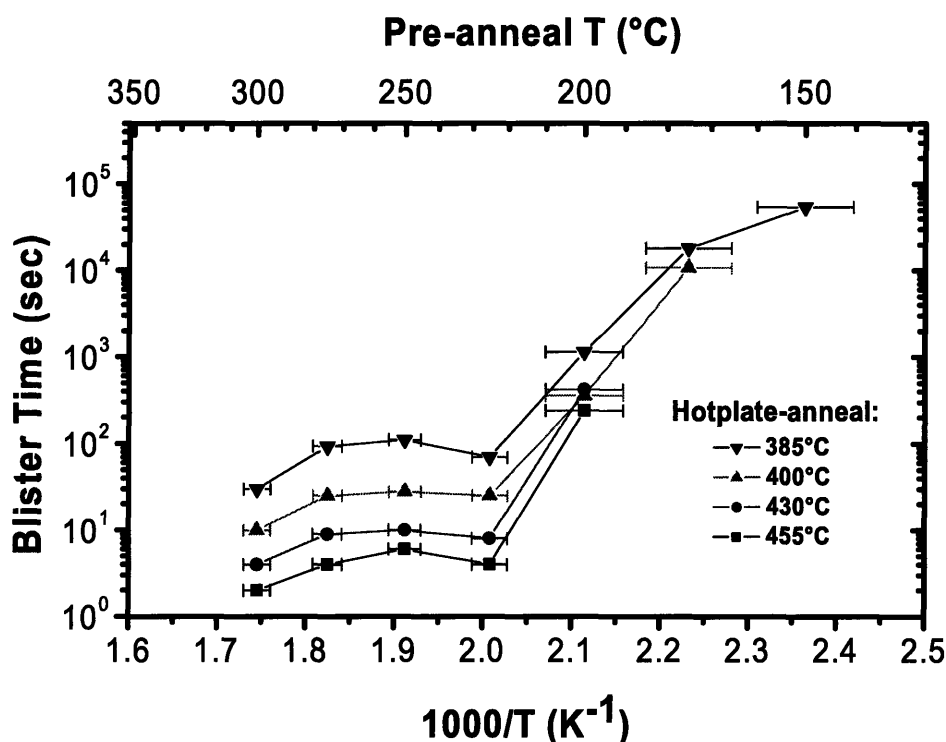


Figure A1.5: Plots of blistering time vs. pre-anneal temperature for Sample 2 subjected to various hotplate anneal temperatures.

Figure A1.5 reemphasizes the critical nature of the pre-anneal treatment to maximize the blistering kinetics of the sample. The blister time is reduced dramatically as the pre-anneal temperature is increased from 150 to 225 °C. Furthermore, as the hotplate annealing temperature is increased, the pre-anneal temperature must also be increased to maintain short blistering times. For example, the sample receiving hotplate anneals of 430 and 455 °C did not blister unless they were previously annealed at a temperature of at least 200 °C. High temperature hotplate anneals result in loss of the

implanted hydrogen to surface effusion and the broadly distributed platelets within the relaxed Ge layers of the sample. Increasing the pre-anneal temperature allows a larger fraction of the implanted hydrogen molecules to become trapped at the large density of platelets confined to the gettering layer, leading to improved blistering kinetics.

A1.3.2: Sample 3 (24 nm $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer)

Figure A1.6 plots the blistering characteristics of Sample 3, containing a 24 nm $\text{Si}_{0.4}\text{Ge}_{0.6}$ gettering layer in close proximity to the maximum concentration of the implanted hydrogen distribution.

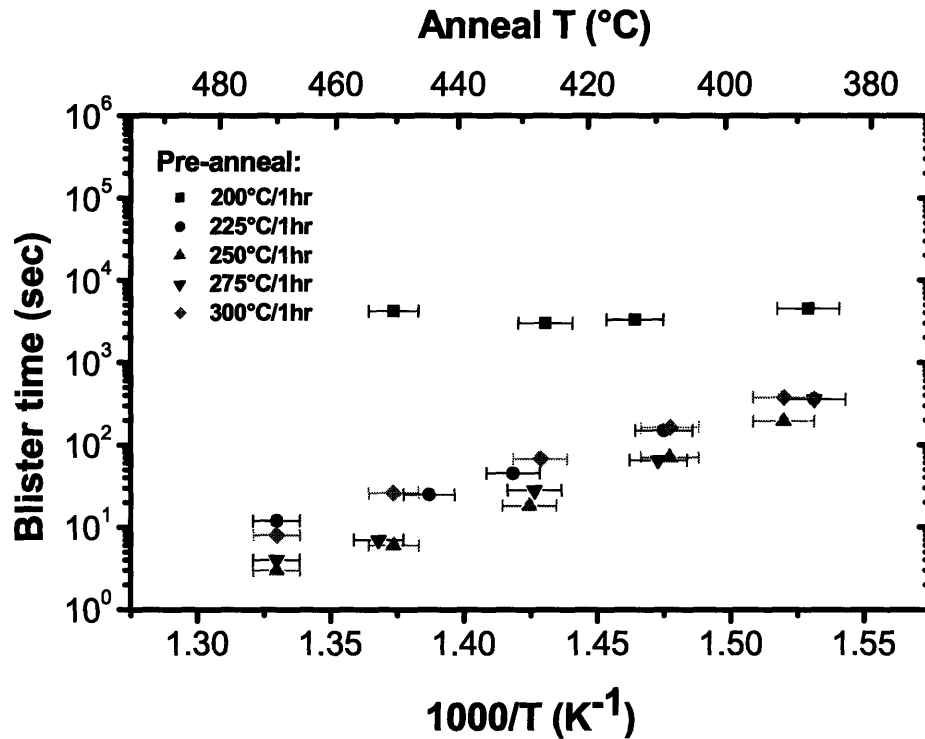


Figure A1.6: Arrhenius plots of blistering time vs. hot-plate annealing temperature for Sample 3 subjected to various pre-anneal temperatures. The pre-anneal time was kept constant at 1 hour.

Appendix A: Data supplement to Chapter 6

Analogous to Figure A1.5, Figure A1.7 plots the blistering data as a function of the pre-anneal temperature. Figure A1.7 illustrates the inferior gettering efficiency of Sample 3 relative to Sample 2. First, blistering of Sample 3 was not observed after pre-annealing at 150 °C. In addition, a gradual increase of the blistering time was observed as the pre-anneal temperature was increased from 250 to 300 °C. This increase becomes more apparent for hotplate annealing temperatures of 430 and 455 °C.

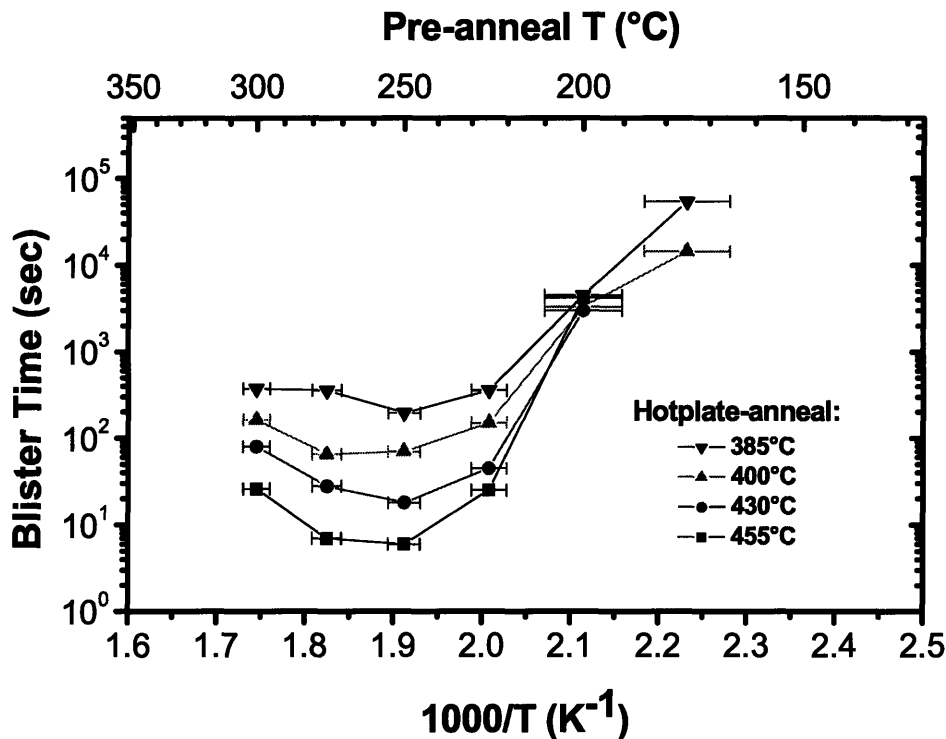


Figure A1.7: Plots of blistering time vs. pre-anneal temperature for Sample 3 subjected to various hotplate anneal temperatures.

The reduction of gettering efficiency in Sample 3 is likely attributable to the proximity of the implanted [H] profile relative to the gettering layer. Since the hydrogen concentration at the gettering layer of Sample 3 is actually higher than that of Sample 2, the vacancies generated during ion implantation are likely contributors to the blistering behavior of the samples. The maximum concentration of vacancies generated during light ion implantation occurs at a depth slightly above the maximum value of [H] as shown in the TRIM simulation result given in Figure 6.6. Furthermore, surface blistering

was not observed in any sample where the maximum implanted hydrogen concentration occurred at a larger depth than that of the gettering layer. This further suggests that ion damage is necessary to produce blistering behavior of $\text{Si}_{0.4}\text{Ge}_{0.6}/\text{Ge}$ gettering structures since the vacancy concentration is negligible in this region (as shown in Figure 6.6). Consequently, the optimal location of the gettering layer for efficient exfoliation kinetics is likely in the region between the maximum concentration of point defects and implanted hydrogen atoms.

Appendix B: Mathematica source code for calculating platelet strain energy

This appendix contains the Mathematica source code for simulating the strain energy of a platelet in a biaxially strained layer. Chapter 7 presents the formulation of the model and provides a discussion of the simulated results. The convention used in this source code is for a (001)-oriented platelet within a layer that is biaxially strained in the [010] and [001] directions. The code was executed using Mathematica Version 6.0.

Platelet Strain Energy Model

This *Mathematica* notebook simulates the strain energy contribution to nucleation and growth of a (001)-oriented platelet in biaxially strained and relaxed semiconductor layers. This version allows forming a smaller mesh near the core of the platelet thus allowing more efficient distribution of computation time. Larger sized mesh elements are used far from the core where strain values change slowly. A comparison is made between platelet growth in relaxed layers and strain-induced growth where the platelet grows within a tensile strained layer. For the case involving strain-induced growth, the platelet is centered in the middle of an epitaxially strained layer that is bound by two relaxed clad layers. The platelet in the strained layer can be modeled with or without confinement to the strained layer. The strain components of the dislocation loop after which the platelet is modeled is defined in a general manner, allowing for the possibility for both prismatic and glide Burgers vector components. This allows the model to also be useful for simulation of layer relaxation by the dislocation mechanism. The solutions to the strain components of the dislocation loop used in this model have been reproduced from T. A. Khraishi, J.P. Hirth, H.M. Zbib, M. A. Khaleel, Int. J. Eng. Sci., 38, 251-266 (2000).

Material constants

The components of the compliance matrix, **C11**, **C12** and **C44Si,Ge** are defined in N/m². The lattice constants, **aSi** and **aGe** are defined in Angstroms.

■ Silicon

```
C11Si = 16.577 × 1010;  
C12Si = 6.393 × 1010;  
C44Si = 7.962 × 1010;  
  
aSi = 5.428;
```

■ Germanium

```
C11Ge = 12.40 × 1010;  
C12Ge = 4.13 × 1010;  
C44Ge = 6.83 × 1010;  
  
aGe = 5.658;
```

■ General

The Poisson's ratio, ν is assumed constant for Si and Ge.

$$\nu = 1 / 3;$$

User-defined parameters

■ Layer structure and calculation parameters

This section defines the geometry and composition of the epitaxial layer structure. The platelet is (001)-oriented (in the x-y plane) with its center at [0, 0, 0]. The strained layer is biaxially strained in the z-y plane and cladded by two relaxed layers of equal thickness. The thickness values of these layers are given by **tFilm** and **tClad**, respectively. The limits of integration in the y and z dimensions are given by **PlateletSpacingY** and **PlateletSpacingZ**, respectively. It is important that the overall dimensions of the matrix are large enough to accomodate the maximum platelet radius evaluated by the simulation. All lengths in this section are defined in Angstrom units.

```
tFilm = 110;  
tClad = 150;  
PlateletSpacingY = 400;  
PlateletSpacingZ = 400;
```

The compositions of the film and caldding layers are defined by **cFilm** and **cClad** (in Ge fraction), respectively. These are used to calculate the elastic and lattice constants of the various layers in the simulated structure. The lattice constants will be further used to calculate the biaxial strain in the buried film.

```
cFilm = 0.6;  
cClad = 1;
```

The degree of film relaxation can be specified by setting **RelaxationFactor** to a non-zero value. This parameter specifies the fraction of the elastic strain that has been relaxed in the strained layer. It can also be used to artificially increase the strain in the layer (by entering a negative number) without changing the composition mismatch between the film and clad layers.

```
RelaxationFactor = 0;
```

The resolution of the matrix in the x, y and z direction is defined by **iStepSize**, **jStepSize** and **kStepSize** respectively. Correspondingly, the stress and strain energy will be calculated in intervals corresponding to a matrix element with a volume given by **iStepSize**×**jStepSize**×**kStepSize**. These, in addition to the overall dimensions of the matrix, will affect the computation time and accuracy of the simulation.


```
iStepSize = 5;  
jStepSize = 5;  
kStepSize = 5;
```

Near the core of the platelet, the resolution is defined in the x, y and z directions by the following respectively:

```
iCoreStepSize = 2;  
jCoreStepSize = 2;  
kCoreStepSize = 2;
```

rAMax defines the maximum platelet radius that the system is evaluated at. **rASize** is the radius interval at which the evaluation is made.

```
rAMax = 120;  
rASize = 4;
```

■ Platelet structure

The following parameters define the Burgers vector, or platelet dilation. **BurgersVectorx**= 0 for a platelet since all the dilation is in the z-axis. A negative **BurgersVectorz** denotes an expansion of the lattice in the direction perpendicular to the platelet habit plane.

```
BurgersVectorx = 0;  
BurgersVectorz = -3;
```

zCutoff is the cutoff distance (corresponding to the cutoff radius in evaluating the strain energy of a dislocation) for which the energy calculation is cutoff near the core of the platelet. Generally for covalently-bonded semiconductors the cutoff radius is taken at **BurgersVectorz**/3. The addition of **kStepSize**/2 accounts for the fact that the strain energy is evaluated at the center of a matrix element. Therefore, it is impossible to evaluate the platelet strain function closer than **zCutoff** without including strain components inside of **BurgersVectorz**/3. Note that **zCutoff** approaches **BurgersVectorz**/3 as the size of the matrix element, **kStepSize** approaches zero.

$$zCutoff = Abs\left[\frac{BurgersVectorz}{3}\right] + \frac{kCoreStepSize}{2};$$

xCutoff and **yCutoff** are cutoff distances in the x and y axes used to avoid the singularities that occur at x, y=0.

```
xCutoff = 1;  
yCutoff = 1;
```

Matrix parameters

This section defines the matrix parameters in terms of the variables already defined in the previous section. Again, all lengths are given in Angstrom units. The compliance and stiffness parameters are given in N/m^2 and m^2/N respectively.

■ Matrix dimensions

The overall dimensions of the matrix in the x, y and z axes are defined by **x**, **y** and **z**, respectively.

```
x = 2 tClad + tFilm;  
y = PlateletSpacingY;  
z = PlateletSpacingZ;
```

The distance from the platelet core in the z-direction at which the mesh size is specified by **iCoreStepSize**, etc. is defined by **CoreLength**. This is typically a few Burgers vectors away from the core.

```
CoreLength = Abs [2 BurgersVectorz] ;
```

Each matrix element is defined by the indices **[[i,j,k]]**. The maximum index is given by **iMax**, **jMax** and **kMax**, respectively. The total number of elements in the matrix is therefore given by **iMax×jMax×kMax**.

```
iMax = Ceiling [  $\frac{\mathbf{x}}{2 \mathbf{iStepSize}}$  ] ;  
jMax = Ceiling [  $\frac{\mathbf{y}}{2 \mathbf{jStepSize}}$  ] ;  
kMax = Ceiling [  $\frac{\mathbf{z}}{2 \mathbf{kStepSize}}$  ] ;  
rMax = Ceiling [  $\frac{\mathbf{rAMax}}{\mathbf{rASize}}$  ] + 1 ;
```

At the core...

```
iCoreMax = Ceiling [  $\frac{\mathbf{x}}{2 \mathbf{iCoreStepSize}}$  ] ;  
jCoreMax = Ceiling [  $\frac{\mathbf{y}}{2 \mathbf{jCoreStepSize}}$  ] ;  
kCoreMax = Ceiling [  $\frac{\mathbf{CoreLength}}{\mathbf{kCoreStepSize}}$  ] ;
```

The following defines the correlation between the matrix elements **[[i,j,k,r]]** and space coordinates **[x,y,z,rA]**.

```
x[i_] := xCutoff + iStepSize (i - 1) ;  
y[j_] := yCutoff + jStepSize (j - 1) ;  
z[k_] := zCutoff + kCoreStepSize  $\left( \mathbf{kCoreMax} - \frac{1}{2} \right)$  + kStepSize  $\left( \mathbf{k} - \frac{1}{2} \right)$  ;  
rA[r_] := rASize (r - 1) ;
```

At the core...

```
xCore[i_] := xCutoff + iCoreStepSize (i - 1) ;  
yCore[j_] := yCutoff + jCoreStepSize (j - 1) ;  
zCore[k_] := zCutoff + kCoreStepSize (k - 1) ;
```

■ Compliance tensor definitions

The Ge fraction c is defined piecewise throughout the matrix. This defines the composition of the epitaxial structure.

$$c[x_, y_, z_] := \text{Which}\left[x \leq -\frac{t_{\text{Film}}}{2}, c_{\text{Clad}}, -\frac{t_{\text{Film}}}{2} < x < \frac{t_{\text{Film}}}{2}, c_{\text{Film}}, x \geq \frac{t_{\text{Film}}}{2}, c_{\text{Clad}}\right];$$

The compliance terms, C_{11} , C_{12} and C_{44} are defined throughout the matrix assuming Vagard's law.

$$\begin{aligned} C_{11}[x_, y_, z_] &:= c[x, y, z] C_{11}\text{Ge} + (1 - c[x, y, z]) C_{11}\text{Si}; \\ C_{12}[x_, y_, z_] &:= c[x, y, z] C_{12}\text{Ge} + (1 - c[x, y, z]) C_{12}\text{Si}; \\ C_{44}[x_, y_, z_] &:= c[x, y, z] C_{44}\text{Ge} + (1 - c[x, y, z]) C_{44}\text{Si}; \end{aligned}$$

The compliance, C_{mn} can now be written in matrix form at every coordinate $[x, y, z]$.

$$C_{mn}[x_, y_, z_] := \begin{pmatrix} C_{11}[x, y, z] & C_{12}[x, y, z] & C_{12}[x, y, z] & 0 & 0 & 0 \\ C_{12}[x, y, z] & C_{11}[x, y, z] & C_{12}[x, y, z] & 0 & 0 & 0 \\ C_{12}[x, y, z] & C_{12}[x, y, z] & C_{11}[x, y, z] & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44}[x, y, z] & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44}[x, y, z] & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{44}[x, y, z] \end{pmatrix};$$

For the case of pure Ge, the compliance matrix is given by $C_{mn}\text{Ge}$.

$$C_{mn}\text{Ge} = \begin{pmatrix} C_{11}\text{Ge} & C_{12}\text{Ge} & C_{12}\text{Ge} & 0 & 0 & 0 \\ C_{12}\text{Ge} & C_{11}\text{Ge} & C_{12}\text{Ge} & 0 & 0 & 0 \\ C_{12}\text{Ge} & C_{12}\text{Ge} & C_{11}\text{Ge} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44}\text{Ge} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44}\text{Ge} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{44}\text{Ge} \end{pmatrix};$$

■ Stiffness tensor definitions

The stiffness matrix components, S_{11} , S_{12} and S_{44} are expressed in terms of the the comppliance terms defined earlier.

$$\begin{aligned} S_{11}[x_, y_, z_] &:= (C_{11}[x, y, z] + C_{12}[x, y, z]) / \\ &\quad ((C_{11}[x, y, z] - C_{12}[x, y, z]) (C_{11}[x, y, z] + 2 C_{12}[x, y, z])); \\ S_{12}[x_, y_, z_] &:= -C_{12}[x, y, z] / \\ &\quad ((C_{11}[x, y, z] - C_{12}[x, y, z]) (C_{11}[x, y, z] + 2 C_{12}[x, y, z])); \\ S_{44}[x_, y_, z_] &:= \frac{1}{C_{44}[x, y, z]}; \end{aligned}$$

The stiffness, S_{mn} can now be written in matrix form at every coordinate $[x, y, z]$.

```

Smn[x_, y_, z_] :=

$$\begin{pmatrix} S11[x, y, z] & S12[x, y, z] & S12[x, y, z] & 0 & 0 & 0 \\ S12[x, y, z] & S11[x, y, z] & S12[x, y, z] & 0 & 0 & 0 \\ S12[x, y, z] & S12[x, y, z] & S11[x, y, z] & 0 & 0 & 0 \\ 0 & 0 & 0 & S44[x, y, z] & 0 & 0 \\ 0 & 0 & 0 & 0 & S44[x, y, z] & 0 \\ 0 & 0 & 0 & 0 & 0 & S44[x, y, z] \end{pmatrix}$$

;

```

Epi-induced strain, stress and energy definitions

■ Layer lattice constants

The lattice constants for the film and clad layer are calculated using Vegard's law.

```

aClad = aGe cClad + aSi (1 - cClad);
aFilm = aGe cFilm + aSi (1 - cFilm);

```

The following defines the lattice constant of the epitaxial structure.

```

aEpiLayers[x_, y_, z_] := aGe c[x, y, z] + aSi (1 - c[x, y, z]);

```

■ Layer strain and strain components

The normal epitaxially-induced strain components in the x, y and z directions are defined by **exxEpiLayers**, **eyyEpiLayers** and **ezzEpiLayers**, respectively.

```

exxEpiLayers[x_, y_, z_] :=
-2 
$$\frac{C12[x, y, z]}{C11[x, y, z]} \frac{(aClad - aEpiLayers[x, y, z]) (1 - RelaxationFactor)}{aEpiLayers[x, y, z]};$$


eyyEpiLayers[x_, y_, z_] := 
$$\frac{(aClad - aEpiLayers[x, y, z]) (1 - RelaxationFactor)}{aEpiLayers[x, y, z]};$$


ezzEpiLayers[x_, y_, z_] := 
$$\frac{(aClad - aEpiLayers[x, y, z]) (1 - RelaxationFactor)}{aEpiLayers[x, y, z]};$$


```

The normal epitaxially-induced stress components are non-zero only the the y and z directions (these are the biaxial directions in the convention used here). They are given by **oyyEpiLayers** and **ozzEpiLayers**, respectively.

```

oyyEpiLayers[x_, y_, z_] :=
  (C11[x, y, z] + C12[x, y, z] -  $\frac{2 C12[x, y, z]^2}{C11[x, y, z]}$ ) eyyEpiLayers[x, y, z];
ozzEpiLayers[x_, y_, z_] :=
  (C11[x, y, z] + C12[x, y, z] -  $\frac{2 C12[x, y, z]^2}{C11[x, y, z]}$ ) ezzEpiLayers[x, y, z];

```

The epitaxially-induced strain and stress components are defined in tensor form as **eEpiLayers** and **σEpiLayers**, respectively.

```

eEpiLayers[x_, y_, z_] :=  $\begin{pmatrix} \text{exxEpiLayers}[x, y, z] \\ \text{eyyEpiLayers}[x, y, z] \\ \text{ezzEpiLayers}[x, y, z] \\ 0 \\ 0 \\ 0 \end{pmatrix};$ 
σEpiLayersT[x_, y_, z_] :=
  ( 0 oyyEpiLayers[x, y, z] ozzEpiLayers[x, y, z] 0 0 0 );

```

Platelet strain and stress field definitions

The platelet strain components are modeled as a Volterra dislocation loop with radius **rA** and Burgers vector **BurgersVectorz**. Closed-form solutions to these strain components have been worked out by T.A. Khraishi, J.P. Hirth, H.M. Zbib and M.A. Khaleel, Int. J. Eng. Sci., 38, 251-266 (2000).

■ General strain parameter definitions

■ Geometric and Elliptic Integral Parameters

```

ρ[x_, y_] :=  $\sqrt{x^2 + y^2}$ ;
rr[x_, y_, z_] :=  $\sqrt{\rho[x, y]^2 + z^2}$ ;
a[x_, y_, z_, rA_] := rr[x, y, z]^2 + rA^2;
b[x_, y_, rA_] := 2 ρ[x, y] rA;
modulusk[x_, y_, z_, rA_] :=  $\sqrt{\frac{2 b[x, y, rA]}{a[x, y, z, rA] + b[x, y, rA]}}$ ;
m[x_, y_, z_, rA_] := modulusk[x, y, z, rA]^2;

```

■ Coefficients (A1-A24)

```

A1[x_, y_, z_, rA_] := x z
(-12 b[x, y, rA]^2 x^2 ρ[x, y]^4 + 3 b[x, y, rA]^4 (x^2 - 6 y^2) + 8 a[x, y, z, rA]^4 (x^2 - 3 y^2) -
4 a[x, y, z, rA] b[x, y, rA]^2 ρ[x, y]^2 (-7 x^2 + y^2 + 2 ν ρ[x, y]^2) +
4 a[x, y, z, rA]^3 ρ[x, y]^2 (-3 x^2 + y^2 + 2 ν ρ[x, y]^2) +
a[x, y, z, rA]^2 (-4 x^2 ρ[x, y]^4 + b[x, y, rA]^2 (-15 x^2 + 42 y^2))) / (16 π ρ[x, y]^6
(-1 + ν) (a[x, y, z, rA] - b[x, y, rA])^2 (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A2[x_, y_, z_, rA_] :=
-x z (8 a[x, y, z, rA]^3 (x^2 - 3 y^2) - 4 b[x, y, rA]^2 ρ[x, y]^2 (-4 x^2 + y^2 + 2 ν ρ[x, y]^2) +
4 a[x, y, z, rA]^2 ρ[x, y]^2 (-3 x^2 + y^2 + 2 ν ρ[x, y]^2) +
a[x, y, z, rA] (-4 x^2 ρ[x, y]^4 + b[x, y, rA]^2 (-9 x^2 + 24 y^2))) / (16 π ρ[x, y]^6
(-1 + ν) (a[x, y, z, rA] - b[x, y, rA]) (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A3[x_, y_, z_, rA_] :=
(x^2 (-4 a[x, y, z, rA]^3 z^2 + 8 a[x, y, z, rA] b[x, y, rA]^2 z^2 + (1 - 2 ν)
(7 a[x, y, z, rA]^2 b[x, y, rA]^2 - 4 a[x, y, z, rA]^4 - 3 b[x, y, rA]^4)) -
2 ρ[x, y]^2 (3 b[x, y, rA]^2 x^2 z^2 + a[x, y, z, rA]^2 (x^2 z^2 + 2 b[x, y, rA]^2 (1 - 2 ν)) +
a[x, y, z, rA] b[x, y, rA]^2 (z^2 + x^2 (1 - 2 ν)) -
(1 - 2 ν) (a[x, y, z, rA]^4 + b[x, y, rA]^4) + a[x, y, z, rA]^3 (-x^2 - z^2 + 2 x^2 ν))) /
(8 π ρ[x, y]^4 (-1 + ν) (a[x, y, z, rA] - b[x, y, rA])^2
(a[x, y, z, rA] + b[x, y, rA])^(3/2));

A4[x_, y_, z_, rA_] :=
(-x^2 (z^2 (5 b[x, y, rA]^2 - 4 a[x, y, z, rA]^2) + 4 a[x, y, z, rA] (1 - 2 ν)
(b[x, y, rA]^2 - a[x, y, z, rA]^2)) +
2 ρ[x, y]^2 (a[x, y, z, rA] (b[x, y, rA]^2 - a[x, y, z, rA]^2) + x^2 (-a[x, y, z, rA]^2 +
b[x, y, rA]^2 + a[x, y, z, rA] z^2) + z^2 (b[x, y, rA]^2 - a[x, y, z, rA]^2) +
2 ν (a[x, y, z, rA]^2 - b[x, y, rA]^2) (a[x, y, z, rA] + x^2))) /
(8 π ρ[x, y]^4 (-1 + ν) (a[x, y, z, rA] - b[x, y, rA])
(a[x, y, z, rA] + b[x, y, rA])^(3/2));

A5[x_, y_, z_, rA_] :=
x z ((32 a[x, y, z, rA]^4 - 57 a[x, y, z, rA]^2 b[x, y, rA]^2 + 21 b[x, y, rA]^4) y^2 -
2 ρ[x, y]^2 (4 a[x, y, z, rA]^4 - 7 a[x, y, z, rA]^2 b[x, y, rA]^2 +
3 b[x, y, rA]^4 + 8 a[x, y, z, rA] y^2 (a[x, y, z, rA]^2 - 2 b[x, y, rA]^2)) +
4 ρ[x, y]^4 (a[x, y, z, rA] (a[x, y, z, rA]^2 - b[x, y, rA]^2) -
y^2 (a[x, y, z, rA]^2 + 3 b[x, y, rA]^2))) / (16 π ρ[x, y]^6 (-1 + ν)
(a[x, y, z, rA] - b[x, y, rA])^2 (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A6[x_, y_, z_, rA_] :=
(x z (-a[x, y, z, rA] y^2 (32 a[x, y, z, rA]^2 - 33 b[x, y, rA]^2) + 4 ρ[x, y]^2
(2 a[x, y, z, rA] (a[x, y, z, rA]^2 - b[x, y, rA]^2) +
4 a[x, y, z, rA]^2 y^2 - 5 b[x, y, rA]^2 y^2) -
4 ρ[x, y]^4 (a[x, y, z, rA]^2 - b[x, y, rA]^2 - a[x, y, z, rA] y^2))) /
(16 π ρ[x, y]^6 (-1 + ν) (a[x, y, z, rA] - b[x, y, rA])
(a[x, y, z, rA] + b[x, y, rA])^(3/2));

```

```

A7[x_, y_, z_, rA_] := (y^2 (-4 a[x, y, z, rA]^3 z^2 + 8 a[x, y, z, rA] b[x, y, rA]^2 z^2 +
  (7 a[x, y, z, rA]^2 b[x, y, rA]^2 - 4 a[x, y, z, rA]^4 - 3 b[x, y, rA]^4) (1 - 2 v)) -
  2 ρ[x, y]^2 (3 b[x, y, rA]^2 y^2 z^2 + a[x, y, z, rA]^2 (y^2 z^2 + 2 b[x, y, rA]^2 (1 - 2 v)) +
  a[x, y, z, rA] b[x, y, rA]^2 (z^2 + y^2 (1 - 2 v)) -
  (a[x, y, z, rA]^4 + b[x, y, rA]^4) (1 - 2 v) + a[x, y, z, rA]^3 (-y^2 - z^2 + 2 y^2 v))) /
  (8 π ρ[x, y]^4 (-1 + v) (a[x, y, z, rA] - b[x, y, rA])^2
  (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A8[x_, y_, z_, rA_] := (-y^2 (-4 a[x, y, z, rA]^2 z^2 + 5 b[x, y, rA]^2 z^2 +
  4 a[x, y, z, rA] (b[x, y, rA]^2 - a[x, y, z, rA]^2) (1 - 2 v)) +
  2 ρ[x, y]^2 (a[x, y, z, rA] (b[x, y, rA]^2 - a[x, y, z, rA]^2) + a[x, y, z, rA] y^2 z^2 +
  (y^2 + z^2 - 2 v (a[x, y, z, rA] + y^2)) (b[x, y, rA]^2 - a[x, y, z, rA]^2))) /
  (8 π ρ[x, y]^4 (-1 + v) (a[x, y, z, rA] - b[x, y, rA])
  (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A9[x_, y_, z_, rA_] := (x z ((a[x, y, z, rA]^2 + 3 b[x, y, rA]^2) z^2 +
  a[x, y, z, rA] (b[x, y, rA]^2 - a[x, y, z, rA]^2) (1 + 2 v))) / (4 π ρ[x, y]^2
  (1 - v) (a[x, y, z, rA] - b[x, y, rA])^2 (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A10[x_, y_, z_, rA_] :=
  (x z (-a[x, y, z, rA] z^2 + (a[x, y, z, rA]^2 - b[x, y, rA]^2) (1 + 2 v))) /
  (4 π ρ[x, y]^2 (1 - v) (a[x, y, z, rA] - b[x, y, rA]) (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A11[x_, y_, z_, rA_] :=
  (b[x, y, rA]^2 (a[x, y, z, rA]^2 - b[x, y, rA]^2 - 4 a[x, y, z, rA] z^2) +
  2 ρ[x, y]^2 (a[x, y, z, rA]^2 z^2 (5 - 4 v) + a[x, y, z, rA] (3 - 2 v)
  (b[x, y, rA]^2 - a[x, y, z, rA]^2) + b[x, y, rA]^2 z^2 (-1 + 4 v)) +
  8 ρ[x, y]^4 (1 - v) (a[x, y, z, rA]^2 - b[x, y, rA]^2)) / (8 π ρ[x, y]^2 (-1 + v)
  (a[x, y, z, rA] - b[x, y, rA])^2 (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A12[x_, y_, z_, rA_] := (b[x, y, rA]^2 z^2 -
  2 ρ[x, y]^2 (a[x, y, z, rA] z^2 + (1 - 2 v) (a[x, y, z, rA]^2 - b[x, y, rA]^2))) / (8 π
  ρ[x, y]^2 (-1 + v) (a[x, y, z, rA] - b[x, y, rA]) (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A13[x_, y_, z_, rA_] := (x y (a[x, y, z, rA] z^2 (a[x, y, z, rA]^2 - 2 b[x, y, rA]^2) +
  v (4 a[x, y, z, rA]^4 - 7 a[x, y, z, rA]^2 b[x, y, rA]^2 + 3 b[x, y, rA]^4) -
  2 ρ[x, y]^2 (a[x, y, z, rA] v (a[x, y, z, rA]^2 + 3 b[x, y, rA]^2)))) /
  (8 π ρ[x, y]^4 (1 - v) (a[x, y, z, rA] - b[x, y, rA])^2
  (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A14[x_, y_, z_, rA_] := (x y (4 a[x, y, z, rA]^2 z^2 -
  5 b[x, y, rA]^2 z^2 + 4 v a[x, y, z, rA] (a[x, y, z, rA]^2 - b[x, y, rA]^2) -
  2 ρ[x, y]^2 (v (a[x, y, z, rA]^2 - b[x, y, rA]^2) - a[x, y, z, rA] z^2))) /
  (8 π ρ[x, y]^4 (-1 + v) (a[x, y, z, rA] - b[x, y, rA])
  (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A15[x_, y_, z_, rA_] := (y z (a[x, y, z, rA] (a[x, y, z, rA]^2 - b[x, y, rA]^2) -
  z^2 (a[x, y, z, rA]^2 + 3 b[x, y, rA]^2))) / (4 π ρ[x, y]^2 (-1 + v)
  (a[x, y, z, rA] - b[x, y, rA])^2 (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A16[x_, y_, z_, rA_] := (y z (a[x, y, z, rA]^2 - b[x, y, rA]^2 - a[x, y, z, rA] z^2)) /
  (4 π ρ[x, y]^2 (1 - v) (a[x, y, z, rA] - b[x, y, rA]) (a[x, y, z, rA] + b[x, y, rA])^(3/2));

```

```

A17[x_, y_, z_, rA_] := (a[x, y, z, rA]^2 z^2 (7 b[x, y, rA]^2 - 3 a[x, y, z, rA]^2) +
  a[x, y, z, rA] y^2 z^2 (a[x, y, z, rA]^2 - b[x, y, rA]^2) +
  v x^2 (-4 a[x, y, z, rA]^4 + 7 a[x, y, z, rA]^2 b[x, y, rA]^2 - 3 b[x, y, rA]^4) +
  ρ[x, y]^2 (z^2 (a[x, y, z, rA] (a[x, y, z, rA]^2 - b[x, y, rA]^2) -
    2 x^2 (a[x, y, z, rA]^2 - b[x, y, rA]^2) -
    2 x^2 (a[x, y, z, rA]^2 + 3 b[x, y, rA]^2)) + 2 v (a[x, y, z, rA]^2 - b[x, y, rA]^2)
    (-b[x, y, rA]^2 + a[x, y, z, rA] (a[x, y, z, rA] + x^2))) +
  2 ρ[x, y]^4 (a[x, y, z, rA]^2 - b[x, y, rA]^2) (-1 + v) (a[x, y, z, rA] - 2 z^2) +
  4 ρ[x, y]^6 (1 - v) (a[x, y, z, rA]^2 - b[x, y, rA]^2) /
  (8 π ρ[x, y]^4 (-1 + v) (a[x, y, z, rA] - b[x, y, rA])^2
    (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A18[x_, y_, z_, rA_] :=
  (x^2 z^2 (3 a[x, y, z, rA]^2 - 4 b[x, y, rA]^2) + y^2 z^2 (b[x, y, rA]^2 - a[x, y, z, rA]^2) +
  4 v a[x, y, z, rA] x^2 (a[x, y, z, rA]^2 - b[x, y, rA]^2) -
  ρ[x, y]^2 (z^2 (a[x, y, z, rA]^2 - b[x, y, rA]^2 - 2 a[x, y, z, rA] x^2) +
    2 v (a[x, y, z, rA]^2 - b[x, y, rA]^2) (a[x, y, z, rA] + x^2)) -
  2 ρ[x, y]^4 (a[x, y, z, rA]^2 - b[x, y, rA]^2) (1 - v)) / (8 π ρ[x, y]^4 (-1 + v)
    (a[x, y, z, rA] - b[x, y, rA]) (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A19[x_, y_, z_, rA_] := (x z (a[x, y, z, rA] (a[x, y, z, rA]^2 - b[x, y, rA]^2) -
  z^2 (a[x, y, z, rA]^2 + 3 b[x, y, rA]^2))) / (4 π ρ[x, y]^2 (-1 + v)
  (a[x, y, z, rA] - b[x, y, rA])^2 (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A20[x_, y_, z_, rA_] := (x z (a[x, y, z, rA]^2 - b[x, y, rA]^2 - a[x, y, z, rA] z^2)) /
  (4 π ρ[x, y]^2 (1 - v) (a[x, y, z, rA] - b[x, y, rA]) (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A21[x_, y_, z_, rA_] :=
  y z (a[x, y, z, rA]^2 x^2 (43 b[x, y, rA]^2 - 24 a[x, y, z, rA]^2) - 15 b[x, y, rA]^4 x^2 +
  y^2 (8 a[x, y, z, rA]^4 - 14 a[x, y, z, rA]^2 b[x, y, rA]^2 + 6 b[x, y, rA]^4) +
  16 a[x, y, z, rA] x^2 ρ[x, y]^2 (a[x, y, z, rA]^2 - 2 b[x, y, rA]^2) -
  4 ρ[x, y]^4 (-x^2 (a[x, y, z, rA]^2 + 3 b[x, y, rA]^2) +
    a[x, y, z, rA] v (a[x, y, z, rA]^2 - b[x, y, rA]^2))) / (16 π ρ[x, y]^6
    (1 - v) (a[x, y, z, rA] - b[x, y, rA])^2 (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A22[x_, y_, z_, rA_] :=
  y z (a[x, y, z, rA] b[x, y, rA]^2 (25 x^2 - 8 y^2) + 8 a[x, y, z, rA]^3 (-3 x^2 + y^2) +
  4 a[x, y, z, rA] x^2 ρ[x, y]^4 + 2 a[x, y, z, rA]^2 ρ[x, y]^2 (8 x^2 - 2 v ρ[x, y]^2) +
  2 b[x, y, rA]^2 ρ[x, y]^2 (-10 x^2 + 2 v ρ[x, y]^2)) / (16 π ρ[x, y]^6 (-1 + v)
    (a[x, y, z, rA] - b[x, y, rA]) (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A23[x_, y_, z_, rA_] :=
  x y ((-1 + 2 v) (4 a[x, y, z, rA]^4 + 3 b[x, y, rA]^4) - 6 b[x, y, rA]^2 z^2 ρ[x, y]^2 +
  a[x, y, z, rA]^2 (7 b[x, y, rA]^2 (1 - 2 v) - 2 z^2 ρ[x, y]^2) +
  2 a[x, y, z, rA]^3 (-2 z^2 + (1 - 2 v) ρ[x, y]^2) +
  2 a[x, y, z, rA] b[x, y, rA]^2 (4 z^2 - (1 - 2 v) ρ[x, y]^2)) / (8 π ρ[x, y]^4
    (-1 + v) (a[x, y, z, rA] - b[x, y, rA])^2 (a[x, y, z, rA] + b[x, y, rA])^(3/2));

A24[x_, y_, z_, rA_] := x y (4 a[x, y, z, rA] (1 - 2 v) (a[x, y, z, rA]^2 - b[x, y, rA]^2) +
  2 a[x, y, z, rA] z^2 ρ[x, y]^2 + b[x, y, rA]^2 (-5 z^2 + 2 ρ[x, y]^2 (1 - 2 v)) +
  2 a[x, y, z, rA]^2 (2 z^2 - ρ[x, y]^2 (1 - 2 v))) / (8 π ρ[x, y]^4 (-1 + v)
    (a[x, y, z, rA] - b[x, y, rA]) (a[x, y, z, rA] + b[x, y, rA])^(3/2));

```


■ Platelet strain and stress tensors: Platelet in a relaxed layer

The platelet strain components are now written in terms of the previously defined coefficients (A1-A24).

```

exxPlatelet[x_, y_, z_, rA_] :=
  BurgersVectorx (A1[x, y, z, rA] EllipticE[m[x, y, z, rA]] +
    A2[x, y, z, rA] EllipticK[m[x, y, z, rA]]) +
  BurgersVectorz (A3[x, y, z, rA] EllipticE[m[x, y, z, rA]] +
    A4[x, y, z, rA] EllipticK[m[x, y, z, rA]]);
eyyPlatelet[x_, y_, z_, rA_] := BurgersVectorx (A5[x, y, z, rA]
  EllipticE[m[x, y, z, rA]] + A6[x, y, z, rA] EllipticK[m[x, y, z, rA]]) +
  BurgersVectorz (A7[x, y, z, rA] EllipticE[m[x, y, z, rA]] +
    A8[x, y, z, rA] EllipticK[m[x, y, z, rA]]);
ezzPlatelet[x_, y_, z_, rA_] := BurgersVectorx (A9[x, y, z, rA]
  EllipticE[m[x, y, z, rA]] + A10[x, y, z, rA] EllipticK[m[x, y, z, rA]]) +
  BurgersVectorz (A11[x, y, z, rA] EllipticE[m[x, y, z, rA]] +
    A12[x, y, z, rA] EllipticK[m[x, y, z, rA]]);
eyzPlatelet[x_, y_, z_, rA_] := BurgersVectorx (A13[x, y, z, rA]
  EllipticE[m[x, y, z, rA]] + A14[x, y, z, rA] EllipticK[m[x, y, z, rA]]) +
  BurgersVectorz (A15[x, y, z, rA] EllipticE[m[x, y, z, rA]] +
    A16[x, y, z, rA] EllipticK[m[x, y, z, rA]]);
exzPlatelet[x_, y_, z_, rA_] := BurgersVectorx (A17[x, y, z, rA]
  EllipticE[m[x, y, z, rA]] + A18[x, y, z, rA] EllipticK[m[x, y, z, rA]]) +
  BurgersVectorz (A19[x, y, z, rA] EllipticE[m[x, y, z, rA]] +
    A20[x, y, z, rA] EllipticK[m[x, y, z, rA]]);
exyPlatelet[x_, y_, z_, rA_] := BurgersVectorx (A21[x, y, z, rA]
  EllipticE[m[x, y, z, rA]] + A22[x, y, z, rA] EllipticK[m[x, y, z, rA]]) +
  BurgersVectorz (A23[x, y, z, rA] EllipticE[m[x, y, z, rA]] +
    A24[x, y, z, rA] EllipticK[m[x, y, z, rA]]);

```

The platelet strain components are defined in matrix form by `ePlatelet`.

$$ePlatelet[x_, y_, z_, rA_] := \begin{pmatrix} exxPlatelet[x, y, z, rA] \\ eyyPlatelet[x, y, z, rA] \\ ezzPlatelet[x, y, z, rA] \\ 2\,eyzPlatelet[x, y, z, rA] \\ 2\,exzPlatelet[x, y, z, rA] \\ 2\,exyPlatelet[x, y, z, rA] \end{pmatrix};$$

The platelet stress matrix, `σPlatelet` is calculated using Hooke's Law. The following is written in transposed form.

```
σPlatelet[x_, y_, z_, rA_] := Cmn[x, y, z].ePlatelet[x, y, z, rA];
```

The stress matrix must be written in 6×1 form to apply matrix multiplication. Therefore it is transposed by the following.

```
σPlateletT[x_, y_, z_, rA_] := Transpose[σPlatelet[x, y, z, rA]]
```

For the specific case of a platelet in relaxed Ge, the stress matrix is given by $\sigma_{\text{PlateletRelaxedGe}}$.

```
 $\sigma_{\text{PlateletRelaxedGe}}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] := \text{CmnGe}.\epsilon_{\text{Platelet}}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}];$ 
```

Again, the stress matrix must be written in 6×1 form to apply matrix multiplication.

```
 $\sigma_{\text{PlateletRelaxedGeT}}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] := \text{Transpose}[\sigma_{\text{PlateletRelaxedGe}}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}]];$ 
```

■ Platelet strain and stress tensors: Unconfined platelet in a strained layer

This section defines the strain and stress tensors of a platelet that is centered in the middle of the strained epitaxial layer. The strained layer is also cladded by two, equally thick relaxed layers. The platelet is not confined to the strained layer in this particular case. Instead, it continues to grow radially beyond the boundaries of the strained layer.

The individual strain components of the combined platelet and epi-induced strain are calculated by superposition.

```
 $\text{exxPlateletFilm}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] :=$   
     $\text{exxPlatelet}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}] + \text{exxEpiLayers}[\mathbf{x}, \mathbf{y}, \mathbf{z}];$   
 $\text{eyyPlateletFilm}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] :=$   
     $\text{eyyPlatelet}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}] + \text{eyyEpiLayers}[\mathbf{x}, \mathbf{y}, \mathbf{z}];$   
 $\text{ezzPlateletFilm}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] :=$   
     $\text{ezzPlatelet}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}] + \text{ezzEpiLayers}[\mathbf{x}, \mathbf{y}, \mathbf{z}];$   
 $\text{eyzPlateletFilm}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] := \text{eyzPlatelet}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}];$   
 $\text{exzPlateletFilm}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] := \text{exzPlatelet}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}];$   
 $\text{exyPlateletFilm}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] := \text{exyPlatelet}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}];$ 
```

The platelet-film strain and stress components are written in matrix form as $\epsilon_{\text{PlateletFilmConfined}}$ and $\sigma_{\text{PlateletFilmConfined}}$, respectively.

```
 $\epsilon_{\text{PlateletFilm}}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] := \epsilon_{\text{Platelet}}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}] + \epsilon_{\text{EpiLayers}}[\mathbf{x}, \mathbf{y}, \mathbf{z}];$   
 $\sigma_{\text{PlateletFilm}}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] := \text{Cmn}[\mathbf{x}, \mathbf{y}, \mathbf{z}].\epsilon_{\text{PlateletFilm}}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}];$   
 $\sigma_{\text{PlateletFilmT}}[\mathbf{x}_-, \mathbf{y}_-, \mathbf{z}_-, \mathbf{rA}_-] := \text{Transpose}[\sigma_{\text{PlateletFilm}}[\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{rA}]];$ 
```

■ Platelet strain and stress tensor: Platelet confined to a strained layer

Similar to the previous case, this section defines the strain and stress mtrices of a platelet that is centered in the middle of the strained epitaxial layer. However, in this case the platelet is confined to the strained layer, growing only along the y-direction upon reaching the SiGe/Ge boundary. For $\mathbf{rA} < \mathbf{tFilm}/2$, the strain definitions for an unconfined platelet hold true. However, for $\mathbf{rA} \geq \mathbf{tFilm}/2$, the strain functions are defined piecewise.

$$\mathbf{yc}[\mathbf{rA}_-] := \sqrt{\mathbf{rA}^2 - \frac{1}{4} \mathbf{tFilm}^2};$$

```

exxPlateletConfined[x_, y_, z_, rA_] :=
  If[rA <  $\frac{tFilm}{2}$ , exxPlatelet[x, y, z, rA], If[Abs[y] > yc[rA],
    exxPlatelet[x, y, z, rA], If[x > 0, exxPlatelet[ $(x + rA - \frac{tFilm}{2})$ , 0, z, rA],
    exxPlatelet[ $(x - rA + \frac{tFilm}{2})$ , 0, z, rA]]];

eyyPlateletConfined[x_, y_, z_, rA_] := If[rA <  $\frac{tFilm}{2}$ , eyyPlatelet[x, y, z, rA],
  If[Abs[y] > yc[rA], eyyPlatelet[x, y, z, rA], If[x > 0, eyyPlatelet[
     $(x + rA - \frac{tFilm}{2})$ , 0, z, rA], eyyPlatelet[ $(x - rA + \frac{tFilm}{2})$ , 0, z, rA]]];

ezzPlateletConfined[x_, y_, z_, rA_] := If[rA <  $\frac{tFilm}{2}$ , ezzPlatelet[x, y, z, rA],
  If[Abs[y] > yc[rA], ezzPlatelet[x, y, z, rA], If[x > 0, ezzPlatelet[
     $(x + rA - \frac{tFilm}{2})$ , 0, z, rA], ezzPlatelet[ $(x - rA + \frac{tFilm}{2})$ , 0, z, rA]]];

eyzPlateletConfined[x_, y_, z_, rA_] := If[rA <  $\frac{tFilm}{2}$ , eyzPlatelet[x, y, z, rA],
  If[Abs[y] > yc[rA], eyzPlatelet[x, y, z, rA], If[x > 0, eyzPlatelet[
     $(x + rA - \frac{tFilm}{2})$ , 0, z, rA], eyzPlatelet[ $(x - rA + \frac{tFilm}{2})$ , 0, z, rA]]];

exzPlateletConfined[x_, y_, z_, rA_] := If[rA <  $\frac{tFilm}{2}$ , exzPlatelet[x, y, z, rA],
  If[Abs[y] > yc[rA], exzPlatelet[x, y, z, rA], If[x > 0, exzPlatelet[
     $(x + rA - \frac{tFilm}{2})$ , 0, z, rA], exzPlatelet[ $(x - rA + \frac{tFilm}{2})$ , 0, z, rA]]];

exyPlateletConfined[x_, y_, z_, rA_] := If[rA <  $\frac{tFilm}{2}$ , exyPlatelet[x, y, z, rA],
  If[Abs[y] > yc[rA], exyPlatelet[x, y, z, rA], If[x > 0, exyPlatelet[
     $(x + rA - \frac{tFilm}{2})$ , 0, z, rA], exyPlatelet[ $(x - rA + \frac{tFilm}{2})$ , 0, z, rA]]];

```

The strain components of a confined platelet are written in matrix form as $\epsilon_{PlateletConfined}$.

$$\epsilon_{PlateletConfined}[x_, y_, z_, rA_] := \begin{pmatrix} \text{exxPlateletConfined}[x, y, z, rA] \\ \text{eyyPlateletConfined}[x, y, z, rA] \\ \text{ezzPlateletConfined}[x, y, z, rA] \\ 2 \text{eyzPlateletConfined}[x, y, z, rA] \\ 2 \text{exzPlateletConfined}[x, y, z, rA] \\ 2 \text{exyPlateletConfined}[x, y, z, rA] \end{pmatrix};$$

Again, the stress componenets are calculated using Hooke's Law.

$$\sigma_{PlateletConfined}[x_, y_, z_, rA_] := \text{Cmn}[x, y, z] . \epsilon_{PlateletConfined}[x, y, z, rA];$$

The individual components of the combined platelet and epi-induced strain are calculated by superposition.

```
exxPlateletFilmConfined[x_, y_, z_, rA_] :=  
  exxPlateletConfined[x, y, z, rA] + exxEpiLayers[x, y, z];  
eyyPlateletFilmConfined[x_, y_, z_, rA_] :=  
  eyyPlateletConfined[x, y, z, rA] + eyyEpiLayers[x, y, z];  
ezzPlateletFilmConfined[x_, y_, z_, rA_] :=  
  ezzPlateletConfined[x, y, z, rA] + ezzEpiLayers[x, y, z];  
eyzPlateletFilmConfined[x_, y_, z_, rA_] := eyzPlateletConfined[x, y, z, rA];  
exzPlateletFilmConfined[x_, y_, z_, rA_] := exzPlateletConfined[x, y, z, rA];  
exyPlateletFilmConfined[x_, y_, z_, rA_] := exyPlateletConfined[x, y, z, rA];
```

The platelet-film strain and stress components are given in matrix form by `ePlateletFilmConfined` and `σPlateletFilmConfined`, respectively.

```
ePlateletFilmConfined[x_, y_, z_, rA_] :=  
  ePlateletConfined[x, y, z, rA] + eEpiLayers[x, y, z];  
  
σPlateletFilmConfined[x_, y_, z_, rA_] :=  
  Cmn[x, y, z].ePlateletFilmConfined[x, y, z, rA];
```

The transposed matrix is given by the following.

```
σPlateletFilmConfinedT[x_, y_, z_, rA_] :=  
  Transpose[σPlateletFilmConfined[x, y, z, rA]];
```

Platelet strain energy calculations

This section executes the platelet strain energy simulations for platelets in the three geometric configurations described above. In order to maximize computation efficiency, the strain energy density functions (`uPlateletRelaxed[x,y,z,rA]`, `uPlateletFilm[x,y,z,rA]` and `uPlateletFilmConfined[x,y,z,rA]`) are all compiled prior to numerical integration. In each case, the strain energy density is integrated over the volume of the simulation matrix for a value of `rA` that ranges between the values specified in the beginning of the code. The resulting values are placed into a table for platelets in relaxed Ge, platelets unconfined to a strained SiGe layer and platelets that are confined to a strained SiGe layer: `UPlateletRelaxedGe[[r]]`, `UPlateletFilm[[r]]` and `UPlateletFilmConfined[[r]]`, respectively. Finally, the net strain energy is plotted as a function of platelet radius in each case.

■ Case 1: Platelet in relaxed Ge

```
uPlateletRelaxed[x_, y_, z_, rA_] :=  
   $\frac{1}{2}$  σPlateletT[x, y, z, rA].ePlatelet[x, y, z, rA] 6.25×10-12;  
  
Clear[uPlateletRelaxedCompiled]  
  
uPlateletRelaxedCompiled =  
  Compile[{x, y, z, rA}, Evaluate[uPlateletRelaxed[x, y, z, rA]]];  
  
UPlateletRelaxedGe = Table[Null, {r, 1, rMax, 1}];
```

```

For[r = 1, r ≤ rMax, r++, {UPlateletRelaxedGe[[r]] =
  Evaluate[ $\left( \sum_{k=1}^{kCoreMax} \sum_{j=1}^{jCoreMax} \sum_{i=1}^{iCoreMax} (uPlateletRelaxedCompiled[xCore[i], yCore[j], zCore[k], rA[r]] \times iCoreStepSize \times jCoreStepSize \times kCoreStepSize) + \sum_{k=1}^{kMax} \sum_{j=1}^{jMax} \sum_{i=1}^{iMax} (uPlateletRelaxedCompiled[x[i], y[j], z[k], rA[r]] \times iStepSize \times jStepSize \times kStepSize) \right) \times 8$ ],
  Print[rA[r], ": ", UPlateletRelaxedGe[[r]]]];
UPlateletRelaxedGe = Transpose[UPlateletRelaxedGe][[1]];
UPlateletRelaxedGe // MatrixForm
UPlateletRelaxedGePlot = ListPlot[UPlateletRelaxedGe, PlotJoined → True]

```

■ **Case 2: Platelet in strained SiGe layer (platelet expands radially)**

```

uPlateletFilm[x_, y_, z_, rA_] :=
 $\frac{1}{2} \sigma_{PlateletFilmT}[x, y, z, rA] . e_{PlateletFilm}[x, y, z, rA] 6.25 \times 10^{-12}$ ;
Clear[uPlateletFilmCompiled];
uPlateletFilmCompiled =
  Compile[{x, y, z, rA}, Evaluate[uPlateletFilm[x, y, z, rA]]];
UPlateletFilm = Table[Null, {r, 1, rMax, 1}];
For[r = 1, r ≤ rMax, r++, {UPlateletFilm[[r]] =
  Evaluate[ $\left( \sum_{k=1}^{kCoreMax} \sum_{j=1}^{jCoreMax} \sum_{i=1}^{iCoreMax} (uPlateletFilmCompiled[xCore[i], yCore[j], zCore[k], rA[r]] \times iCoreStepSize \times jCoreStepSize \times kCoreStepSize) + \sum_{k=1}^{kMax} \sum_{j=1}^{jMax} \sum_{i=1}^{iMax} (uPlateletFilmCompiled[x[i], y[j], z[k], rA[r]] \times iStepSize \times jStepSize \times kStepSize) \right) \times 8$ ], Print[rA[r], ": ", UPlateletFilm[[r]]]];
UPlateletFilm = Transpose[UPlateletFilm][[1]];
UPlateletFilmNet = Table[UPlateletFilm[[r]] - UPlateletFilm[[1]], {r, 1, rMax, 1}];
UPlateletFilmNet // MatrixForm
UPlateletFilmNetPlot = ListPlot[UPlateletFilmNet, PlotJoined → True];

```

■ Case 3: Platelet confined to strained layer

```

uPlateletFilmConfined[x_, y_, z_, rA_] :=  $\frac{1}{2}$  σPlateletFilmConfinedT[x, y, z, rA].
      ePlateletFilmConfined[x, y, z, rA] 6.25×10-12;

Clear[uPlateletFilmConfinedCompiled];

uPlateletFilmConfinedCompiled =
  Compile[{x, y, z, rA}, Evaluate[uPlateletFilmConfined[x, y, z, rA]]];

UPlateletFilmConfined = Table[Null, {r, 1, rMax, 1}];

For[r = 1, r ≤ rMax, r++, {UPlateletFilmConfined[[r]] = Evaluate[
   $\left( \sum_{k=1}^{kCoreMax} \sum_{j=1}^{jCoreMax} \sum_{i=1}^{iCoreMax} (uPlateletFilmConfinedCompiled[xCore[i], yCore[j], \right.$ 
    zCore[k], rA[r]] × iCoreStepSize × jCoreStepSize × kCoreStepSize) +
     $\sum_{k=1}^{kMax} \sum_{j=1}^{jMax} \sum_{i=1}^{iMax} (uPlateletFilmConfinedCompiled[x[i], y[j], z[k], rA[r]] \times$ 
    iStepSize × jStepSize × kStepSize)  $\left. \right) \times 8$ ],
  Print[rA[r], ": ", UPlateletFilmConfined[[r]]]};

UPlateletFilmConfined = Transpose[UPlateletFilmConfined][[1]];

UPlateletFilmConfinedNet =
  Table[UPlateletFilmConfined[[r]] - UPlateletFilmConfined[[1]], {r, 1, rMax, 1}];

UPlateletFilmConfinedNet // MatrixForm

UPlateletFilmConfinedNetPlot =
  ListPlot[UPlateletFilmConfinedNet, PlotJoined → True];

```

■ Comparison of Cases 1-3

```

Show[UPlateletRelaxedGePlot,
  UPlateletFilmNetPlot, UPlateletFilmConfinedNetPlot];

```

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